

Learning Activity Sheets for General Mathematics

CORE SUBJECT



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Learning Activity Sheets for General Mathematics

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LEARNING ACTIVITY SHEET

Name:		Grade & Section:	
Learning Area:	General Mathematics	Term:	1

Unit 1: Business and Finance

Lesson 1.1. Computing Salary, Wage, and Commission

Objectives:	At the end of the lesson, you should be able to: 1. differentiate among annual salary, hourly wage, commission, and piecework income; 2. accurately compute weekly and monthly wages from an annual salary and hourly rate including overtime pay and allowance; 3. calculate earnings from commission and piecework; and 4. solve word problem involving weekly and monthly wages from an annual salary and hourly rate including overtime pay and allowance
Key Ideas and Examples:	Wage means the remuneration or earnings, however designated, capable of being expressed in terms of money, whether fixed or ascertained on a time, task, piece, or commission basis, which is payable by an employer to an employee under a written or unwritten contract of employment for work done or to be done, or for services rendered or to be rendered. • Hourly wage is the amount earned for each hour worked. • Daily wages are the amount of money a worker earns for each day of actual work. It does not include payment for days the employee did not work (e.g., rest days, holidays, or leaves), unless otherwise required by law or employer policy. Salary refers to a fixed, regular payment—typically distributed monthly, semi-monthly, or weekly—offered to employees regardless of the exact number of hours worked. It is standard for professional, managerial, supervisory, or administrative positions. • Annual salary is the total gross amount of money an employee earns in a full calendar year before tax deductions. • Monthly Salary is the fixed amount an employee earns each calendar month. $\text{Monthly Salary} = \frac{\text{Annual Salary}}{12 \text{ months}}$ $\text{Monthly Salary} = \frac{\text{Daily Wage} \times \text{Factor}}{12 \text{ Months}}$ (DOLE use the Factor 313 days for a 6-day workweek, and 261 days for a 5-day workweek). • A semi-monthly salary is the amount of money paid to a worker twice a month. By law, the payment intervals must not exceed sixteen (16) days. $\text{Semi-Monthly Salary} = \frac{\text{Monthly Salary}}{2}$ • Weekly salary is the income received or computed for a single workweek. $\text{Weekly Salary} = \frac{\text{Annual Salary}}{52 \text{ weeks}}$ $\text{Weekly Salary} = \text{Daily Wage} \times \text{No. of Workdays per Week}$ Premium Pay refers to the additional compensation required by law for work performed within the first eight regular hours on non-working days, such as a scheduled Rest Day, Special Non-Working Day, and Regular Holiday. For work performed during regular 8-hour shifts on non-working days, multiply the basic daily rate by the following standard factors from the DOLE Labor Code rules on premium pay: • Work on a Scheduled Rest Day: $\text{Daily Rate} \times 1.30$ • Work on a Special Non-Working Day: $\text{Daily Rate} \times 1.30$

- Work on a Special Non-Working Day falling on a Rest Day:
 $\text{Daily Rate} \times 1.50$
- Work on a Regular Holiday:
 $\text{Daily Rate} \times 2.00$
- Work on a Regular Holiday falling on a Rest Day:
 $\text{Daily Rate} \times 2.60$
- Work on Double Special Non-Working Day:
 $\text{Daily Rate} \times 1.50$
- Work on a Double Special Non-Working Day falling on a Rest Day:
 $\text{Daily Rate} \times 1.95$
- No Work on a Double Regular Holiday:
 $\text{Daily Rate} \times 2.00$
- Work on a Double Regular Holiday:
 $\text{Daily Rate} \times 3.00$
- Work on a Double Regular Holiday falling on a Rest Day:
 $\text{Daily Rate} \times 3.90$

Overtime Pay refers to the additional compensation an employee must receive for work performed beyond the standard eight (8) hours in a single workday. When work exceeds 8 hours, multiply the hourly premium rate of that specific day by an additional 125% (1.25) for ordinary days, and by an additional 130% (1.30) for Rest Day, Special Non-Working Day and Regular Holiday.

- Overtime on an Ordinary Day:
 $\text{Regular Hourly Rate} \times 1.25$
- Overtime on a Rest Day or Special Non-Working Day:
 $\text{Regular Hourly Rate} \times 1.69$
- Overtime on a Special Non-Working Day that Falls on the Rest Day:
 $\text{Regular Hourly Rate} \times 1.95$
- Overtime on a Regular Holiday:
 $\text{Regular Hourly Rate} \times 2.60$
- Overtime on a Regular Holiday falling on a Rest Day:
 $\text{Regular Hourly Rate} \times 3.38$
- Overtime on a Double Regular Holiday:
 $\text{Regular Hourly Rate} \times 3.38$
- Overtime on a Double Special Non-Working Day:
 $\text{Regular Hourly Rate} \times 1.95$

Night Shift Differential (NSD) - often called night differential or graveyard pay - is the mandatory additional compensation paid to employees who perform work during late-night and early-morning hours. For any work performed between 10:00 PM and 6:00 AM, the employee receives an additional 10% on top of the applicable hourly rate for that day.

$$\text{Hourly Rate} \times 1.10$$

Commission Income refers to earnings calculated as a percentage of total sales. This can be paid as a straight commission or a combination of a base salary plus commission.

Piecework Earnings refer to compensation paid to an employee based on the specific quantity of items or products they complete.

Examples on Computing Weekly Wages from an Annual Salary

1. An office clerk has an annual salary of ₱312,000. What is his/her weekly wage?
Solutions:

$$\begin{aligned} \text{Weekly Wage} &= \frac{\text{Annual Salary}}{52 \text{ weeks}} \\ &= \frac{₱312,000}{52 \text{ weeks}} \\ &= ₱6,000 \end{aligned}$$

2. A manager is offered an annual salary of ₱936,000 for a role requiring a 6-day workweek. To assess the value of the offer, calculate the weekly wage and then determine the equivalent daily wage.

Solutions:

A. Weekly Wage

$$\begin{aligned}\text{Weekly Wage} &= \frac{\text{Annual Salary}}{52 \text{ weeks}} \\ &= \frac{₱936,000}{52 \text{ weeks}} \\ &= ₱18,000\end{aligned}$$

B. Daily Wage

$$\begin{aligned}\text{Weekly Wage} &= \text{Daily Wage} \times \text{No. of Workdays per Week} \\ \text{Daily Wage} &= \frac{\text{Weekly Salary}}{\text{No. of Workdays per Week}} \\ &= \frac{₱18,000}{6 \text{ days}} \\ &= ₱3,000\end{aligned}$$

3. A junior supervisor earns a semi-monthly salary of ₱19,500. The company is transitioning from this semi-monthly pay cycle to a weekly cycle. What will be his/her new weekly wage?

Solutions:

A. Monthly Salary

$$\begin{aligned}\text{Semi-Monthly Salary} &= \frac{\text{Monthly Salary}}{2} \\ \text{Monthly Salary} &= 2 \times \text{Semi-Monthly Salary} \\ &= 2 \times ₱19,500 \\ &= ₱39,000\end{aligned}$$

B. Annual Salary

$$\begin{aligned}\text{Monthly Salary} &= \frac{\text{Annual Salary}}{12 \text{ months}} \\ \text{Annual Salary} &= 12 \text{ Months} \times \text{Monthly Salary} \\ &= 12 \text{ Months} \times ₱39,000 \\ &= ₱468,000\end{aligned}$$

C. Weekly Wage

$$\begin{aligned}\text{Weekly Wage} &= \frac{\text{Annual Salary}}{52 \text{ weeks}} \\ &= \frac{₱468,000}{52 \text{ weeks}} \\ &= ₱9,000\end{aligned}$$

Examples on Computing Weekly Wages from an Hourly Rate

1. A part-time worker earns an hourly wage of ₱100. In one week, he works on a standard 40-hour schedule. What is his total weekly wage?

Solutions:

$$\begin{aligned}\text{Weekly Wage} &= \text{Hourly Rate} \times \text{No. of Hours} \\ &= ₱100 \times 40 \text{ hours} \\ &= ₱4,000\end{aligned}$$

2. An employee earns ₱150 per hour and worked 45 hours this week. The first 40 hours are regular time, and 5 hours are overtime on an ordinary day. He also receives a fixed ₱500 weekly allowance. What is the employee's total weekly wage?

Solutions:

A. Regular Pay

$$\begin{aligned}\text{Regular Pay} &= \text{Hourly Rate} \times \text{No. of Hours} \\ &= ₱150 \times 40 \text{ Hours} \\ &= ₱6,000\end{aligned}$$

B. Overtime Pay

$$\begin{aligned}\text{Overtime Pay} &= \text{Regular Hourly Rate} \times 1.25 \times \text{No. of Hours} \\ &= ₱150 \times 1.25 \times 5 \text{ hours} \\ &= ₱937.50\end{aligned}$$

C. Weekly Wage

$$\begin{aligned}\text{Weekly Wage} &= \text{Regular Pay} + \text{Overtime Pay} + \text{Allowance} \\ &= \text{₱}6,000 + \text{₱}937.50 + 500 \\ &= \text{₱}7,437.50\end{aligned}$$

3. A technician has a ₱200 hourly rate. This week he worked 40 regular hours plus 4 hours of overtime on his scheduled rest day. He also receives a ₱1,000 weekly allowance. Overtime on a rest day is calculated at 1.69 times the regular hourly rate. What is the technician's total wage for the week?

Solutions:

A. Regular Pay

$$\begin{aligned}\text{Regular Pay} &= \text{Hourly Rate} \times \text{No. of Hours} \\ &= \text{₱}200 \times 40 \text{ Hours} \\ &= \text{₱}8,000\end{aligned}$$

B. Overtime Pay

$$\begin{aligned}\text{Overtime Pay} &= \text{Regular Hourly Rate} \times 1.69 \times \text{No. of Hours} \\ &= \text{₱}200 \times 1.69 \times 4 \text{ hours} \\ &= \text{₱}1,352\end{aligned}$$

C. Weekly Wage

$$\begin{aligned}\text{Weekly Wage} &= \text{Regular Pay} + \text{Overtime Pay} + \text{Allowance} \\ &= \text{₱}8,000 + \text{₱}1,352 + 1,000 \\ &= \text{₱}10,352\end{aligned}$$

Examples on Computing Monthly Wages from an Annual Salary

1. An employee has a fixed annual salary of ₱240,000. What is the employee's monthly wage?

Solutions:

$$\begin{aligned}\text{Monthly Wage} &= \frac{\text{Annual Salary}}{12 \text{ months}} \\ &= \frac{\text{₱}240,000}{12 \text{ months}} \\ &= \text{₱}20,000\end{aligned}$$

2. A teacher earns an annual salary of ₱480,000. Calculate his equivalent monthly wage and determine the gross amount he will receive in each of the two semi-monthly pay periods.

Solutions:

A. Monthly Wage

$$\begin{aligned}\text{Monthly Wage} &= \frac{\text{Annual Salary}}{12 \text{ months}} \\ &= \frac{\text{₱}480,000}{12 \text{ months}} \\ &= \text{₱}40,000\end{aligned}$$

Semi-Monthly Wage

$$\begin{aligned}\text{Semi-Monthly Wage} &= \frac{\text{Monthly Wage}}{2} \\ &= \frac{\text{₱}40,000}{2} \\ &= \text{₱}20,000\end{aligned}$$

3. An executive is considering a job with an annual salary of ₱1,200,000. To compare this to their current daily rate of ₱4,000 (based on a 5-day workweek with a 261-day factor), calculate the monthly wage for the new offer and the monthly equivalent of their current daily rate.

Solutions:

A. New Monthly Wage

$$\begin{aligned}\text{Monthly Wage} &= \frac{\text{Annual Salary}}{12 \text{ months}} \\ &= \frac{\text{₱}1,200,000}{12 \text{ months}} \\ &= \text{₱}100,000\end{aligned}$$

B. Current Monthly Wage (Daily based)

$$\begin{aligned} \text{Monthly Wage} &= \frac{\text{Daily Wage} \times \text{Factor}}{12 \text{ Months}} \\ &= \frac{\text{P}4,000 \times 261}{12 \text{ Months}} \\ &= \text{P}87,000 \end{aligned}$$

Examples on Computing Monthly Wages from an Hourly Rate

1. An employee has an hourly rate of P150 (Daily Rate: P1,200) and works a 5-day workweek (261-day factor). During the month, the employee worked 20 hours that fell between 10:00 PM and 6:00 AM. Find the employee's total monthly wage.

Solutions:

A. Base Monthly Wage

$$\begin{aligned} \text{Monthly Wage} &= \frac{\text{Daily Wage} \times \text{Factor}}{12 \text{ Months}} \\ &= \frac{\text{P}1,200 \times 261}{12 \text{ Months}} \\ &= \text{P}26,100 \end{aligned}$$

B. Night Shift Differential (NSD)

$$\begin{aligned} \text{NSD} &= \text{Hourly Rate} \times 0.10 \times \text{No. of Hours} \\ &= \text{P}150 \times 0.10 \times 20 \text{ hours} \\ &= \text{P}300 \end{aligned}$$

C. Total Monthly Wage

$$\begin{aligned} \text{Monthly Wage} &= \text{Base Monthly Wage} + \text{NSD} \\ &= \text{P}26,100 + \text{P}300 \\ &= \text{P}26,400 \end{aligned}$$

2. A technician earns P125 per hour on a 6-day workweek (313-day factor). In a specific month, he was required to work 10 hours on his scheduled Rest Day. What is the technician's total monthly wage?

Solutions:

A. Base Monthly Wage

$$\begin{aligned} \text{Monthly Wage} &= \frac{\text{Daily Wage} \times \text{Factor}}{12 \text{ Months}} \\ &= \frac{\text{P}1,000 \times 313}{12 \text{ Months}} \\ &= \text{P}26,083.33 \end{aligned}$$

B. Rest Day Premium Pay

$$\begin{aligned} \text{Premium Pay} &= \text{Daily Rate} \times 1.30 \\ &= \text{P}1,000 \times 1.30 \\ &= \text{P}1,300 \end{aligned}$$

C. Rest Day Overtime Pay

$$\begin{aligned} \text{Overtime Pay} &= \text{Hourly Rate} \times 1.69 \times \text{No. of Hours} \\ &= \text{P}125 \times 1.69 \times 2 \text{ hours} \\ &= \text{P}422.50 \end{aligned}$$

Total Monthly Wage

$$\begin{aligned} \text{Monthly Wage} &= \text{Base Monthly Wage} + \text{Premium Pay} + \text{Overtime Pay} \\ &= \text{P}26,083.33 + \text{P}1,300 + \text{P}422.50 \\ &= \text{P}27,805.83 \end{aligned}$$

3. A specialized nurse earns P200 per hour with a 261-day factor. This month, he worked 12 hours on a Regular Holiday that fell on his Rest Day. The last 4 hours (which were all overtime) occurred between 10:00 PM and 2:00 AM (Night Shift window). What is the nurse's total wage for the month?

Solutions:

A. Base Monthly Wage

$$\begin{aligned} \text{Monthly Wage} &= \frac{\text{Daily Wage} \times \text{Factor}}{12 \text{ Months}} \\ &= \frac{\text{P}1,600 \times 261}{12 \text{ Months}} \\ &= \text{P}34,800 \end{aligned}$$

B. Premium Pay on Regular Holiday Falling on Rest Day

$$\text{Premium Pay} = \text{Daily Rate} \times 2.60$$

	$= ₱1,600 \times 2.60$ $= ₱4,160$ C. Overtime Pay on Regular Holiday Falling on Rest Day $\text{Overtime Pay} = \text{Hourly Rate} \times 3.38 \times \text{No. of Hours}$ $= ₱200 \times 3.38 \times 4 \text{ hours}$ $= ₱2,704$ D. Night Shift Differential (NSD) on Overtime Pay $\text{Overtime Hourly Rate} = \text{Hourly Rate} \times 3.38, \text{ thus}$ $\text{NSD} = \text{Overtime Hourly Rate} \times 0.10 \times \text{No. of Hours}$ $= \text{Hourly Rate} \times 3.38 \times 0.10 \times \text{No. of Hours}$ $= ₱200 \times 3.38 \times 0.10 \times 4$ $= ₱270.40$ E. Total Monthly Wage $\text{Monthly Wage} = \text{Base Monthly Wage} + \text{Premium Pay} + \text{Overtime Pay} + \text{NSD}$ $= ₱34,800 + ₱4,160 + ₱2,704 + ₱270.40$ $= ₱41,934.40$ Examples on Calculating Earnings from Commission and Piecework 1. A real estate agent earns a 5% straight commission on all sales. This month, he closed a deal worth ₱2,000,000. How much is the agent's commission? Solutions: $\text{Commission Amount} = \text{Total Sales} \times \text{Commission Rate}$ $= ₱2,000,000 \times 0.05$ $= ₱100,000$ 2. A sales representative has a base monthly salary of ₱15,000 plus a 3% commission on all sales made. His total sales for the month reached ₱500,000. What are his total earnings for the month? Solutions: A. Commission Amount $\text{Commission Amount} = \text{Total Sales} \times \text{Commission Rate}$ $= ₱500,000 \times 0.03$ $= ₱15,000$ B. Total Earnings $\text{Total Earnings} = \text{Base Monthly Salary} + \text{Commission Amount}$ $= ₱15,000 + ₱15,000$ $= ₱30,000$ 3. A factory worker is paid on a piecework basis. He earns ₱20 per unit for the first 500 units produced in a week, and ₱25 per unit for any units produced beyond that as an incentive. This week, he produced 650 units. What is the worker's total earnings? Solutions: A. Base Piecework $\text{Base Earning} = \text{Base Quota} \times \text{Base Rate}$ $= 500 \text{ units} \times ₱20$ $= ₱10,000$ B. Incentive Piecework $\text{Incentive Amount} = (\text{Total Units} - \text{Base Quota}) \times \text{Incentive Rate}$ $= (650 \text{ units} - 500 \text{ units}) \times ₱25$ $= 150 \text{ units} \times ₱25$ $= ₱3,750$ C. Total Earnings $\text{Total Earnings} = \text{Base Earning} + \text{Incentive Amount}$ $= ₱10,000 + ₱3,750$ $= ₱13,750$
Video Lesson:)	Understanding Salary, Wage, and Commission Prof D. (2025, June 24). <i>Understanding Salary, Wage, and Commission</i> Grade 11 General Mathematics Q1 W1. https://www.youtube.com/watch?v=IlJw0mACR9g Calculating Overtime Pay

	WOW MATH. (2025, June 30). <i>Calculating Overtime Pay</i>. https://www.youtube.com/watch?v=e66FJLWxeNg AGES Explained! Salary, Commission, Piecework (Easy Math Guide) Cris Cruz. (2026, April 8). <i>AGES Explained! Salary, Commission, Piecework (Easy Math Guide)</i>. https://www.youtube.com/watch?v=visu3NeSY_Q
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I. Activity 1:	Practicing Learned Skills																																										
II. Materials:	Pen and paper, Calculators (Scientific or Basic)																																										
III. Instructions:	<i>Instructions:</i> Complete the master table by calculating the missing payroll values. To ensure perfectly consistent calculations throughout this activity, always base your math on a strict 52-week year and adjust daily calculations using the employee's specific workweek schedule (5-day or 6-day). <table><tr><th>Employee</th><th>Annual Salary</th><th>Monthly Salary</th><th>Weekly Salary</th><th>Daily Wage</th><th>Hourly Rate (8hrs)</th><th>Workweek Schedule</th></tr><tr><td>1</td><td>₱312,000</td><td></td><td></td><td></td><td>₱150</td><td>5-Day</td></tr><tr><td>2</td><td></td><td></td><td></td><td>₱800</td><td></td><td>6-Day</td></tr><tr><td>3</td><td></td><td>₱39,000</td><td></td><td></td><td></td><td>5-Day</td></tr><tr><td>4</td><td></td><td></td><td>₱7,200</td><td></td><td></td><td>6-Day</td></tr><tr><td>5</td><td></td><td></td><td></td><td></td><td>₱225</td><td>5-Day</td></tr></table> Read the short passage below regarding a worker's choice between two different pay structures and answer the questions. Mateo is a freelance courier torn between two assignments this Friday, which has been officially declared a Special Non-Working Day: Offer A is a one-time, fixed-rate independent contract that guarantees a flat ₱1,300 for a standard 8-hour shift, with no additional benefits or overtime provisions; and Offer B requires him to work a 10-hour shift at his regular logistics agency, where his standard base rate is ₱800 per day (equivalent to ₱100 per hour). Because the assignment falls on a Special Non-Working Day, Mateo knows his agency is legally mandated to pay him a 1.30 premium multiplier on his hourly rate for the first 8 hours of work. Furthermore, any hours worked beyond the standard 8-hour threshold on this day are classified as overtime, which commands an additional 1.30 multiplier on top of the day's premium rate—resulting in a composite 1.69 overtime multiplier for the final 2 hours. To make the most financially sound decision for his Friday schedule, Mateo must calculate and compare the total net earnings of both offers. - What is the flat, guaranteed payout for completing the 8-hour shift under Offer A? ₱1,040 ₱800 ₱1,300 ₱1,450 - Why is the hourly rate for the final two hours of Offer B higher than the hourly rate of the first eight hours? - Because the agency applies a compounding overtime rate on top of the special non-working day premium. - Because independent contracts are legally required to pay double after sunset. - Because Mateo receives a fixed weekly allowance allocation for travel expenses. - Because the logistics company offers a flat bonus for working past a 5-day schedule. - Based on the text, what can you infer about independent contracts like Offer A compared to regular agency labor laws? - Independent contracts always pay more per hour than agency jobs.	Employee	Annual Salary	Monthly Salary	Weekly Salary	Daily Wage	Hourly Rate (8hrs)	Workweek Schedule	1	₱312,000				₱150	5-Day	2				₱800		6-Day	3		₱39,000				5-Day	4			₱7,200			6-Day	5					₱225	5-Day
Employee	Annual Salary	Monthly Salary	Weekly Salary	Daily Wage	Hourly Rate (8hrs)	Workweek Schedule																																					
1	₱312,000				₱150	5-Day																																					
2				₱800		6-Day																																					
3		₱39,000				5-Day																																					
4			₱7,200			6-Day																																					
5					₱225	5-Day																																					

	B. Independent contracts might offer simpler, flat pricing but do not automatically include legal overtime or holiday premium structures. C. Independent contracts are regulated by standard 261-day payroll factors. D. Independent contracts must always base their math on a 6-day workweek schedule. 4. Which offer should he accept if he wants to maximize his earnings for Friday? A. Offer A, because it pays ₱1,300, which is higher than Offer B's total of ₱1,245. B. Offer B, because it pays a total of ₱1,450, which is higher than Offer A. C. Offer A, because Offer B only totals ₱1,040 after calculations. D. Offer B, because it pays a total of ₱1,378, which is higher than Offer A. 5. If the government suddenly upgrades Friday from a <i>Special Non-Working Day</i> to a <i>Regular Holiday</i>, how would this alter the values of Offer B? A. Offer B's total earnings would rise to ₱2,120, making it even more profitable than Offer A. B. Offer B's total earnings would drop below Offer A due to flat taxation factors. C. Offer B's total earnings would remain exactly ₱1,365 because agency contracts are fixed. D. Offer B's total earnings would shift to exactly ₱1,600 for the 10-hour shift.
V. Reflection:	To synthesize your gains, provide your insights to these questions: 1. As you worked through the Master Payroll Conversion Table and the Mateo case study, which specific variable—such as switching between 5-day and 6-day workweek schedules or calculating the 1.69 composite overtime multiplier—did you find the most challenging to keep consistent, and how did you overcome that difficulty? 2. What did you discover about the financial difference between working on a Special Non-Working Day versus a Regular Holiday, particularly when comparing how 'multipliers' (like the 1.30 premium for regular hours or the 3.38 rate for holiday-rest day overtime) can dramatically increase a worker's total earnings for a single shift? 3. If you were in Mateo's position in the real world, would you prefer the simplicity of a flat-rate independent contract (Offer A) or the complex, multiplier-based earnings of an agency assignment (Offer B)? How does knowing how to calculate these values change your perspective on what makes a job offer 'fair' or 'profitable'?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	<i>Instructions:</i> Choose the letter of the correct answer. 1. Which of the following refers to a fixed, regular payment offered to employees regardless of the exact number of hours worked, typically standard for managerial or administrative positions? A. Hourly Wage C. Salary B. Commission D. Piecework Earnings 2. Ella is a government clerk with an annual salary of ₱360,000. What is her fixed monthly salary? A. ₱30,000 C. ₱36,000 B. ₱25,000 D. ₱12,000 3. A retail representative receives a base monthly salary of ₱15,000 plus a 4% straight commission on all sales. If their total sales for the month reached ₱600,000, what are their total monthly earnings? A. ₱15,000 C. ₱39,000 B. ₱24,000 D. ₱45,000 4. An office assistant earns ₱150 per hour. Last week, they worked their standard 40-hour shift plus 6 hours of overtime on an ordinary workday. What is his/her total weekly wage? A. ₱6,900 C. ₱7,500 B. ₱7,125 D. ₱8,125

	Transportation: ₱1,000 Mobile load: ₱500 School needs: ₱1,000 She wants to save ₱1,500 every month. a. Can Anna save ₱1,500 with her current budget? b. What can she change to meet her savings goal? FOR ENHANCEMENT <i>Instructions:</i> Analyze and solve each of the following problems. - Daniel is a junior engineer in Quezon City with an annual salary of ₱480,000. What is his monthly salary? - Trixie works for a BPO company. Her rate is ₱120 per hour, and overtime is paid at ₱150 per hour. She also receives a night differential of ₱25 per hour for night shifts. This week, she worked: 38 regular hours 6 overtime hours 10 of the regular hours were night shift hours. How much did Trixie earn for the week? - In a Commission with Base Salary and Bonus, Leah sells insurance and earns: ₱12,000 base salary 5% commission on total monthly sales A ₱3,000 bonus if sales exceed ₱200,000 This month, she sold ₱230,000 worth of insurance. What is Leah's total income for the month? - Job A offers a Base Salary of ₱18,000 per month, plus ₱2,000 allowance. Job B offers ₱21,000 per month with Performance-Based Bonuses (₱0 to ₱3,000 depending on monthly targets). a. Which job gives a more consistent income? b. Which job could offer higher pay with better performance? c. If you prefer stability over high risk, which job would you choose? - Carlos earns ₱35,000 per month as a freelance video editor. His monthly expenses are: Rent: ₱8,000 Food: ₱6,500 Utilities: ₱2,500 Transportation: ₱2,000 Miscellaneous: ₱4,000 He wants to buy a ₱18,000 camera in 3 months without taking a loan. a. How much can Carlos save monthly on his current budget? b. Will he afford the camera in 3 months? If not, what changes can he make?
IV. Reflection:	To synthesize your gains, provide your insights to these questions - Which step in breaking down Trixie's multi-layered BPO earnings or Carlos's camera savings goals caused the most confusion during your initial analysis? - How does adding a fixed monthly allowance or a conditional performance bonus change the mathematical strategy you use to find an employee's total income? - How does mastering the process of tracking expenses against income help a person determine if their personal financial goals are mathematically realistic? *****END OF LESSON 1.1*****
Lesson 1.2: Computing Gross and Net Pay	
Objectives:	At the end of the lesson, you should be able to: - differentiate gross pay from net pay; - use technology to compute tax, SSS or GSIS, Pag-IBIG, PhilHealth contributions, gross and overtime pay; and - solve word problems involving gross and net pay.
Key Ideas and Examples:	Computing Net Income a. Key Concepts and Formulas

Gross income is the total earnings before any deductions are made. Includes base salary, overtime, commission, and piecework pay.

$$\text{Gross income} = \text{Salary} + \text{Allowances}$$

Net Pay is also called "take-home pay." This is the amount left after deductions are subtracted from gross pay.

$$\text{Net Pay} = \text{Gross Income} - \text{Total Deductions}$$

Deductions are the amounts subtracted from gross pay by law or agreement. In the Philippines, common deductions include:

1. Social Security System (SSS)

Applicable to: Private sector employees

Purpose: Retirement, disability, maternity, sickness, and death benefits

Rate: Based on a schedule tiered to the employee's monthly gross income.

The total SSS contribution rate is 15% of the employee's designated Monthly Salary Credit (MSC). This total percentage is split between the worker and the company:

- Employee Share: 5%
- Employer Share: 10%

SSS has strict boundaries that dictate how wages are categorized.

The Minimum Cap (Income Floor): ₱5,000.00 MSC

- If an employee's gross monthly salary is less than ₱5,250.00, their wage defaults to the lowest bracket base of ₱5,000.00.
- Absolute Minimum Employee Share: $\text{₱5,000.00} \times 5\% = \text{₱250.00}$

The Maximum Cap (Income Ceiling): ₱35,000.00 MSC

- If an employee earns an exceptionally high salary (e.g., ₱50,000.00 or ₱100,000.00), their premium computation hits the maximum bracket base of ₱35,000.00.
- Absolute Maximum Employee Share: $\text{₱35,000.00} \times 5\% = \text{₱1,750.00}$

Step-by-Step Payroll Computation

To determine an employee's exact deduction during payroll processing, follow this workflow:

1. *Calculate the Gross Monthly Earnings:* Sum the base salary and any taxable allowances or overtime pay received during the month.
2. *Find the Income Bracket:* Locate the official SSS Contribution Table and find where the gross income fits along the "Range of Compensation" column.
<https://www.sss.gov.ph/wp-content/uploads/2024/12/2025-SSS-Contribution-Table-rev.pdf>
3. *Identify the MSC:* Look at the corresponding Monthly Salary Credit (MSC) assigned to that specific bracket.
4. *Multiply by the Employee Rate:* Multiply that specific MSC by 5% to find the exact salary deduction.

2. Government Service Insurance System (GSIS)

Applicable to: Government employees only

Purpose: Life insurance, retirement, disability, and survivorship benefits

Rate: 9% of monthly salary rounded to the nearest peso (employee share)

Under Republic Act No. 8291 (the GSIS Act), mandatory monthly contributions are computed as a direct percentage of the employee's Actual Monthly Basic Salary (MBS). The total mandatory contribution rate is 21%, split as follows:

- Employee (Personal) Share: 9% of the actual Monthly Basic Salary.
- Employer (Government) Share: 12% of the actual Monthly Basic Salary.

There is no maximum salary ceiling or income cap for regular GSIS members. Additionally, there is no fixed statutory "floor amount."

GSIS membership is compulsory for all government personnel who receive compensation and hold a recognized appointment, regardless of their employment status. This includes regular employees and casual employees. Job Orders (JO), Contract of Service (COS) workers, and uniformed personnel are not covered by GSIS.

3. Pag-IBIG Fund (HDMF)

Applicable to: All employees

Purpose: Housing loans, savings, and calamity loans

Rate: 2% of monthly salary rounded to the nearest peso (employee share).

Pag-IBIG (HDMF) has a strict minimum baseline and a maximum cap. The parameters are governed by Pag-IBIG Circular No. 460, which outlines the mandatory contribution rules based on an employee's Monthly Fund Salary (MFS) or Regular Income.

The Maximum Fund Salary cap is exactly ₱10,000.00.

- Any income earned *above* ₱10,000.00 is completely ignored for the mandatory calculation.
- Because the standard rate for incomes over ₱1,500 is 2%, the maximum deduction is locked:
 - Maximum Employee Share: $10,000.00 \times 2\% = \text{₱}200.00$
 - Maximum Employer Share: $10,000.00 \times 2\% = \text{₱}200.00$
 - *Total Maximum Remittance: ₱400.00*

The Minimum Cap

- There isn't a "floor" that forces low-income earners to pay a higher flat amount (unlike PhilHealth's ₱10,000 floor). Instead, the minimum rules change the percentage rate applied to actual earnings:
 - Employees earning ₱1,500.00 or less per month qualify for a reduced Pag-IBIG deduction rate of 1% of their monthly basic contract pay. The employer is still legally required to pay a matching share of 2%.

Despite the set cap, Pag-IBIG allows employees to voluntarily contribute extra money beyond the standard limit to build up personal savings and retirement investments.

4. PhilHealth

Applicable to: All employees

Purpose: Healthcare benefits and hospitalization coverage

Rate: 5% of monthly salary, shared between employee and employer

The total premium rate for PhilHealth Contribution is 5% of the Monthly Basic Salary (MBS) or the Regular Income, which is split equally (2.5% each) between the employee and the employer. However, the calculation is strictly bound by these limits:

The Minimum Cap (Income Floor): ₱10,000.00

- If an employee earns *less* than ₱10,000.00 a month, their contribution is calculated as if they earned exactly ₱10,000.00.
 - Minimum Total Premium: ₱500.00 per month
 - Minimum Employee Share: ₱250.00

The Maximum Cap (Income Ceiling): ₱100,000.00

- If an employee earns *more* than ₱100,000.00 a month, their contribution is capped exactly at the ₱100,000 limit.
 - Maximum Total Premium: ₱5,000.00 per month
 - Maximum Employee Share: ₱2,500.00

Republic Act No. 11223 (Universal Health Care Act) is the current legal mandate that explicitly governs the progressive increase of PhilHealth premium collections up to the 5% rate along with the designated salary floor and ceiling limits.

5. Withholding Tax (BIR)

Applicable to: All employees

Purpose: To deduct and remit an employee's income tax liabilities progressively

Rate: Based on monthly salary bracket and applicable exemptions for tax rate. Refer to this link <https://mpm.ph/bir-tax-table/#incometaxtable>.

The tax charged directly to a person's income, salary, or wages is called the Withholding Tax on Compensation (often referred to broadly as Individual Income Tax). It is called a *withholding* tax because the employee doesn't have to manually pay it to the Bureau of Internal Revenue (BIR) every month. Instead, the employer is legally mandated to deduct (withhold) the tax directly from the employee's paycheck and remit it to the government on their behalf.

The Philippines uses a progressive tax system regulated by the TRAIN Law (RA 10963- Tax Reform for Acceleration and Inclusion). Under this system, those who earn more are taxed at higher percentage rates, while lower earners are exempted entirely.

The tax is calculated using a structured tiered table.

Payroll Frequency	Bracket/ Column	From	To	Fix Tax	Tax Rate on Excess
Monthly	1	0	20,833	0	0
	2	20,833	33,332	0	15%
	3	33,333	66,666	1,875	20%
	4	66,667	166,666	8,541.80	25%
	5	166,667	666,666	33,541.80	30%
	6	666,667	and above	183,541.80	35%

Step-by-Step Payroll Computation

1. *Determine Gross Compensation Income:* Base Salary + Overtime + Allowances.
2. *Deduct Non-Taxable Items:* Subtract the employee's statutory contributions (SSS or GSIS, PhilHealth, and Pag-IBIG shares). These are strictly non-taxable.
3. *Compute Taxable Income:*
Taxable Income = Gross Income - (SSS or GSIS + PhilHealth + Pag-IBIG)
4. *Apply the Bracket Table:* Look up the Taxable Income amount in the monthly tax bracket table to calculate the exact deduction.

6. Other Deductions (Optional or Conditional)

These deductions depend on agreements between the employee and employer

- Loans
- Cash Advances
- Insurance Premiums
- Union Dues
- Deductions for Absences or Tardiness

b. Computing the Net Income or Net Pay

Examples

Problem 1.

Anna works in a factory and earns ₱550 per day. She worked for 22 days this month. What is Anna's gross income for the month? Considering her deductions for SSS, PhilHealth, Pag-IBIG, and tax, compute her net income for the month.

Solutions

Items	Computation	Amount (P)
1. Gross Income	550.00 × 22	12,100.00
2. SSS	12,000.00 × 5%	600.00
3. PhilHealth	12,100.00 × 2.5%	302.50
4. Pag-IBIG	Max: 200.00	200.00
5. Tax	Exempted	0.00
Deductions	Sum of items 2 to 5	1,102.50
Net Pay	Gross Income – Deductions	10,997.50

Note:

- A Gross Income of ₱12,100.00 falls under a Monthly Salary Credit (MSC) of ₱12,000.00. To compute Anna's SSS Personal Share, multiply the MSC by 5%. (Refer to the 2025 SSS Contribution Table under SSS Circular No. 2024-006).
- In computing the PhilHealth premium, the total 5% contribution is divided equally between the employee and the employer. Therefore, Anna will only contribute to her personal share of 2.5%.
- The Pag-IBIG employee share is computed at 2% of the monthly compensation but is strictly capped at a maximum of ₱200.00.

Problem 2.

James earns ₱75 per hour. He worked 160 regular hours and 10 hours Over Time (125% rate). What is James's total gross income, including OT? Considering his mandatory government deductions, compute his net income for the month.

Solutions:

Items	Computation	Amount (P)
1. Regular	160 × 75.00	12,000.00
2. Overtime	10 × 75.00 X 1.25	937.50
3. Gross Income	Sum of items 1 and 2	12,937.50
4. SSS	13,000.00 × 5%	650.00
5. PhilHealth	12,000.00 × 2.5%	300.00
6. Pag-IBIG	Max: 200.00	200.00
7. Tax Withheld	Exempted	0.00
Deductions	Sum of items 4 to 7	1,150.00
Net pay	Gross Income - Deductions	11,787.50

Note:

- A Gross Income of ₱12,937.50 falls under a Monthly Salary Credit (MSC) of ₱13,000.00. To compute Jame's SSS Personal Share, multiply the MSC by 5%.

Problem 3.

Ms. Rivera is a public-school teacher in the Philippines with a Salary Grade (SG) 22, Step 1. She receives a monthly salary of ₱71,511.00 and non-taxable allowance of ₱2,000.00, but she also has deductions including taxes, contributions, and a GSIS loan of ₱5,000 per month. Find her net pay after all deductions.

Solutions:

Items	Computation	Amount (P)
1. Regular Salary	Given	71,511.00
2. Allowance	Given	2,000.00
3. Gross Pay	Sum of items 1 to 2	73,511.00
4. GSIS	71,511.00 × 9%	6,435.99
5. PhilHealth	71,511.00 × 2.5%	1,787.78
6. Pag-IBIG	Max: 200.00	200.00

	1,875.00 + (Taxable Income – 33,333.00) × 7,825.85 7. Tax 20% 8. Loan Given 5,000.00 Deductions Sum of items 4 to 8 21,249.62 Net Pay Gross Pay - Deductions 52,261.38 <i>Note:</i> • The contributions for GSIS, PhilHealth, and Pag-IBIG are computed strictly based on the Monthly Basic Salary, not the Gross Pay. Allowances, such as the ₱2,000.00 ACA/PERA, are non-taxable and excluded from these computations. • Under the TRAIN Law, the withholding tax is determined using the Taxable Income (Basic Salary minus mandatory GSIS, PhilHealth, and Pag-IBIG contributions). • Since the taxable income for this specific bracket falls between ₱33,333.00 and ₱66,667.00, the tax is computed as a fixed ₱1,875.00 plus 20% of the amount exceeding ₱33,333.00. • The Pag-IBIG employee share is computed at 2% of the monthly compensation but is strictly capped at a maximum of ₱200.00.
Video Lesson:)	Computing Gross Income and Net Pay Prof. D. (2025, June 27). <i>Computing Gross Income and Net Pay Grade 11 General Mathematics Q1 W1</i>. https://www.youtube.com/watch?v=3KGa1NKj0gI Gross Income and Net Pay WOW MATH. (2025, August 7). <i>Gross Income and Net Pay</i>. https://www.youtube.com/watch?v=jGDXkK4-w4 Computing Gross and Net Pay (PART 1: SSS and GSIS Deductions) PETER PAUL CARREON. (2025, June 29). <i>Lesson 1.2: Computing Gross and Net Pay (PART 1: SSS and GSIS Deductions) / GENERAL MATHEMATICS</i>. https://www.youtube.com/watch?v=OoDLlcRmm8M Computing Gross and Net Pay (PART 2: Pag-ibig and PhilHealth) PETER PAUL CARREON. (2025, July 2). <i>Lesson 1.2: Computing Gross and Net Pay (PART 2: Pag-ibig and PhilHealth) / GENERAL MATHEMATICS</i>. https://www.youtube.com/watch?v=KkdAOxsU1pI Computing Gross and Net Pay (PART 3: Withholding tax and Tax Income) PETER PAUL CARREON. (2025, July 3). <i>Lesson 1.2: Computing Gross and Net Pay (PART 3: Withholding tax and Tax Income) / GENERAL MATHEMATICS</i>. https://www.youtube.com/watch?v=DTAuyfINEJms
I. Activity 1:	Practicing Learned Skills
II. Materials:	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	<i>Instructions:</i> Analyze and solve the following problems. <i>Problem 1.</i> Mia works in a factory and earns ₱610 per day. She worked for 20 days this month. Calculate Mia’s gross monthly income, determine her mandatory deductions (SSS, PhilHealth, and Pag-IBIG) and withholding tax, and solve for her final net pay. <i>Problem 2.</i> Liam earns ₱90 per hour. In a month, he worked 176 regular hours and 12 overtime (OT) hours at a 125% rate. Calculate Liam’s gross income, determine his mandatory government deductions (SSS, PhilHealth, and Pag-IBIG) and withholding tax using the standard formulas, and solve for his final net income. <i>Problem 3.</i> Carla is a government employee with a monthly basic salary of ₱38,150.00 and a non-taxable allowance of ₱2,000.00. She also pays a monthly GSIS loan of ₱4,000.00. Calculate Carla’s gross income, determine her mandatory government deductions

(GSIS, PhilHealth, and Pag-IBIG) and withholding tax using the standard formulas, and solve for her final net pay.

Problem 4.

Ariel earns a monthly basic salary of ₱10,000.00 and a 10% commission on his total sales of ₱80,000.00. Calculate Ariel's gross income, determine his mandatory government deductions (SSS, PhilHealth, and Pag-IBIG) and withholding tax using the standard formulas, and solve for his final net income.

Problem 5.

You are choosing between two job offers in the private sector:

- Job A: A monthly basic salary of ₱25,000.00 plus a ₱3,000.00 non-taxable allowance.
- Job B: A monthly basic salary of ₱28,000.00 with no allowance.

Calculate the gross income, determine the mandatory government deductions (SSS, PhilHealth, and Pag-IBIG) and withholding tax for each job using the standard formulas, and evaluate both offers to solve for which job provides the higher final net income.

Instructions: Read the following passage and answer the given questions.

After weeks of interviews, Alex, a young professional in the private sector, finally received two job offers on the same day. At first glance, both appeared identical because they both promised a Total Gross Income of ₱28,000.00. However, Alex remembered a lesson from school: the amount written on a contract is rarely the amount that ends up in one's pocket.

The first offer, Job A, proposed a Monthly Basic Salary of ₱25,000.00 plus a ₱3,000.00 non-taxable allowance. The second offer, Job B, was simpler: a straight Monthly Basic Salary of ₱28,000.00 with no additional allowances.

Alex sat down with a calculator to look closer. For both jobs, the SSS contribution would be the same (₱1,400.00) because their total gross income fell into the same salary bracket. But when Alex looked at PhilHealth, Job B required a higher deduction of ₱700.00 compared to Job A's ₱625.00, because PhilHealth is computed based on the *basic salary* rather than the total gross.

The biggest surprise came with the Withholding Tax. Because Job A's ₱3,000 allowance was non-taxable, Alex's taxable income was lower, resulting in a tax of only ₱291.30. In contrast, the entire ₱28,000 in Job B was subject to the TRAIN law's progressive tax, leading to a much higher deduction of ₱730.05.

When Alex finished the math, the "take-home pay" for Job A was ₱25,483.70, while Job B was only ₱24,969.95. Alex realized that while both companies offered the same gross amount, the *structure* of the pay made one offer significantly better than the other.

Questions:

1. According to the text, what is the exact amount of the SSS contribution for both Job A and Job B, and why is this amount the same?
2. What was the specific Net Pay calculated for Job A and Job B?
3. The text mentions that PhilHealth is computed based on the "basic salary" while SSS is based on the "total gross income." How does this difference in calculation rules specifically disadvantage the employee in Job B?
4. Why does the text refer to the ₱3,000.00 allowance in Job A as the "biggest surprise" regarding the Withholding Tax?
5. If a company wants to give an employee a ₱2,000 raise, would it be more beneficial for the employee's "take-home pay" if that raise were added to their Basic Salary or given as a Non-taxable Allowance? Justify your answer using the tax and PhilHealth principles found in the text.
6. Beyond the numerical net pay, why is it important for a job seeker to ask for a breakdown of their compensation package (basic salary vs. allowances) before signing a contract? How might this knowledge affect their long-term financial planning, such as when applying for loans that are based on basic salary?

V. Reflection:	Express your thoughts on each question in 2-3 sentences. 1. Which part of computing the varying deduction rules—such as figuring out that PhilHealth and withholding taxes are tied to basic salary rather than gross income—was the most eye-opening to work through? 2. What mathematical principle explains why two job offers with the exact same gross income of ₱28,000.00 can end up producing entirely different final net take-home pay? 3. How does knowing how to "read between the paylines" empower a future job seeker to evaluate a compensation package based on actual disposable income rather than just the initial face-value offer?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	<i>Instructions:</i> Analyze and solve each of the following problems: <i>Problem 1.</i> Ella works as a clerk with a daily wage of ₱570.00. This month, she worked only 15 days due to a family emergency. Compute her gross pay, deductions, and net pay. <i>Problem 2.</i> Jayson is a technician earning ₱185.00 an hour. In a month, he worked: • 152 regular hours • 5 overtime hours (paid at 125%) • 8 holiday hours (paid at 200%) Compute his gross income, deductions, and net pay. <i>Problem 3.</i> Grace has a monthly salary of ₱30,000.00. She also received a ₱5,000.00 performance bonus this month. Compute her gross income, government-mandated deductions, and net pay. <i>Problem 4.</i> Nikki earns a ₱20,000.00 base salary and receives a 12% commission on ₱50,000.00 monthly sales. Compute total earnings, apply deductions, and determine net income. <i>Problem 5.</i> You are offered: Job X: ₱22,000.00 monthly + ₱2,500.00 food allowance Job Y: ₱25,000.00 monthly, no allowance Calculate net pay for both offers and decide which is better.
V. Reflection:	Express your thoughts on each question in 2-3 sentences. 1. Which calculation variable—such as Jayson's hourly overtime and holiday multipliers, or Nikki's commission percentage—proved to be the trickiest to navigate when structuring your gross income equations? 2. What key structural differences between a basic salary and a non-taxable allowance account for the variation in final net pay when evaluating Job X and Job Y? 3. How does learning to independently calculate payroll deductions alter your understanding of the actual take-home value of an advertised gross salary?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	<i>Instructions:</i> Analyze and solve the following problems: Remediation <i>Problem 1.</i> Ella earns ₱500 per day. She worked for 22 days this month. Compute her gross pay, mandatory deductions, and net pay. <i>Problem 2.</i> Jay earns ₱65 per hour and worked 150 hours for the month. Compute his gross pay, mandatory deductions, and net pay. <i>Problem 3.</i> Arnel is paid ₱30 per unit and completed 180 units in a month. Compute his gross pay, mandatory deductions, and net pay. <i>Problem 4.</i>

	Mina receives a monthly salary of ₱12,500. Compute her monthly gross pay, mandatory deductions, and net pay. <i>Problem 5.</i> Mila earns a monthly salary ₱10,000 and an 8% commission of ₱30,000 sales. Compute her monthly gross pay, mandatory deductions, and net pay. Enhancement <i>Problem 1.</i> Ariel's monthly salary is ₱30,000. What is his monthly net pay after deducting all mandatory deductions? <i>Problem 2.</i> Ben has ₱25,000 monthly salary and he received a performance bonus of ₱5,000. Compute his monthly gross pay, deductions, and net pay. <i>Problem 3.</i> Celine earns ₱95 per hour. For the month, she worked 160 hours with 10 hours of Overtime (OT). Compute her monthly gross pay, deductions, and net pay. <i>Problem 4.</i> Compare the two jobs below, which gives better earnings? Job A: ₱26,000 Monthly Salary and ₱3,000 allowance (assume taxed) Job B: ₱29,000 Monthly Salary <i>Problem 5.</i> You want to take home ₱25,000 after mandatory deductions and 15% tax bracket. Compute required gross x, where $net = x - (SSS + PhilHealth + Pag - IBIG + Tax) = 25,000$. Use spreadsheet iteration or formula.
IV. Reflection:	Express your thoughts on each question in 2-3 sentences. 1. Which problem in this set—such as computing Celine's overtime hours or using algebraic formulas to reverse-calculate the required gross income for a target net pay of ₱25,000.00—required the most mental effort for you to set up? 2. What key change occurs in your mathematical approach when you transition from simply multiplying rates by time or units to performing a backward iteration to find a missing gross salary? 3. How does developing the ability to mathematically reverse-calculate your gross income from a desired net pay empower you when negotiating salary expectations with future employers?

*****END OF LESSON 1.2*****

Lesson 1.3: Applying Percentages in Various Contexts

Objectives:	By the end of the lesson, you will be able to: 1. explain the concept of percentage increase and decrease; 2. apply percentage calculations in various contexts such as inflation, VAT, discounts, and profit/loss; and 3. solve problems involving percentage.
Key Ideas and Examples:	Percentage in the Various Contexts 1. Understanding Percentage Change a. <i>Introduction to Percentage Increase and Decrease</i> Percentage change (PC) indicates the extent to which a quantity has increased or decreased in relation to its original value. It is often utilized in finance, sales, budgeting, and data analysis. b. <i>Formula for Computing Percentage Change</i> $\text{Percentage Change} = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100$ c. <i>Determining Percentage Change Direction</i> • If the result is positive, it indicates a percentage increase. • If the result is negative, it indicates a percentage decrease. d. <i>Interpreting Positive and Negative Percentages</i> • +15%: The value increased by 15% from the original. • -10%: The value decreased by 10% from the original.

Examples:

1. An item was priced at ₱600 last week. Today, it is being sold for ₱650. By what percentage did the price change, and in what direction?

Given:

Original Value= ₱600

New Value= ₱650

Percentage Change=?

Direction=?

Solutions:

$$PC = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100$$
$$= \frac{650 - 600}{600} \times 100$$
$$= 8.33\%$$

Direction: Percentage Increase

2. A stock's value dropped from ₱120 to ₱90. By what percentage did the value change?

Given:

Original Value= ₱120

New Value= ₱90

Percentage Change=?

Direction=?

Solutions:

$$PC = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100$$
$$= \frac{90 - 120}{120} \times 100$$
$$= -25\%$$

Direction: Percentage Decrease

2. Inflation and Its Effects

- a. *What is Inflation?*

Inflation is the general increase in prices of goods and services over time, resulting in decreased purchasing power.

- b. *How Inflation Affects the Cost of Goods*

When inflation occurs, everyday items like rice, bread, or gasoline become more expensive, meaning your money buys less.

- c. *Adjusting Wages for Inflation*

To maintain the same standard of living, wages need to increase at the same rate as inflation (or keep pace with inflation). If wages stay the same while prices rise, people will effectively earn less.

- d. *Rate of Inflation*

The Inflation Rate measures the percentage change in the overall price level of a country, typically tracked using the Consumer Price Index (CPI) over a specific period (usually a year).

- e. *Formula*

$$\text{Inflation Rate} = \frac{CPI_{\text{Current Year}} - CPI_{\text{Previous Year}}}{CPI_{\text{Previous Year}}} \times 100$$

- f. *Understanding the Consumer Price Index (CPI)*

CPI represents the average price of a standard "basket" of goods and services (like food, housing, transportation, and utilities) consumed by households.

Examples

1. The average Consumer Price Index (CPI) of the Philippines was 120.0 in 2023 and rose to 124.8 in 2024. What is the country's inflation rate for 2024?

Given:

$$CPI_{\text{Previous Year}} = 120.0$$

$$CPI_{\text{Current Year}} = 124.8$$

Inflation rate =?

Solutions:

$$\begin{aligned} \text{Inflation Rate} &= \frac{CPI_{\text{Current Year}} - CPI_{\text{Previous Year}}}{CPI_{\text{Previous Year}}} \times 100 \\ &= \frac{124.8 - 120.0}{120.0} \times 100 \\ &= 4.0\% \end{aligned}$$

2. If the Consumer Price Index (CPI) for food items jumped from 115 to 122 due to agricultural supply shortages, what is the food inflation rate?

Given:

$$\begin{aligned} CPI_{\text{Previous Year}} &= 115 \\ CPI_{\text{Current Year}} &= 122 \\ \text{Inflation rate} &=? \end{aligned}$$

Solutions:

$$\begin{aligned} \text{Inflation Rate} &= \frac{CPI_{\text{Current Year}} - CPI_{\text{Previous Year}}}{CPI_{\text{Previous Year}}} \times 100 \\ &= \frac{122 - 115}{115} \times 100 \\ &= 6.09\% \end{aligned}$$

3. Mark-ups and Discounts in Business

a. *Difference Between Selling Price and Cost Price*

Cost Price (CP): The original amount a business pays to buy, manufacture, or produce a product.

Selling Price (SP): The final retail price tag set for the consumer. This amount must cover the cost price plus operational expenses (like rent and salaries) to yield a profit.

b. *Computing Mark-up Percentage*

The Mark-up Percentage (MP) indicates how much a business increases the cost price to determine its retail selling price.

$$\text{Mark-up Percentage} = \frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}} \times 100$$

The top part of the formula (*Selling Price – Cost Price*) is simply the Mark-up Amount in cash value.

c. *Calculating Final Prices after Discounts*

A discount reduces the selling price. To find the final price (Discounted Price or DP) in a single step, multiply the original selling price by the remaining percentage ($1 - \text{Discount Rate}$ or $1 - r$) the customer has to pay.

$$\text{Discounted Price} = \text{Selling Price} \times (1 - \text{Discount Rate})$$

Examples:

1. If the cost price of an item is ₱50 and its selling price is ₱75, then what is its mark-up percentage?

Given:

Cost Price= ₱50

Selling Price= ₱75

Mark-Up Percentage=?

Solutions:

$$\begin{aligned} MP &= \frac{SP - CP}{CP} \times 100 \\ &= \frac{75 - 50}{50} \times 100 \\ &= 50\% \end{aligned}$$

2. If the cost price of a material is ₱200 and is mark-up to 30%, then what is its selling price?

Given:

Cost Price= ₱200

Mark-Up Percentage= 30

Selling Price=?

Solutions:

$$\begin{aligned}
 MP &= \frac{SP - CP}{CP} \times 100 \\
 \frac{CP}{100} \left(MP = \frac{SP - CP}{CP} \times 100 \right) \frac{CP}{100} \\
 \frac{MP \times CP}{100} + CP &= SP - CP + CP \\
 SP &= \frac{MP \times CP}{100} + CP \\
 &= \frac{30 \times 200}{100} + 200 \\
 &= 260
 \end{aligned}$$

3. A product is discounted to 20%. If its selling price is ₱1,000, then what is its discounted price?

Given:

Selling Price= ₱1,000

Discount Rate= 20%

Discounted Price=?

Solutions:

$$\begin{aligned}
 DP &= SP(1 - r) \\
 &= 1000(1 - 0.20) \\
 &= 1000(0.80) \\
 &= 800
 \end{aligned}$$

4. Value Added Tax (VAT)

a. Introduction to Value Added Tax (VAT)

- VAT is an indirect consumption tax levied on the sale, barter, exchange, or lease of goods, properties, and services in the Philippines, as well as on items imported into the country.
- Under the National Internal Revenue Code, the standard tax rate is 12%.
- It is an indirect tax. The consumer bears the economic burden of the tax, but businesses act as collection agents that remit the funds directly to the national government.
- VAT-Inclusive Price: The final retail price tag shown to the customer, which already includes the 12% tax. In the Philippines, businesses are legally required to display VAT-inclusive prices on price tags.
- VAT-Exclusive Price: The base value or net price of an item before the 12% tax is added. This represents the actual revenue the business keeps for itself.

b. Converting Between VAT-Inclusive and VAT-Exclusive Prices

VAT can be separated or added using these mathematical formulas:

- Finding the Price Without VAT (VAT-Exclusive Price): To strip the 12% tax out of a retail price tag, divide by 1.12.

$$VAT\text{-Exclusive Price} = \frac{VAT\text{-Inclusive Price}}{1.12}$$

- Finding the Retail Price with VAT (VAT-Inclusive Price): To add the 12% tax to a wholesale or net baseline price, multiply by 1.12.

$$VAT\text{-Inclusive Price} = VAT\text{-Exclusive Price} \times 1.12$$

c. Computing the VAT Amount

To determine the exact cash value of the tax collected for the government, use either of these methods.

- Subtraction Method

$$VAT\text{ Amount} = (VAT\text{-Inclusive Price}) - (VAT\text{-Exclusive Price})$$

- Direct Percentage Method

$$VAT\text{ Amount} = (VAT\text{-Exclusive Price}) \times 0.12$$

Examples:

1. If the VAT-Inclusive Price is ₱112, then what is the VAT-Exclusive Price and the VAT Amount?

Solutions:

	$\text{VAT-Exclusive Price} = \frac{\text{VAT-Inclusive Price}}{1.12}$ $= \frac{112}{1.12}$ $= 100$ $\text{VAT Amount} = (\text{VAT- Exclusive Price}) \times 0.12$ $= 100 \times 0.12$ $= 12$ 2. If the VAT-Inclusive Price is ₱560, determine the VAT-Exclusive Price and the VAT Amount. Solutions: $\text{VAT-Exclusive Price} = \frac{\text{VAT-Inclusive Price}}{1.12}$ $= \frac{560}{1.12}$ $= 500$ $\text{VAT Amount} = (\text{VAT- Exclusive Price}) \times 0.12$ $= 500 \times 0.12$ $= 60$ 3. If the VAT-Exclusive Price is ₱1,200, then what is the VAT Amount and the VAT-Inclusive Price? Solutions: $\text{VAT Amount} = (\text{VAT- Exclusive Price}) \times 0.12$ $= 1,200 \times 0.12$ $= 144$ $\text{VAT-Inclusive Price} = \text{VAT-Exclusive Price} \times 1.12$ $= 1,200 \times 1.12$ $= 1,344$																				
Video Lesson:)	Understanding Percentage Change Percentage Increase and Decrease Prof. D. (2025, June 29). <i>Understanding Percentage Change Percentage Increase and Decrease</i>. https://www.youtube.com/watch?v=abxD1OLbztQ Example question calculating CPI and inflation AP Macroeconomics Khan Academy. (2018, February 22). <i>Example question calculating CPI and inflation AP Macroeconomics</i>. https://www.youtube.com/watch?v=qpgzgkrB6A Mark-Up and Discount WOW MATH. (2025, July 8). <i>Mark-Up and Discount</i>. https://www.youtube.com/watch?v=lJ4Ndtwbuj8 Value Added Tax (VAT) Applying Percentages in Various Contexts Prof D. (2025, June 29). <i>Value Added Tax (VAT) Applying Percentages in Various Contexts Grade 11 General Mathematics Q1</i>. https://www.youtube.com/watch?v=YvFgHoF9m9o																				
I. Activity 1:	Practicing Learned Skills																				
II. Materials:	Pen and paper, Calculators (Scientific or Basic)																				
III. Instructions:	Use the appropriate formulas to compute the missing values for each row. Round off your final answers to two decimal places when necessary. <i>Part A. Percentage Change, Markups, and Discounts</i> <table><tr><th>Item/ Scenario</th><th>Original/ Cost Value</th><th>Final Value</th><th>Rate/ Change</th><th>Type of Change</th></tr><tr><td>1. Wholesale Calculator</td><td>₱800</td><td>₱1,040</td><td>_____</td><td>☐ Markup ☐ Discount</td></tr><tr><td>2. Smart Watch Sale</td><td>₱2,500</td><td>_____</td><td>15%</td><td>☐ Markup ☐ Discount</td></tr><tr><td>3. Leather Shoes</td><td>_____</td><td>₱1,950</td><td>30%</td><td>☐ Markup ☐ Discount</td></tr></table>	Item/ Scenario	Original/ Cost Value	Final Value	Rate/ Change	Type of Change	1. Wholesale Calculator	₱800	₱1,040	_____	☐ Markup ☐ Discount	2. Smart Watch Sale	₱2,500	_____	15%	☐ Markup ☐ Discount	3. Leather Shoes	_____	₱1,950	30%	☐ Markup ☐ Discount
Item/ Scenario	Original/ Cost Value	Final Value	Rate/ Change	Type of Change																	
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3. Leather Shoes	_____	₱1,950	30%	☐ Markup ☐ Discount																	

4. Stock Price Shift	₱160	₱112	_____	☐ Increase ☐ Decrease
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Part B. Value-Added Tax

Item/ Scenario	VAT-Exclusive Price	VAT Amount	VAT-Inclusive Price
5. Desk Organizer	₱450	_____	_____
6. Computer Monitor	_____	_____	₱8,960
7. Wireless Mouse	_____	₱112	_____

Part C. CPI and Inflation Rate

Year / Period	Previous Year CPI	Current Year CPI	Inflation Rate
8. Period X	110	115.5	_____
9. Period Y	125	_____	4.80%
10. Period Z (<i>New</i>)	_____	145.6	4.00%

Analyze and solve the following word problems below.

- Mariel visits a bookstore during its back-to-school sale and decides to buy a notebook for ₱60 with a 20% discount, a ballpen set for ₱120 with a 10% discount, and a backpack for ₱850 with a 25% discount. Given these sale prices and rates, calculate:
 - total discount amount Mariel receives across all three items; and
 - final amount she must pay the cashier.
- A store sells a calculator for ₱1,344, which already includes a 12% Value-Added Tax (VAT). Based on this selling price, find:
 - the VAT-Exclusive Price of the calculator; and
 - the VAT Amount.
- Jomar bought 10 sets of pencils for a total of ₱150, costing him ₱15 per set. He originally planned to sell each set for ₱20, but to sell them faster, he decided to reduce the selling price to ₱13 per set. After selling all 10 sets at this reduced price, calculate his:
 - total sales;
 - amount of gained or lost money;
 - percentage change; and
 - selling price per set should have been if he wanted to earn a 30% profit.
- The Consumer Price Index (CPI) of a country was 110 in the year 2020 and rose to 132 by the year 2024. During this same period, a worker's weekly salary increased from ₱3,000 to ₱3,300.
 - Calculate the country's overall inflation rate between 2020 and 2024.
 - Find the percentage increase in the worker's salary.
 - Determine if the wage increase successfully kept up with the official inflation rate.
- Your class is organizing a learner council planning event with an initial budget of ₱5,000 and needs to estimate the final costs for three main expenses, while also fundraising to support the event. First, food items that cost ₱2,000 in 2022 must be adjusted assuming a 10% inflation rate. Second, event materials with an original cost of ₱1,200 are available at a 15% discount. Third, the printing and sound system services are quoted at a total of ₱1,500, which is already inclusive of a 12% Value-Added Tax (VAT). To help cover these expenses, the council makes and sells custom keychains for ₱75 each, which cost ₱50 each to produce. Given this scenario:
 - adjust the food price for inflation;
 - compute the discounted price of the materials;
 - determine the VAT-exclusive cost of the services; and
 - calculate the total profit earned if the council sells exactly 30 keychains to see how it can financially support the event.

Instructions: Read the following passage and answer the given questions.

Growing up in a city, siblings Cara and Tomas always dreamed of starting their own enterprise. Last year, they finally launched *Balik-Eskwela Supplies*, a small retail shop specializing in school essentials and custom keychains. To get their footing, they spent months studying how pricing, consumer habits, and the broader economy affect a small business.

Initially, managing inventory taught them about markups and discounts. They realized that to earn a profit, they had to add a strategic markup to the cost of their wholesale items. However, they also learned that retail requires flexibility. During a slow month, Tomas decided to reduce the selling price of their premium pencil sets from ₱20 down to ₱13 to clear stock quickly. While calculating the percentage change showed a significant markdown, it successfully caught the attention of budget-conscious students.

As the business grew, the siblings faced macroscopic challenges they couldn't control inflation and taxes. When looking at the national economy, they monitored the Consumer Price Index (CPI), which tracks the average cost of a standard basket of household goods. They noticed that a rise in the CPI meant that their money bought less, eroding their purchasing power. To ensure their part-time delivery rider wasn't effectively earning less due to these rising costs; they adjusted his weekly wages to match the country's annual inflation rate.

Furthermore, navigating local regulations meant understanding the 12% Value-Added Tax (VAT). Cara managed the bookkeeping, making sure to distinguish between a product's retail price tag and its VAT-exclusive value. When purchasing store equipment, she knew that a VAT-inclusive price of ₱1,344 meant that ₱1,200 went to the manufacturer, while ₱144 was accounted for as the government tax amount.

Through careful calculation and a deep understanding of economic trends, Cara and Tomas kept *Balik-Eskwela Supplies* thriving, proving that literacy in business math is just as vital as passion.

Questions:

- Which economic metric do Cara and Tomas track to monitor the average cost of a standard basket of household goods and services?
 - Value-Added Tax (VAT)
 - Consumer Price Index (CPI)
 - Percentage Change (PC)
 - Wholesale Markup Rate
- The passage states that if wages stay the same while prices rise, people "effectively earn less." What does this reveal about the relationship between inflation and a worker's salary?
 - Inflation automatically causes employers to lower the numerical amount on a worker's paycheck.
 - Rising inflation increases the value of cash, making wage adjustments unnecessary.
 - Workers can buy more everyday goods even if their salary remains unchanged during inflation.
 - If a salary does not grow at the same rate as inflation, the worker's purchasing power decreases.
- Why must Cara calculate a "VAT-exclusive value" when evaluating the store's equipment purchase instead of just recording the retail price tag?
 - Because the 12% tax amount is an additional cost that goes entirely to the manufacturer.
 - Because the retail price tag does not legally include the standard local tax yet.
 - Because a portion of the retail payment represents a tax amount meant for the government, not store asset value.
 - Because calculating the tax allows the store to automatically lower the wholesale cost of its inventory.

	- Imagine Cara and Tomas are looking to expand their inventory. If their wholesale supplier increases the cost of notebook raw materials by 15% due to a paper shortage, which strategy best applies the concepts from the text to keep their business thriving? - Calculate a strategic markup based on the new wholesale cost to ensure they still earn a profit while maintaining flexibility - Keep the retail price exactly the same to maximize their profit margins. - Immediately drop the selling price to clear out all existing notebook stock at a deep loss. - Ignore the 15% cost change because the 12% Value-Added Tax will automatically cover the price difference. - Suppose the national inflation rate spikes drastically next year. Based on how the siblings adjusted their delivery rider's pay in the story, what step should they take to construct a fair compensation plan for their team? - Freeze all wages permanently to prevent the business from losing its standard budget. - Calculate the percentage change in the CPI and increase nominal wages accordingly to protect their workers' standard of living. - Switch all workers to a commission-only system based entirely on custom keychain production costs. - Reduce retail prices for customers while lowering worker salaries at the exact same percentage rate. - If a student entrepreneur wants to replicate the success of <i>Balik-Eskwela Supplies</i>, which synthesis of business math and economic literacy is most vital based on the text? - Relying solely on business passion without tracking taxes, markup costs, or macroeconomic trends. - Knowing how to run promotional discounts while ignoring how the consumer price index erodes everyday purchasing power. - Balancing internal pricing strategies (like markups and discounts) with external economic forces (like inflation and tax regulations). - Prioritizing high initial retail prices over consumer demand flexibility and worker wage protection.
V. Reflection:	Share your experiences in performing the task by responding to the questions provided. - Which section of the activity—such as managing the multiple variables in the learner council event problem or navigating the missing base values in the CPI tables—challenged your mathematical thinking the most? - What fundamental rule distinguishes how you calculate the 12% Value-Added Tax (VAT) from an existing retail price versus how you apply a standard percentage markup or discount to a wholesale cost? - How does analyzing the relationship between the Consumer Price Index (CPI) and employee wages explain why nominal salary numbers do not always reflect a worker's actual purchasing power?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	<i>Instructions:</i> Analyze and solve each problem. - Althea visits a tech store during its weekend clearance sale. She decides to buy a computer mouse for ₱450 with a 15% discount, a gaming keyboard for ₱1,200 with a 20% discount, and a wireless headset for ₱2,400 with a 30% discount. Given these sale prices and rates, calculate: - The total discount amount Althea receives across all three items; and - The final amount she must pay the cashier. - A home appliance store sells an air purifier for ₱4,480. This retail price already includes a 12% Value-Added Tax (VAT). Based on this selling price, find: - The VAT-Exclusive Price of the air purifier; and - The actual VAT Amount embedded in the price.

	3. Kevin bought 20 pieces of custom phone cases for a total cost of ₱2,000, which means each case cost him ₱100. He originally planned to sell each piece for ₱180. To clear his stock faster, he decided to reduce his actual selling price to ₱120 per piece. After selling all 20 pieces at this reduced price, calculate his: - Total sales revenue generated; - Net amount of gained or lost money; - Percentage change relative to his initial investment cost; and - Target selling price per piece if he had wanted to earn a 40% profit instead. 4. The Consumer Price Index (CPI) of an economy was 120 in the year 2021. It rose steadily to 138 by the year 2025. During this exact same period, a worker's monthly base salary increased from ₱15,000 to ₱16,500. - Calculate the country's overall inflation rate between 2021 and 2025. - Find the percentage increase in the worker's salary. - Determine if the wage increase successfully kept up with the official inflation rate. 5. A local baker makes a specialized customized cake with a total raw ingredient cost of ₱800. She applies a 60% regular markup to determine her baseline retail selling price. During a weekend store anniversary promo, she offers a 10% discount off that baseline retail selling price. Calculate: - The regular retail selling price after the initial markup; - The promotional selling price after the discount is applied; and - The final net profit amount earned per cake sold during the promo.
V. Reflection:	Share your experiences in performing the task by responding to the questions provided. - Which multi-step calculation—such as separating the base cost from the embedded tax in the VAT problem or balancing sequential markups and discounts in the baking problem—required you to map out your formulas most carefully? - What specific mathematical relationship explains why reducing a price to ₱120.00 from an intended ₱180.00 still results in a net monetary gain relative to Kevin's original ₱100.00 production cost? - How does comparing a salary's growth percentage directly against an economy's inflation rate help an individual measure their actual changes in financial purchasing power over time?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	FOR REMEDIATION <i>Instructions:</i> Analyze and solve each of the following problems. - A ₱1,000 phone is on sale for 10% off. How much is the discount? - A toy costs ₱500. VAT is 12%. What is the total cost after VAT? - A dress costs ₱800 but is now sold at 25% off. How much will the customer pay? - The product cost 600 and is sold for ₱750. What is the profit? - A bag costs ₱1,200. There is a 50% discount. What is the new price of the bag? FOR ENHANCEMENT <i>Instructions:</i> Analyze and solve each of the following problems. - A premium laptop was priced at ₱50,000 in 2025. By 2026, a market-wide inflation spike caused its baseline retail price to increase by 8%. To clear out inventory later that year, the store offered a 15% holiday discount on the newly adjusted 2026 price. - What was the retail price of the laptop after the 8% inflation increase? - What was the final promotional price after the 15% discount was applied? - What is the net percentage change from the original 2025 price (₱50,000) to the final promotional price? - A showroom refrigerator is sold at a total VAT-inclusive price of ₱28,000 (which already includes the standard fixed 12% VAT). The store offers a special 20%

	discount on the floor model, but store policy dictates that the discount applies strictly to the VAT-exclusive price (the base price before tax). - Determine the original VAT-exclusive price of the refrigerator. - Calculate the new discounted base price after the 20% markdown is subtracted. - If the 12% VAT is reapplied to this new discounted base price, what is the final total amount the customer pays at the counter? 3. A local bakeshop buys bulk baking supplies for an initial baseline cost of ₱12,500. Due to a sudden global supply chain inflation shock, the raw cost of these materials jumps by 6%. When the supplier generates the final invoice, they apply the standard 12% VAT on top of this newly inflated base price. - Calculate the new cost of the baking supplies after the 6% inflation spike. - Determine the final invoice total after the 12% VAT is added to the inflated price. - Find the total percentage change from the original ₱12,500 raw cost to the final VAT-inclusive invoice total.
IV. Reflection:	Share your experiences in performing the task by responding to the questions provided. - Which multi-step calculation in the enhancement section—such as tracking the three-part laptop pricing shift or stripping away and reapplying the 12% VAT on the refrigerator—required you to be the most precise when mapping out your steps? - What mathematical flaw occurs if a discount is accidentally calculated using a total VAT-inclusive price instead of the base VAT-exclusive cost? - How does learning to calculate the net percentage change across sequential price adjustments help you understand if a retail "sale" is saving a consumer money? *****END OF LESSON 1.3*****

Unit 2. Sequence and Series

Lesson 2.1. Sequence

Objectives:	At the end of the lesson, you should be able to: - identify patterns found in art and nature, - describe a sequence, - classify sequences as finite or infinite based on their structure and number of terms, - determine the next term of a given sequence, including a Fibonacci sequence, and - describe the rule for finding the next term in a sequence
Key Ideas and Examples:	Sequence - a list of numbers arranged in a specific order, following a rule. Example: 3, 6, 12, 24, . . . Term - each number in a sequence. Example: In sequence 3, 6, 12, 24, each of the numbers 3, 6, 12, and 24 is a term. Position (or Index) - the place of a term in the sequence, usually counted as 1st term, 2nd term, and so on Notation: a_1 is the first term, a_2 is the second term, etc. In general, a_n denotes the nth term Rule (or Pattern Rule) - a description of how the term is formed or how they change from one to the next. Example: "Add 2" in the sequence 2, 4, 6, 8, ... Fibonacci Sequence - a sequence where each term after the first two is the sum of the two preceding terms. Example: 1, 1, 2, 3, 5, 8, 13, ... Classification of Sequences (based on the number of terms)

Finite Sequence

A finite sequence has a specific number of terms – it has a beginning and an end.
You can count all the terms.
It stops after a certain point.

Example: 2, 4, 6, 8, 10

(This sequence has 5 terms – it ends at 10)

Infinite Sequence

An infinite sequence has no end- it goes on forever.

It keeps following the same rule endlessly.

You can describe the rules, but you can't list all the terms.

Example: 1, 2, 4, 8, 16, 32, ...

(This keeps doubling and never stops.)

In mathematical notation, an infinite sequence often includes ellipsis “...” to show it continues.

Sample Exercise 1: Identifying a Sequence

Observe the Pattern

Figure 1 ● ●

Figure 2 ● ● ● ●

Figure 3 ● ● ● ● ● ●

Guide Questions

How many dots are in Figure 1? 2

How many dots are in Figure 2? 4

How many dots are in Figure 3? 6

What sequence is formed? 2, 4, 6, ...

What pattern do you observe? Add 2 each time

Sample Exercise 2: Consider the sequence: 5, 10, 15, 20, 25, ...

Question

Answer

What is the first term? 5

What is the third term? 15

What is the fifth term? 25

What is a_2 ? 10

What is a_4 ? 20

Sample Exercise 3: Staircase Sequence Investigation

Count the number of squares in each figure and answer the questions that follow.
Pattern

Figure 1



Figure 2



Figure 3



Figure 4



Complete the Table

Figure Number	Number of Squares
1	(1)
2	(3)
3	(6)
4	(10)

Guide Questions

1. What sequence is formed by the number of squares?

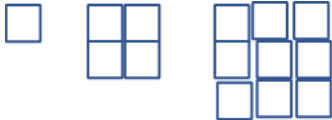
2. What is the first term (a_1) of the sequence?

3. What is the fourth term (a_4) of the sequence?

4. How many squares are added from Figure 1 to Figure 2?

5. How many squares are added from Figure 2 to Figure 3?

6. Predict the number of squares in Figure 5.

	7. Is the sequence finite or infinite? Explain your answer. Answer: - Sequence: 1, 3, 6, 10, ... $a_1 = 1$ $a_4 = 10$ - Added 2 squares - Added 3 squares - Figure 5 = 15 squares ($10 + 5$) - Infinite, if the pattern continues indefinitely.
Video Lesson:	"Sequences and Series." YouTube video, n.d. https://www.youtube.com/watch?v=REDVjUTvXvk&t=368s. "Types of Sequences." YouTube Shorts video, n.d. https://www.youtube.com/shorts/qW0yF2_-LEc..
I. Activity 1:	Practicing Learned Skills
II. Materials:	Pen or pencil, paper, Calculator (optional), Formula reference sheet (optional)
III. Instructions:	Perform the following activities: - Give the next term of the given sequences. - Identify the rules. 81, 27, 9, 3, ____  7, 14, 28, 56, ____ 10, 21, 31, ____  $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \underline{\hspace{1cm}}$ 8, 27, 64, ____ 3, 6, 10, 15, ____ $1 \times 2, 2 \times 3, 3 \times 4, \underline{\hspace{1cm}}$ 5, 10, 15, 20, ____
V. Reflection:	Answer each question. - How did identifying the rule help you determine the next term in each sequence? - What patterns did you observe in the given sequences? - Which sequence was the easiest or most difficult to solve? Why? - What mathematical operations were commonly used in forming sequences? - Why is recognizing patterns important in solving real-life mathematical problems?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen or pencil, paper, Calculator (optional), Formula reference sheet (optional)
III. Instructions:	Read each item carefully and write the correct answer. - What is true about the sequence represented by the pattern below? - Each term in the sequence is the sum of all counting numbers from 1 up to that term's position in the sequence - Each term in the sequence is the square of all counting numbers from 1 up to that term's position in the sequence - Each term in the sequence is the product of all numbers from 1 up to that term's position in the sequence - Each term in the sequence is the thrice the sum of all counting numbers from 1 up to that term's position in the sequence - What is the next term in the sequence: 1, 1, 2, 3, 5? 6

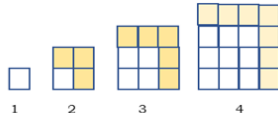
	b. 7 c. 8 d. 10 3. What is the rule in the sequence: 3, 6, 12, 24, ____? a. Add 3 b. Add 6 c. Multiply by 2 d. Multiply by 3 4. A sequence has a pattern: add 4 each time. The first term is 2, what is the 5th term? a. 10 b. 18 c. 20 d. 22 5. A group of learners is arranging chairs in a theater where each row has 3 more chairs than the previous one. The first row has 10 chairs. How many chairs are there in the fifth row? a. 13 b. 16 c. 19 d. 22
V. Reflection:	Answer each question. 1. What strategies did you use to identify the pattern or rule in each sequence? 2. Which item was the easiest or most difficult for you to answer? Why? 3. How did recognizing patterns help you determine the next term in a sequence? 4. What real-life situations can you relate to sequences and patterns? 5. Why is it important to understand rules and relationships in sequences in mathematics?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials	Pen or pencil, paper, Calculator (optional), Visual aids or number pattern cards
III. Instructions:	<i>Remediation</i> For each sequence: - Look carefully at the numbers. - Complete the guide questions - Find the next term A. Guided Steps for Each Item 10, 7, 4, 1, -2, ____ Step 1: What is happening from 10 → 7? ____ Step 2: What is happening from 7 → 4? ____ Step 3: What is the pattern? (Add or subtract?) ____ Step 4: Next term: ____ 1, 3, 5, 7, 9, ____ Step 1: 1 → 3 = ____ Step 2: 3 → 5 = ____ Step 3: Pattern: ____ Step 4: Next term: ____ 100, 50, 25, 12.5, ____ Step 1: 100 → 50 = ____ Step 2: 50 → 25 = ____ Step 3: Pattern: ____ Step 4: Next term: ____ 1000, 2000, 4000, ____ Step 1: 1000 → 2000 = ____ Step 2: 2000 → 4000 = ____ Step 3: Pattern: ____ Step 4: Next term: ____ 1000, 2000, 4000, 8000, ____ Step 1: 1000 → 2000 = ____

	Step 2: $2000 \rightarrow 4000 = \underline{\hspace{2cm}}$ Step 3: Pattern: $\underline{\hspace{2cm}}$ Step 4: Next term: $\underline{\hspace{2cm}}$ 6. 10, 10, 20, 30, 50, $\underline{\hspace{2cm}}$ Step 1: $10 \rightarrow 10 = \underline{\hspace{2cm}}$ Step 2: $10 \rightarrow 20 = \underline{\hspace{2cm}}$ Step 3: $20 \rightarrow 30 = \underline{\hspace{2cm}}$ Step 4: Pattern is: $\underline{\hspace{2cm}}$ Step 5: Next term: $\underline{\hspace{2cm}}$ 7. 4, 6, 10, 16, 24, 34, $\underline{\hspace{2cm}}$ Step 1: $4 \rightarrow 6 = \underline{\hspace{2cm}}$ Step 2: $6 \rightarrow 10 = \underline{\hspace{2cm}}$ Step 3: $10 \rightarrow 16 = \underline{\hspace{2cm}}$ Step 4: Pattern is: $\underline{\hspace{2cm}}$ Step 5: Next term: $\underline{\hspace{2cm}}$ <i>Enhancement</i> Read each situation carefully. - Identify the pattern - Find the required value(s) - Answer all questions completely - Explain your answer in 1–2 sentences - Daily Step Challenge A student walks 2,000 steps on Day 1 and increases his steps by 500 each day. - How many steps will he walk on Day 7? - How many total steps will he have walked from Day 1 to Day 7? - Investment Growth A student invests ₱5,000. The amount becomes twice as much every year. - What is the amount after 4 years? - What do you notice about how the amount changes each year? - Pattern in Nature Consider the sequence: 1, 1, 2, 3, 5, 8, 13, ... - What is the next 3 numbers in the sequence? - Describe how the pattern is formed in your own words. - Saving for a Bike Luis saves ₱100 in Week 1 and adds ₱100 more each week. - How much does he save in Week 6? - How much total savings will he have after 6 weeks? - If the bike costs ₱3,000, will he be able to buy it after 6 weeks? Explain briefly. - Number Pattern Puzzle 1, 4, 9, 16, 25, ... - What are the next two numbers? - What pattern do you observe in the sequence?
IV. Reflection:	Answer each question. - What common characteristics did you observe in all the number of patterns and situations presented in the activities? - How did identifying the pattern help you solve the given tasks? - What strategies did you use to determine the pattern in a sequence? - How are sequences and series used in real-life situations such as saving money, investing, or tracking daily activities? - What important understanding about sequences and series did you gain from the activities?

****END OF LESSON 2.1****

Lesson 2.2. Sequence and Series	
Objectives:	At the end of the lesson, you should be able to: 1. describe a series; 2. identify whether a given list of numbers is a sequence or a series; 3. explain the key difference between a sequence and a series; and 4. write a given series in sigma notation and vice versa
Key Ideas and Examples:	Key Terms and Definitions Series - the sum of the terms of a sequence. <i>Example:</i> For the sequence 1, 3, 5, the series is $1 + 3 + 5 = 9$ Finite Series - a series with a definite number of terms. <i>Example:</i> $2 + 4 + 6 + 8$ (only 4 terms) Infinite Series - a series that continues without end. <i>Example:</i> $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ The nth term of a sequence is usually denoted by a_n Partial Sum (S_n) - the sum of the first n terms of a sequence. If the nth term of the sequences is a_n, then $S_n = a_1 + a_2 + a_3 + \dots + a_n$ <i>Example:</i> In the sequence, 1, 3, 5, 7, 9, . . . , $S_4 = 1 + 3 + 5 + 7 = 16$ S_n is the nth term of the series Sigma Notation (Σ) - a shorthand symbol used to express a series. To write $a_1 + a_2 + a_3 + \dots + a_n$ in-sigma notation: $\sum_{n=1}^n a_n$ In general, $\sum_{n=start}^{k=end} a_n \text{ (expression in terms of } a_n \text{)}$ Example: $1 + 2 + 3 + 4 + 5$ in sigma notation is $\sum_{n=1}^5 a_n$
Video Lesson:	“Sequences and Series.” YouTube video, n.d. https://www.youtube.com/watch?v=REDVjUTvXvk&t=368s. “Types of Sequences.” YouTube Shorts video, n.d. https://www.youtube.com/shorts/qW0yF2_-LEc..
I. Activity 1:	Practicing Learned Skills
II. Materials:	Pen or pencil, paper, Calculator (optional), Formula sheet
III. Instructions:	Do the following: 1. Find the sum of the first 4 terms of the sequence, 3, 7, 11, 15, . . . 2. Find the sum of the sequence: 5, 10, 20, 40, 50 3. Write the series, $2 + 4 + 6 + 8 + 10$ in sigma notation. 4. Evaluate the series written in sigma notation: $\sum_{n=1}^4 (3n + 1)$

	5. The figure shows a growing pattern of tiles. Count the number of tiles in each figure. a. How many tiles are there in the 5th figure? b. What is the rule used to determine the number of tiles in each figure? c. How many tiles are there in total from the 1st to the 7th figure? 
V. Reflection:	Answer each question. 1. What is a series? 2. What does S_n represent? 3. Why is S_4 called a partial sum and not a total sum? 4. How do you differentiate between a finite series from an infinite series? 5. Why do mathematicians use sigma notation instead of writing all the terms of a long series?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen or pencil, paper, Calculator (optional for computation)
III. Instructions:	Choose the letter of the correct answer. Write your answers on a sheet of paper. 1. Which of the following best describes a <i>series</i>? a. A sum of numbers in a pattern b. A list of numbers in a specific order c. A variable expression with a pattern d. A formula for a term 2. Which of the following is a series? a. 2, 4, 6, 8 b. 3, 6, 9, 12 c. $2 + 4 + 6 + 8$ d. 1, 3, 5, 7 3. What does the sigma symbol (Σ) represent in mathematics? a. A product b. A term c. A square root d. A sum 4. What is the sigma notation of the series $5 + 10 + 15 + 20$? a. $\sum_{n=1}^4 5n$ b. $\sum_{n=1}^4 5n + 1$ c. $\sum_{n=1}^4 5n - 1$ d. $\sum_{n=1}^4 5n + 5$ 5. What is the sigma notation for $1 + 3 + 9 + 27$? a. $\sum_{n=1}^4 3^n$ b. $\sum_{n=1}^4 3^{n-1}$ c. $\sum_{n=1}^4 3n$ d. $\sum_{n=1}^4 3n - 1$
V. Reflection:	Answer each question. 1. How did you identify whether a sequence is geometric or not in the multiple-choice items? 2. What steps did you follow to compute the common ratio in a geometric sequence? 3. Which formula did you use in solving geometric series problems, and why? 4. How did you approach solving real-life word problems involving sequences and series?

	5. What difficulties did you encounter while solving the problems, and how did you address them? 6. How can understanding geometric sequences and series help in real-life decision-making such as savings or investments?
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I. Activity 3:	Extending and Reinforcing Learning
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II. Materials	Pen or pencil, paper, Calculator (optional for computation), Formula reference sheet
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III. Instructions:	Remedial: A. Sort each of the following into Sequence or Series: <table border="1"> <thead> <tr> <th>Expressions</th> <th>Type (Sequence/Series)</th> </tr> </thead> <tbody> <tr> <td>4, 8, 12, 16</td> <td></td> </tr> <tr> <td>$4 + 12 + 16$</td> <td></td> </tr> <tr> <td>10,20,30,40</td> <td></td> </tr> </tbody> </table> B. Write the sigma notation for each series. $4 + 8 + 16 + 32$ $3 + 6 + 9 + 12$ C. Expand the Sigma Notation and calculate the sum. $\sum_{n=1}^4 5n$ = _____ $\sum_{k=0}^7 (2k + 1)$ = _____ Enhancement: Read the scenario carefully. Analyze the pattern in the weekly savings and complete all the tasks that follow. Show your solutions clearly and write your answers neatly on a separate sheet of paper. - Determine the pattern in the weekly savings. - Write the first eight terms of the sequence. - Express the sequence as a series by writing the sum of the first eight terms. - Solve the total amount saved after eight weeks and show your complete solution. - Answer the reflection question using 2–3 complete sentences to explain how the savings pattern helps achieve a financial goal over time. Scenario: You decided to start a savings plan to prepare for your future studies or personal goals. In Week 1, you save ₱50, and each week you increase your savings by ₱20. Tasks: - Write the Sequence: List the amount of money saved for the first 8 weeks - Form the Series: Write the corresponding series by expressing the sum of the first 8 terms. Example format: ₱50 + ₱70 + ₱90 + ... - Find the Total Savings Add all 8 terms to find the total amount saved. Show your complete solution. - Understanding the Pattern (Short Explanation) Answer in 2–3 sentences:	Expressions	Type (Sequence/Series)	4, 8, 12, 16		$4 + 12 + 16$		10,20,30,40	
Expressions	Type (Sequence/Series)								
4, 8, 12, 16									
$4 + 12 + 16$									
10,20,30,40									

IV. Reflection:	Reflect on this: How does the pattern in your savings help you reach your financial goal over time?
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*******END OF LESSON 2.2*******

Lesson 2.3. Arithmetic Sequence and Series

Objectives:	By the end of the lesson, you will be able to: - illustrate sequence and series, - define arithmetic sequence and series and their key components, - find the nth term of an arithmetic sequence, - use the appropriate formula $S_n = \frac{n}{2}(a_1 + a_n)$ or $S_n = \frac{n}{2}[2a + (n - 1)d]$ in calculating the sum of an arithmetic series, - solve real world problem involving arithmetic sequence and series, and
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	6. demonstrate appreciation of the importance of arithmetic sequences and series by explaining or presenting how they are used in making informed decisions in real-life situations.
Key Ideas and Examples:	Sequence A sequence is a set of numbers arranged in a specific order or pattern. Example: 2, 4, 6, 8, 10, ... Each number in the sequence is called a term. Series A series is the sum of the terms of a sequence. Example: $2 + 4 + 6 + 8 + 10$ Term A term refers to each individual element in a sequence or series. Example: In the sequence 3, 7, 11, 15, ... • 3 is the 1st term • 7 is the 2nd term • 11 is the 3rd term Arithmetic Sequence An arithmetic sequence is a special type of sequence in which the difference between any two consecutive terms is constant. This constant difference is called the common difference (d). Example: 5, 8, 11, 14, 17, ... Observe the pattern: $8 - 5 = 3$ $11 - 8 = 3$ $14 - 11 = 3$ Since the difference between consecutive terms is always 3, the sequence is an arithmetic sequence with: $d = 3$ Important Parts of an Arithmetic Sequence <i>First Term (a_1)</i> The first term is the starting value of the sequence. Example: 7, 10, 13, 16, ... The first term is: $a_1 = 7$ <i>Common Difference (d)</i> The common difference is the constant number added to each term to get the next term. Example: 12, 17, 22, 27, ... Find the difference between consecutive terms: $17 - 12 = 5$ Thus, $d = 5$ <i>nth Term (a_n)</i> The nth term refers to the term found in the nth position of the sequence. To determine the nth term of an arithmetic sequence, use the formula: $a_n = a_1 + (n - 1) d$ where: • a_n = nth term • a_1 = first term • d = common difference • n = position of the term Example 1: Find the 10th term of the arithmetic sequence: 5, 8, 11, 14, ... Solution: Step 1: Identify the value of the variables of the formula. $a_1 = 5$

$$\begin{aligned}d &= 3 \\n &= 10 \\a_n &= ?\end{aligned}$$

Step 2:

Substitute the values to the formula. Then, solve for a_n .

$$\begin{aligned}a_{10} &= 5 + (10 - 1)3 \\&= 5 + (9)3 \\&= 5 + 27 \\a_{10} &= 32\end{aligned}$$

Therefore, the 10th term in the arithmetic sequence is 32.

Example 2. In the arithmetic sequence: 3, 9, 15, 21, ...; find the value of n if $a_n = 75$.

Solution:

Step 1: Identify the value of the variables of the formula.

$$\begin{aligned}a_1 &= 3 \\d &= 6 \\a_n &= 75 \\n &= ?\end{aligned}$$

Substitute the values to the formula. Then, solve for n .

$$\begin{aligned}75 &= 3 + (n - 1)6 \\75 &= 3 + 6n - 6 \\75 - 3 + 6 &= 6n \\78 &= 6n \\n &= 13\end{aligned}$$

Therefore, 75 is in the 13th term of the arithmetic sequence.

Example 3: The 5th term of an arithmetic sequence is 20, and the 12th term is 48. Find the following: (a) common difference, (b) first term, and (c) the 30th term.

Solution:

Step 1: Identify the given values.

$$\begin{aligned}a_5 &= 20 \\a_{12} &= 48\end{aligned}$$

Step 2: Substitute the given values to the formula

For $a_5 = 20$

$$\begin{aligned}20 &= a_1 + (5 - 1)d \\20 &= a_1 + 4d \text{ (Equation 1)}\end{aligned}$$

For $a_{12} = 48$

$$\begin{aligned}48 &= a_1 + (12 - 1)d \\48 &= a_1 + 11d \text{ (Equation 2)}\end{aligned}$$

Step 3: Subtract the equations

$$\begin{aligned}20 &= a_1 + 4d \\48 &= a_1 + 11d\end{aligned}$$

$$\begin{array}{r} -28 = -7d \\ d = 4 \end{array}$$

Step 4: Substitute the value of $d = 4$ to equation 1:

$$\begin{aligned}20 &= a_1 + 4(4) \\20 &= a_1 + 16 \\20 - 16 &= a_1 \\a_1 &= 4\end{aligned}$$

Step 5. Solve for the 30th term.

$$\begin{aligned}a_{30} &= 4 + (30 - 1)4 \\&= 4 + (29)4 \\&= 4 + 116 \\a_{30} &= 120\end{aligned}$$

Therefore, the common difference is 4, the first term is 4, and the 30th term is 120.

Example 4: During the dry season, a farmer in your town is testing a new water irrigation system for his rice field. After 1 minute of turning on the water pump, the water level in the irrigation canal rose to 30 centimeters. After 2 minutes, it reached 85 centimeters, and after 3 minutes, the level was 140 centimeters. If the water level continues to rise at this constant rate, how high will the water level be after 16 minutes?

Solution:

Let a_n = the water level after n minutes

Step 1: Identify the given.

$$a_1 = 30\text{cm}$$

$$d = 55\text{ cm}$$

$$n = 16\text{ min}$$

$$a_n = ?$$

Step 2:

Substitute the given values to the formula.

$$a_{16} = 30 + (16 - 1)55$$

$$= 30 + (15) \cdot 55$$

$$= 30 + 825$$

$$a_{16} = 855\text{ cm}$$

Therefore, after 16 minutes, the water level will be 855 centimeters high in the irrigation canal.

Arithmetic Series

Arithmetic Series is the sum of terms of an arithmetic sequence. It represents adding the terms of the sequence together.

Let's say we have the arithmetic sequence: 3, 6, 9, 12. This is an arithmetic sequence where the common difference $d = 3$. Now, let's add the terms: $3 + 6 + 9 + 12 = 30$. The sum is an example of arithmetic series.

Let us compute the sum of the first n terms of an arithmetic series.

$$S_n = a_1 + a_2 + \dots + a_n$$

$$S_n = a_1 + (a_1 + d) + (a_1 + 2d) + \dots + [a_1 + (n - 1)d]$$

There are n terms, so we add together all the a_1 's

$$S_n = na_1 + [1 + 2 + \dots + (n - 1)]d$$

We have already derived the formula for the sum of the numbers from 1 to $n-1$.

$$S_n = na_1 + (n - 1)nd/2 = n[2a_1 + (n - 1)d]/2$$

Now we know that $a_1 + (n - 1)d = a_n$. Therefore,

$$S_n = n(a_1 + a_n)/2$$

We therefore have the following formulas to compute the sum of the first n terms of an arithmetic series:

If the last n th term (a_n) is given: $S_n = \frac{n}{2}(a_1 + a_n)$, or

If the last n th term (a_n) is not given: $S_n = \frac{n}{2}[2a_1 + (n - 1)d]$

where: S_n = the sum of first n terms

a_1 = first term

a_n = last n th term

d = common difference

Example 1: Find the sum of the arithmetic sequence with 10 terms: 3, 10, 17, ..., 66.

Solution:

Since the last n th term is given, use the 1st formula.

Step 1: Identify the given values.

$$a_1 = 3$$

$$a_n = 66$$

$$n = 10$$

	$S_n = ?$ Step 2: Substitute the values to the formula. $S_{10} = \frac{10}{2} (3 + 66)$ $= 5 (69)$ $S_{10} = 345$ Therefore, the sum of the arithmetic sequence is 345. Example 2. Find the sum of the first 15 terms of the arithmetic sequence: 4, 9, 14, 19, ... Solution: Since the last nth term is not given, use the 2nd formula. Step 1: Identify the given values. $a_1 = 4$ $n = 15$ $d = 5$ $S_n = ?$ Step 2: Substitute the given values to the formula. $S_{12} = \frac{15}{2} [2(4) + (15 - 1)5]$ $= 7.5 [8 + (14)5]$ $= 7.5 [8 + 70]$ $= 7.5 [78]$ $S_{12} = 585$ Therefore, the sum of the arithmetic sequence is 585. Example 3: Mang Tonyo is a kalesa driver in the historic streets of Vigan, Ilocos Sur. On his first day, he earns ₱200. As more tourists visit throughout the week, his earnings increase by ₱100 each day. If this pattern continues for 15 days, how much will Mang Tonyo earn in total? Solution: Since the last earning or nth term is not given, use the 2nd formula. Let a_n = the earnings of Mang Tonyo on day n. Step 1: Identify the given values. $a_1 = ₱200$ $d = ₱100$ $n = 15$ days $S_n = ?$ Step 2: Substitute the given values to the formula. $S_{15} = \frac{15}{2} [2(200) + (15 - 1)100]$ $= 7.5 [400 + (14)100]$ $= 7.5 [400 + 1400]$ $= 7.5 [1800]$ $S_{15} = ₱13,500$ Therefore, Mang Tonyo will earn a total of ₱13,500 in 15 days.
Video Lesson:	“Sequences and Series.” YouTube video, n.d. https://www.youtube.com/watch?v=REDVjUTvXvk&t=368s. “Types of Sequences.” YouTube Shorts video, n.d. https://www.youtube.com/shorts/qW0yF2_-LEc. “Arithmetic Sequence.” YouTube video, n.d. https://www.youtube.com/watch?v=jExpsJTU9o8. “Arithmetic Series.” YouTube video, n.d. https://www.youtube.com/watch?v=GZH68SubgRE.
I. Activity 1:	Practicing Learned Skills
II. Materials:	Pen or pencil, Calculator (optional), Paper, Formula reference sheet (optional)
III. Instructions:	1. Read each problem carefully and identify what is being asked. a. Find the 8th term of the arithmetic sequence: 5, 11, 17, 23, ...

	b. The 5th term of an arithmetic sequence is 40, and the common difference is 6. Find the 1st term of the sequence. c. Find the sum of the first 12 terms of the arithmetic sequence: 4, 7, 10, ... d. What is the sum of the first 15 terms of the sequence: 3, 6, 9, ...? e. Mang Ernesto, a jeepney driver in Manila, starts with a fare of ₱20 on his first trip of the day. For each succeeding trip, he earns ₱5 more than the previous trip. How much will he earn in total after 10 trips? 2. Determine whether the problem involves finding a specific term or the sum of terms in a number pattern. 3. Identify the given values such as the first term, common difference, number of terms, or specific term. 4. Choose the appropriate formula for solving each problem. 5. Show complete and organized solutions for each item. 6. Check answers for accuracy by substituting values back into the formula when needed.
V. Reflection:	Reflect on the following question: 1. How did you determine which formula to use in each problem? 2. What steps helped you identify the first term and common difference in the sequence problems? 3. How does knowing the common difference make solving for missing terms easier? 4. In what way does solving for the sum differ from solving for a specific term? 5. Which problem did you find most challenging and why? 6. How can these types of problems be applied in real-life situations such as transportation fares or savings?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen or pencil, paper, Calculator (optional), Formula reference sheet (optional)
III. Instructions:	1. Read each item carefully before answering. 2. For Part I, choose the letter of the best answer. 3. For Part II, solve each problem completely and show clear, step-by-step solutions. 4. For Part III, write a reflective essay with at least 200 words. 5. Use appropriate formulas and ensure all solutions are properly organized. 6. Review your answers before submitting. Part I. Multiple Choice. Choose the best answer. 1. What is the main difference between a sequence and a series? A. A sequence has a formula, while a series does not B. A sequence is an ordered group of terms, while a series is their sum C. A sequence includes variables, while a series includes constants D. A sequence is always infinite, while a series is finite 2. Which of the following best defines a sequence? A. The difference between two terms B. A list of numbers without any order C. A set of numbers arranged in a specific order or pattern D. The sum of terms in an equation 3. What does the term “series” refer to in mathematics? A. A pattern that repeats B. A group of equations C. The sum of the elements in a sequence D. The position of an element in a list 4. Which formula is used to find the nth term of an arithmetic sequence? a. $a_n = a_1 + (n - 1)d$ b. $a_n = a_1 + (n + 1)d$ c. $S_n = \frac{n}{2}(a_1 + a_n)$ d. $S_n = \frac{n}{2}[2a + (n - 1)d]$ 5. What is the common difference of the arithmetic sequence: 6, 10, 14, 18, ...? A. 2 B. 4 C. 5 D. 3

	6. What is the sum of the first 8 terms of the arithmetic series: 4, 8, 12, 16, ...? A. 120 B. 136 C. 144 D. 160 7. In an arithmetic sequence, the 5th term is 12 and the 12th term is 40. What is the common difference (d)? A. 2 B. 3 C. 4 D. 5 8. Which of the following represents the sum of the first n terms of an arithmetic sequence? A. $S_n = a_1 + (n - 1)d$ B. $S_n = \frac{n}{2} (2a_1 + (n - 1)d)$ C. $S_n = \frac{n}{2} (a_1 + a_n)$ D. Both B and C Part II. Solve the problem. Show your solution. Maria is a student who wants to buy a new bike. She saves ₱50 on the first day, and every day after that, she saves an additional ₱20. She continues to save for 15 days. Questions: 1. How much money will Maria save on the 15th day? 2. What is the total amount saved after the 15 days? Part III. Reflective Essay. Write a reflective essay of at least 200 words discussing the following: How do you plan to use the knowledge of arithmetic sequences and series in future lessons or in your daily life? <table><tr><th>Criteria</th><th>Excellent (4)</th><th>Proficient (3)</th><th>Developing (2)</th><th>Beginning (1)</th></tr><tr><td>Clarity of Expression</td><td>Ideas are clearly and logically expressed; reflection is easy to understand and well-organized throughout.</td><td>Ideas are mostly clear with minor lapses in organization or flow.</td><td>Ideas are somewhat unclear; organization is inconsistent and may require reader interpretation.</td><td>Ideas are unclear, disorganized, and difficult to understand.</td></tr><tr><td>Depth of Thought / Insight</td><td>Demonstrates deep, meaningful reflection with critical thinking, personal insight, and strong connections to experiences or concepts.</td><td>Shows good reflection with some insight and connections, though not fully developed.</td><td>Limited reflection; ideas are mostly descriptive with minimal insight or analysis.</td><td>Lacks reflection; purely descriptive or off-topic with no evidence of insight.</td></tr><tr><td>Relevance to Prompt / Topic</td><td>All ideas are highly relevant and directly address the prompt with strong focus throughout.</td><td>Most ideas are relevant with minor digressions from the topic.</td><td>Some ideas are relevant, but there are noticeable off-topic sections.</td><td>Response is largely irrelevant or does not address the prompt.</td></tr><tr><td>Support / Use of Experience or Evidence</td><td>Provides strong, meaningful examples, experiences, or evidence to support reflections.</td><td>Provides some examples or evidence, but not fully developed or specific.</td><td>Few or vague examples; support is weak or general.</td><td>No supporting examples or evidence provided.</td></tr><tr><td>Language Use (Grammar as Support)</td><td>Language is clear and accurate; minor or no errors that do not affect meaning.</td><td>Some errors present but meaning remains clear.</td><td>Frequent errors that sometimes affect clarity.</td><td>Persistent errors that significantly hinder understanding.</td></tr></table>	Criteria	Excellent (4)	Proficient (3)	Developing (2)	Beginning (1)	Clarity of Expression	Ideas are clearly and logically expressed; reflection is easy to understand and well-organized throughout.	Ideas are mostly clear with minor lapses in organization or flow.	Ideas are somewhat unclear; organization is inconsistent and may require reader interpretation.	Ideas are unclear, disorganized, and difficult to understand.	Depth of Thought / Insight	Demonstrates deep, meaningful reflection with critical thinking, personal insight, and strong connections to experiences or concepts.	Shows good reflection with some insight and connections, though not fully developed.	Limited reflection; ideas are mostly descriptive with minimal insight or analysis.	Lacks reflection; purely descriptive or off-topic with no evidence of insight.	Relevance to Prompt / Topic	All ideas are highly relevant and directly address the prompt with strong focus throughout.	Most ideas are relevant with minor digressions from the topic.	Some ideas are relevant, but there are noticeable off-topic sections.	Response is largely irrelevant or does not address the prompt.	Support / Use of Experience or Evidence	Provides strong, meaningful examples, experiences, or evidence to support reflections.	Provides some examples or evidence, but not fully developed or specific.	Few or vague examples; support is weak or general.	No supporting examples or evidence provided.	Language Use (Grammar as Support)	Language is clear and accurate; minor or no errors that do not affect meaning.	Some errors present but meaning remains clear.	Frequent errors that sometimes affect clarity.	Persistent errors that significantly hinder understanding.
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V. Reflection:	Reflect on the following question. - How did you determine whether a problem involves a sequence or a series? - What strategy did you use to find the common difference in a sequence? - How did you decide which formula to use for solving the problems in Part II? - What steps helped you organize your solutions clearly and accurately? - How can arithmetic sequences and series be applied in real-life situations such as savings or planning expenses? - What challenges did you encounter while answering the test, and how did you address them?																														
I. Activity 3:	Extending and Reinforcing Learning																														
II. Materials	Pen or pencil, paper, Calculator (optional), Visual aids or number pattern cards																														
III. Instructions:	<i>Remediation</i> Instructions: Group Formation: Form groups of 3–4 members with mixed ability levels to promote peer support and collaboration. Warm-Up: Pattern Recognition:																														

	Each group is given number patterns and will determine whether each is an arithmetic sequence. If yes, identify the common difference. Examples: • 5, 8, 11, 14, ____ • 20, 16, 12, 8, ____ 3. Group Task: Complete the Sequence Each group will analyze 2–3 given problems involving arithmetic sequences. Tasks include: • Identifying the first term (a_1) • Finding the common difference (d) • Writing the general formula: 1. $a_n = a_1 + (n - 1) d$ • Computing the required term (e.g., 10th or 15th term) Example Task: • The 3rd term of a sequence is 10 and the 6th term is 19. Find the first term and common difference. • Find the 12th term of the sequence. 4. Peer Teaching and Feedback: Each group presents one solution. Other groups may ask questions, clarify steps, or suggest improvements. <i>Enhancement</i> 1. Read each problem carefully and understand what is being asked before solving. 2. Identify the key information in each situation, such as the first term, common difference, and number of terms. 3. Determine whether the problem requires finding a specific term, the total number of items, or both. 4. Use the appropriate formula for arithmetic sequences or series: • $a_n = a_1 + (n - 1) d$ for finding a specific term • $S_n = \frac{n}{2}[a_1 + a_n]$ or $S_n = \frac{n}{2}[2a_1 + (n - 1)d]$ for finding the sum 5. Show complete solutions for each problem, including substitution and computation. 6. Clearly label final answers and include correct units (e.g., desks, steps, toothpicks). 7. Double-check answers for accuracy and completeness before submission. Solve the following. 1. In a classroom, the first row has 5 desks, the second row has 7 desks, the third row has 9 desks, and so on, increasing by 2 desks per row. If there are 10 rows in total, how many desks are there altogether? 2. A school building has stairs with decreasing steps on each level. The top level has 10 steps, the second has 12 steps, the third has 14 steps, and so on. If there are 8 levels, how many steps are there in total from the ground floor to the top? 3. Percy is making patterns with toothpicks as follows: a. How many toothpicks will Percy need for Pattern 15? b. How many toothpicks will Percy have used after doing Patterns 1 to 15?
IV. Reflection:	Reflect on the following questions: 1. How did your group determine whether a sequence is arithmetic or not? 2. What strategies helped you solve for missing terms and values in the sequence? 3. How does identifying the common difference help in solving sequence problems faster? 4. What was the most effective part of your group discussion in solving the tasks?

	5. How do arithmetic sequences help in modeling real-life situations like seating arrangements or patterns? 6. How does teamwork improve your understanding of mathematical concepts? *****END OF LESSON 2.3*****
Lesson 2.4. Geometric Sequence and Series	
Objectives:	By the end of the lesson, you shall be able to: 1. illustrate the geometric sequence, 2. define geometric sequences and series, including their key components, 3. calculate the nth term of a geometric sequence, 4. compute the geometric series, 5. solve problems involving geometric sequences and series, and 6. demonstrate appreciation of the importance of arithmetic sequences and series by explaining or presenting how they are used in making informed decisions in real-life situations.
Key Ideas and Examples:	Geometric Sequence and Series Geometric Sequence is a special type of sequence where each term after the first is obtained by multiplying the preceding term by a fixed non-zero number. The fixed multiplier is called the common ratio, denoted by r. It is called the common ratio because it is the ratio of any two consecutive terms in the sequence. $r = \frac{a_{n+1}}{a_n}$ The formula in calculating the nth term in a geometric sequence is $a_n = a_1 r^{n-1}$ where: a_n = the last nth term a_1 = first term r = common ratio n = the position of the term To calculate the nth term of a geometric sequence, let's follow the steps: 1. Identify the value of the first term. 2. Find the value of the common ratio. 3. Apply the formula in calculating the nth term. Example 1: Find the 5th term of a geometric sequence where the first term is 5 and the common ratio is 3. Solution: $a_1 = 5$ $r = 3$ $n = 5$ $a_n = ?$ $a_5 = 5(3)^{5-1}$ $= 5(3)^4$ $= 5(81)$ $a_5 = 405$ Therefore, 405 is the value of the 5th term in the sequence. Example 2: Find the 10th term of the geometric sequence: 4, 12, 36, ... Solution: $a_1 = 4$ $r = 3$ $n = 10$ $a_n = ?$ $a_{10} = 4(3)^{10-1}$ $= 4(3)^9$ $= 4(19683)$ $a_{10} = 78732$

Example 3: The 4th term of a geometric sequence is 64, and the 9th term is 486. Find (a) the common ratio, (b) the first term, and (c) the 7th term.

Solution:

From a_4 :

$$64 = a_1 r^{4-1}$$

$$16 = a_1 r^3 \text{ (Equation 1)}$$

From a_9 :

$$486 = a_1 r^{9-1}$$

$$486 = a_1 r^8 \text{ (Equation 2)}$$

Divide the equation 2 by the equation 1:

Solving for the common ratio (r).

$$\frac{486}{64} = \frac{a_1 r^8}{a_1 r^3}$$

$$\frac{243}{32} = r^5$$

$$\frac{3}{2} = r$$

Substitute $r = 3/2$ in equation 2:

Solving for the first term (a_1)

$$64 = a_1 \left(\frac{3}{2}\right)^3$$

$$\frac{512}{27} = a_1$$

Solve for the 7th term:

Solution:

$$a_1 = \frac{512}{27}$$

$$r = \frac{3}{2}$$

$$n = 7$$

$$a_n = ?$$

$$a_7 = \frac{512}{27} \left(\frac{3}{2}\right)^{7-1}$$

$$= \frac{512}{27} \left(\frac{3}{2}\right)^6$$

$$= 216$$

Example 4: A local farmer named Mang Pedro started raising native chickens. In his first month, he had 3 chickens. Every month, the number of chickens tripled as each chicken hatched more chicks. The number of chickens each month follows a geometric sequence.

Problem:

1. How many chickens does Mang Pedro have by the 5th month?
2. How many chickens does he have by the 8th month?
3. Write the general formula to find the number of chickens in the nth month.

Solutions:

Given:

First term $a_1 = 3$

Common ratio $r = 3$

1. 5th month

$$a_5 = 3 (3)^{5-1}$$

$$= 3 (3)^4$$

$$= 3 (81)$$

$$a_5 = 243 \text{ chickens}$$

2. 8th month

$$a_8 = 3 (3)^{8-1}$$

$$= 3 (3)^7$$

$$= 3 (2187)$$

$a_8 = 6561$ chickens

3. General Formula

$$a_n = 3(3)^{n-1}$$

$$a_n = 3^n$$

Geometric Series

Review of terms:

1. First Term (a_1): This is the very first term of the sequence. It's the starting value from which all other terms are derived by multiplication with the common ratio.

2. Common Ratio (r): This is the constant factor by which each term in the sequence is multiplied to get the next term. It is the ratio of all consecutive terms in the series.

3. S_n : This is the sum of the first n terms of a geometric sequence.

To derive the formula for the sum of the first n terms of a geometric sequence.

A geometric sequence has:

First term: a_1

Common ratio: r

Number of terms: n

When $r \neq 1$, let:

$$S_n = a_1 + a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^{n-1} \quad \text{eq. (1)}$$

Now, multiply both sides of equation (1) by r :

$$rS_n = a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^n \quad \text{eq. (2)}$$

Next, subtract equation (2) from equation (1):

$$\begin{array}{r} S_n = a_1 + a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^{n-1} \\ - \\ rS_n = a_1r + a_1r^2 + a_1r^3 + \dots + a_1r^n \\ \hline S_n - rS_n = a_1 - a_1r^n \end{array}$$

Extract S_n as a common factor on the left-hand side of the equation and a_1 on the right-hand side.

$$S_n(1 - r) = a_1(1 - r^n)$$

Now, divide both sides $1 - r$.

$$S_n = \frac{a_1(1 - r^n)}{(1 - r)}$$

This gives us a formula for a finite geometric series with n terms.

What happens if the series is infinite? This means that n becomes very large. If we look at the expression on the right side of the equation, the part affected by n is r^n . If r is "large" ($r > 1$ or $r < -1$), r^n also becomes large and we don't get a real number value for S_n . If r is "small" ($-1 < r < 1$), then r^n becomes very small for large values of n .

We can therefore say that as the number of terms go to infinity ($n \rightarrow \infty$), the values of r^n become very small ($r^n \rightarrow 0$). Which means that the numerator in the expression for S_n will go to $a_1(1 - 0) = a_1$.

Therefore, if $-1 < r < 1$, the sum of an infinite geometric series is

$$S = \frac{a_1}{1 - r}$$

Example 1: Find the sum of the first five terms of the geometric series: $2 + 6 + 18 + 54 + \dots$

Solution:

$$a_1 = 2$$

$$r = 3$$

$$n = 5$$

	$S_n = ?$ $S_5 = 2 \left(\frac{1 - 3^5}{1 - 3} \right)$ $S_5 = 2 \left(\frac{1 - 243}{1 - 3} \right)$ $S_5 = 2 \left(\frac{-242}{-2} \right)$ $S_5 = 2 (121)$ $S_5 = 242$ Example 2: Consider the geometric series: $10 + 5 + 2.5 + 1.25 + \dots$. Find the sum of the infinite series. Solution: $a_1 = 10$ $r = \frac{1}{2}$ $S = ?$ $S_n = \frac{10}{1 - \frac{1}{2}}$ $S_n = \frac{10}{\frac{1}{2}}$ $S_n = 10 \times 2$ $S_n = 20$ Example 3: Mang Tonyo owns a small sari-sari store in the barangay. One day, he decided to start saving coins in an alkansya (piggy bank). On the first day, he puts ₱20. Each following day, he doubles the amount he saves from the previous day. He plans to continue this for 7 days. How much total money will Mang Tonyo have saved in the alkansya after 7 days? Solution: Let a_n = The money Mang Tonyo will save in the alkansya on the nth day. $a_1 = 20$ $r = 2$ $n = 7$ Use the Geometric Series Formula: $S_7 = a_1 \frac{1 - r^n}{1 - r}$ $S_7 = 20 \left(\frac{1 - 2^7}{1 - 2} \right)$ $S_7 = 20 \left(\frac{1 - 128}{1 - 2} \right)$ $S_7 = 20 \left(\frac{-127}{-1} \right)$ $S_7 = 20(127)$ $S_7 = ₱2540$ Therefore, Mang Tonyo will save ₱2540 in 7 days.
Video Lesson:	"Geometric Sequence." YouTube video, n.d. https://www.youtube.com/watch?v=DVpfJC_-waw&t=76s "Geometric Series." YouTube video, n.d. https://www.youtube.com/watch?v=kygPMFN4baM
I. Activity 1:	Practicing Learned Skills
II. Materials:	Pen or pencil, paper, Calculator (optional), Formula sheet for geometric sequence and series
III. Instructions:	1. Solve each problem carefully and show complete solutions.

	2. Identify the given values in each item (first term, common ratio, and number of terms). 3. Apply the appropriate formula for geometric sequences or series. 4. Simplify and compute accurately. 5. Write final answers clearly and in simplest form. Problems 1. Given the geometric sequence: 3, 6, 12, 24, ... Find the 7th term of the sequence. 2. In a geometric sequence, the first term is 5 and the common ratio is 3. Find the value of the 10th term. 3. The 4th term of a geometric sequence is 16, and the 7th term is 128. Find the: - Common ratio - First term 9th term 4. Find the sum of the first 6 terms of the geometric series: $2 + 4 + 8 + 16 + \dots$ 5. A Forestry student is conducting research on sustainable bamboo production as part of an Entrepreneurship and Employment SHS Exit Pathway. He observed that a bamboo tree grows such that each week its height is multiplied by 1.5. If its initial height is 1 meter, find the total height gained after 5 weeks.
V. Reflection:	Respond to these questions. - What steps did you follow in identifying the correct formula for each problem? - How do you determine whether a problem requires finding a term or a sum of a series? - What role does the common ratio play in solving geometric sequence problems? - Which problems required working backward to find missing values, and how did you approach them? - How can geometric sequences and series be applied to real-life growth situations like population or plant growth? - What strategies helped you avoid errors in computation?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen or pencil, paper, Calculator (optional for computation)
III. Instructions:	- Read each question carefully before answering. - For Part I, choose the letter of the correct answer for each item. - For Part II, solve each problem completely and show all necessary solutions. - For Part III, write a short reflective essay consisting of 3–5 sentences. - Ensure answers are clear, organized, and properly labeled. - Review your work before submitting it. <i>Part I. Multiple Choices.</i> Choose the letter of the best answer. - Which of the following is a geometric sequence? A. 2, 4, 6, 8, ... B. 3, 6, 12, 24, ... C. 1, 3, 6, 10, ... D. 10, 9, 8, 7, ... - What is the common ratio in the geometric sequence 5, 15, 45, 135, ...? A. 3 B. 10 C. 15 D. 5 - In the geometric sequence, what does the variable a_1 represent? A. nth term B. number of term C. first term D. sum of terms - What does a geometric series represent? A. A list of repeated numbers B. A pattern that increases by addition C. The sum of terms in a geometric sequence D. The difference between terms in a pattern - Which formula is used to find the sum of the first n terms in a geometric series when $r \neq 1$. A. $S_n = a_1 \frac{1-r^n}{1-r}$ B. $S_n = \frac{a_1}{1-r}$

	C. $S_n = a_1 r^{n-1}$ D. $S_n = \frac{n}{2}(a_1 + a_n)$ <i>Part II: Problem Solving</i> Solve and show your complete solution. 1. Geometric Sequence Jorge saves money by doubling his savings every week. In the first week, he saves ₱50. How much will he have saved in total by the 6th week? 2. Geometric Series A ball is dropped and it bounces back to $\frac{3}{4}$ of its previous height. If the initial drop is 16 meters, what is the total vertical distance the ball travels at the 4th bounce? Part III: Reflective Essay. Answer the question below in 3–5 sentences. <i>Why do you think understanding geometric sequences and series is important in real-life situations like savings, investments, or technology? Give one example where this concept could help you in the future.</i> Rubric is to be presented by the teacher.
V. Reflection:	Respond to these question. - How did you identify whether a sequence is geometric or not in the multiple-choice items? - What steps did you follow to compute the common ratio in a geometric sequence? - Which formula did you use in solving geometric series problems, and why? - How did you approach solving real-life word problems involving sequences and series? - What difficulties did you encounter while solving the problems, and how did you address them? - How can understanding geometric sequences and series help in real-life decision-making such as savings or investments?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	Pen or pencil, paper, Calculator (optional for computation), Formula reference sheet
III. Instructions:	- Read each problem carefully and determine what is being asked. - For the <i>Remediation</i> section, review basic concepts before answering each item. - For the <i>Enhancement</i> section, apply higher-order thinking and problem-solving strategies. - Identify whether each task involves finding the common ratio, nth term, or sum of a series. - Show complete solutions for all problems, including necessary formulas and computations. - Check answers for accuracy and clarity before submitting. <i>For Remediation</i> Problem 1: Identifying the Common Ratio Given: 2, 6, 18, 54, ... Question: What is the common ratio of the geometric sequence? Problem 2: Finding the nth Term Given: The geometric sequence is 5, 10, 20, 40, ... Question: What is the 6th term? Problem 3: Finding the Sum of a Geometric Series Given: Find the sum of the first 4 terms of the geometric series: 3 + 6 + 12 + 24 + ... <i>For Enhancement</i> Problem 1: Finding the Term Position (n) Given: In a geometric sequence, the first term is 2 and the common ratio is 3. Question: Which term is equal to 486? Problem 2: Word Problem – Real-Life Context

	Scenario: A small sari-sari store sells bottled water. On day 1, it sold 2 bottles. Each day, the number of bottles sold has tripled. Question: How many bottles were sold on the 7th day? Problem 3: Sum of Geometric Series Given: A geometric series has a first term of 8 and a common ratio of $\frac{1}{2}$. Question: What is the sum of the infinite geometric series?
IV. Reflection:	Respond to these questions. 1. How do you determine the common ratio in a geometric sequence? 2. What steps did you use to find the nth term of a geometric sequence? 3. How did you decide which formula to use when solving sum problems? 4. What strategies helped you solve word problems involving geometric growth? 5. How does exponential growth appear in real-life situations such as sales or population increase? 6. Why is understanding geometric sequences and series important in analyzing real-world trends?

*****END OF LESSON 2.4*****

Lesson 2.5. Applications of Sequences and Series in Financial Mathematics: Loans, Mortgages, and Investment Growth

Objectives:	By the end of the lesson, you are expected: 1. apply arithmetic and geometric sequences and series in solving financial problems involving periodic loan payments, mortgages, and investment growth; 2. demonstrate appreciation of responsible financial decision-making in managing loans, mortgages, and investments; and 3. create a digital financial poster using EdTech tools (Canva/PowerPoint/Google Slides) presenting solutions and interpretations of a real-life financial situation involving loans or investments
Key Ideas and Examples:	When solving financial problems involving loans, savings, investments, and mortgages, first determine how the amount changes over time. • Use an arithmetic sequence or arithmetic series if it involves a constant difference (d), • Use a geometric sequence or geometric series if it involves a constant ratio (r) Common Applications of Sequences and Series are: • Loan problems → Arithmetic series (sum of constant payments) • Savings problems → Arithmetic sequence (constant increase or decrease) • Investment problems → Geometric sequence (compound growth) Example: 1. Arithmetic Sequence in Savings Carlo saves ₱2,000 during the first month and increases his monthly savings by ₱500 every month. How much will Carlo save during the 6th month? Solution: $a_n = a_1 + (n-1)d$ $a_6 = 2,000 + (6-1)(500)$ $a_6 = 2,000 + 2,500$ $a_6 = ₱4,500$ Answer: Carlo will save ₱4,500 during the 6th month. 2. Geometric Sequence in Investment Angela invests ₱2,000 in an account earning 5% annual interest compounded yearly. How much will the investment become after 3 years? Solution: $a_n = a_1 r^{n-1}$ $a_3 = 2100(1.05)^{3-1} = ₱2,315.25$ Answer: Angela's investment will grow to ₱2,315.25 after 3 years.

Video Lesson:	KhanAcademy: https://www.khanacademy.org/math/grade-8-math-tx/x42e41b058fcf4059:one-variable-equations-inequalities-and/x42e41b058fcf4059:simple-and-compound-interest/v/calculating-simple-compound-interest . Accessed 8 May 2025.
I. Activity 1:	Practicing Learned Skills
II. Materials:	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	Instructions: Solve the following financial problems involving sequences and series. Show your complete solution and final answer. 1. Loan Payment Problem A borrower pays ₱12,000 monthly for 3 years for a motorcycle loan. Guide Questions: a. How many months will the borrower pay? b. What is the total amount paid after 3 years? 2. Savings Growth Problem John saves ₱2,000 during the first month and increases his savings by ₱500 every month. Guide Questions: a. What is the common difference? b. How much will John save during the 8th month? 3. Investment Growth Problem ₱10,000 is invested in an account earning 5% annual interest compounded yearly for 4 years. Guide Questions: a. What type of sequence is involved? b. Find the accumulated amount after 4 years. 4. Mortgage Payment Problem A family pays ₱18,000 monthly for a housing loan for 10 years. Guide Questions: a. How many monthly payments will be made? b. Find the total payment. 5: Accumulated Investment Ana, an investor, begins a start-up to revitalize Camela Homes. She begins with P1,000,000, making her investor 1. Each investor that joins will invest P500,000 more than the previous investor. Guide Questions: 1. How much does the 10th investor invest in the project? 2. With that 10th investor, what is the total amount invested in the project?
V. Reflection:	Reflect on what you learned from the activity. Answer the following questions in 2–3 sentences each. 1. Why is it important to know the total amount paid when taking out a loan? 2. If you were planning your own savings or investment strategy, which concept from this activity would be most useful to you? Explain your answer. 3. What real-life situations in your community or family can be represented using arithmetic or geometric sequences?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	Part I: Identification Identify whether each situation represents an arithmetic sequence, geometric sequence, or series. 1. A borrower pays ₱8,000 every month for a car loan for 5 years. 2. A student saves ₱2,000 in the first month and adds ₱500 every month. 3. A bank account earns 4% interest compounded annually. 4. A family calculates the total amount paid after 10 years of monthly mortgage payments. Part II: Computation and Application Solve each problem. Show your solution and write your final answer clearly.

	1. A loan requires a payment of ₱10,000 per month for 3 years. a. How many total payments will be made? b. What is the total amount paid? 2. A student saves ₱3,000 in the first month and increases savings by ₱500 monthly. a. Find the amount saved in the 6th month. b. Identify the sequence type. 3. A family pays ₱18,500 monthly for a housing mortgage for 8 years. a. How many monthly payments will be made? b. What is the total mortgage payment after 8 years? c. Why is this situation related to a financial series? 4. A worker deposits ₱2,500 monthly into a savings program for 3 years. a. How many deposits will be made? b. What is the total accumulated investment? c. How can this investment help in future financial planning? Part III: Application and Interpretation Instructions: Analyze the situation and answer the guide questions. Support your answers with computations and explanation Maria chooses between: Option A: Pay ₱12,000 monthly for a loan for 4 years Option B: Save ₱12,000 monthly for 4 years Guide Questions: a. What is the total amount for each option? b. Which option is better for financial growth? Explain your answer. c. How does mathematics help in making this decision?
V. Reflection:	Reflect on what you learned from the activity. Answer the following questions in 2–3 sentences each. 1. What did you learn about the use of sequences and series in financial situations such as loans, savings, and investments? 2. How can understanding mathematical patterns help you make better financial decisions in the future? 3. Among the financial situations discussed (loan, savings, mortgage, or investment), which do you think is most relevant to your life? Why?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	Pen and paper, Calculators (Scientific or Basic)
III. Instructions:	Activity C.4: Enrichment and Remediation A. Remediation Activity: “Identifying Financial Patterns” Identify whether each situation shows an arithmetic sequence, geometric sequence, or series. Then briefly explain your answer. 1. A person pays ₱5,000 every month for a personal loan. 2. A student saves ₱1,000 and adds ₱200 every month. 3. A bank account earns 4% interest compounded annually. Guide Questions: 1. Is the pattern increasing or constant? 2. Does it involve addition or multiplication? 3. What type of sequence is it? B. Enrichment Activity: Title: “Smart Loan vs Investment Decision” Carefully analyze the situation below and complete the required tasks. A person has ₱10,000 per month for 3 years. He may choose: • Option A: Pay a loan (₱10,000 monthly) • Option B: Invest ₱10,000 monthly at 6% annual interest Guide Tasks: 1. Compute total loan payments 2. Compute total investment growth (conceptual use of compound interest) 3. Compare both options 4. Which is financially better in the long run? Explain briefly
IV. Reflection:	Reflect on what you learned from the activity. Answer the following questions in 2–3 sentences each.

1. How did the remediation activity help strengthen your understanding of arithmetic sequences, geometric sequences, and series in financial situations?
2. What did you learn from the enrichment activity about comparing loans and investments when making financial decisions?
3. How can the concepts of sequences and series help you become a more responsible saver, borrower, or investor in the future?

*****END OF LESSON 2.5*****

Unit 3. Conversion of Measurement

Lesson 3.1. Quantities of Different Magnitudes: Choosing Appropriate Units and Applying Scientific Notation

Objectives:

At the end of the lesson, you should be able to:

1. identify the appropriate metric units for measuring small and large quantities
2. apply scientific notation to express small or large numbers and vice versa.
3. solve real-life problems by applying the appropriate metric units for small and large quantities and scientific measurements.

Key Ideas and Examples:

METRIC UNITS

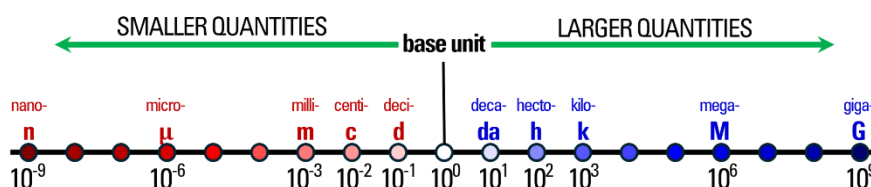
Metric units are standard units used to measure length, mass, time, temperature, and volume.

Examples:

- Length → centimeter (cm), kilometer (km), millimeter (mm)
- Mass → gram (g), milligram (mg), metric ton (t)
- Volume → liter (L), milliliter (mL)
- Time → minute (min), hour (h)
- Electricity → watt (W), kilowatt (kW)

METRIC PREFIXES

Metric prefixes are added to base units to show very small or very large quantities. They help make measurements easier to read and understand.



Examples:

- **Nano (n)** → nanometer (nm), used for structures inside cells that cannot be seen with the naked eye
- **Micro (μ)** → micrometer (μm), used for very thin materials such as the **thickness of laminating film or paper**
- **Milli (m)** → millimeter (mm), used for small lengths such as the thickness of a sheet of paper or small objects
- **Kilo (k)** → kilometer (km), used for distance between towns or cities
- **Mega (M)** → megawatt (MW), used for electricity produced by power plants

SCIENTIFIC NOTATION

Scientific notation is a mathematical method used to express very large or very small numbers in a shorter and more manageable form.

It is written as:

$$m \times 10^n$$

where:

- m is a number from 1 to less than 10
- n is an integer exponent indicating the power of 10.

Converting Standard Form to Scientific Notation

To convert a number to scientific notation:

1. Move the decimal point, so the number becomes from 1 to less than 10.
2. Count how many places the decimal point moved.
3. If the decimal moves to the left, the exponent is positive.
4. If the decimal moves to the right, the exponent is negative.

Examples:

- A virus has a diameter of **0.00000006 meter**. Express the measurement in scientific notation.

$$0.\underbrace{00000000}_{1\ 2\ 3\ 4\ 5\ 6\ 7\ 8}6 = 6 \times 10^{-8} \text{ meters}$$

This is a very small measurement, so scientific notation makes it easier to write and understand. A more appropriate unit is **nanometer (nm)** because it is used for extremely small measurements. This can also be expressed as **60 nm**.

- A fully loaded truck has a mass of **12,000,000 grams**. Express the mass in scientific notation.

$$\underbrace{12}_{7\ 6\ 5\ 4\ 3\ 2\ 1}000000 = 1.2 \times 10^7 \text{ g}$$

The mass of the loaded truck is 1.2×10^7 grams, which is close to the mega (10^6) scale, so it can be expressed in megagrams. In daily life, this is commonly written as a **metric ton (t)** when referring to large vehicle loads or heavy objects. Thus, the measurement can be expressed more conveniently as **12 megagrams (Mg) or 12 metric tons (t)**.

Converting Scientific Notation to Standard Form**To convert scientific notation to standard form:**

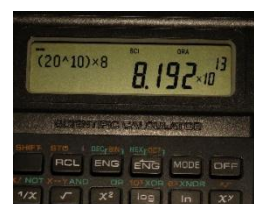
1. Identify the exponent.
2. Move the decimal point to the right if the exponent is positive.
3. Move the decimal point to the left if the exponent is negative.
4. Write the resulting number.

Examples:

- A scientific calculator displays the result of a computation as: 8.192×10^{13} . Express the number in the standard form.

$$8.192 \times 10^{13} = \underbrace{81920000000000}_{1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12\ 13}$$

$$= 81,920,000,000,000$$



- A laboratory report shows that a blood sample contains: 4.7×10^{-9} g/mL of a certain hormone. Express the measurement in standard form.

$$4.7 \times 10^{-9} = \underbrace{0.0000000047}_{9\ 8\ 7\ 6\ 5\ 4\ 3\ 2\ 1}$$

$$= 0.0000000047$$

Video Lesson:)

Math with Mr. J. 2025. "How to Convert Scientific Notation to Standard Form | Math with Mr. J." Video. *YouTube*. <https://www.youtube.com/watch?v=2HliVCyaC5E>.
mathantics. 2018. "Math antics - Intro to the metric system." Video. *YouTube*.
<https://www.youtube.com/watch?v=ZNX-a-5jGeM>.

I. Activity 1:**Practicing Learned Skills****II. Materials:****For Online Completion**

- Mobile phone, tablet, laptop, or desktop computer
- Calculator (optional)
- Scratch paper or notebook for computations

For Printed or Manual Completion

	• Pen or pencil • Notebook or answer sheet • Calculator (optional)																
III. Instructions:	Choose one of the following ways to complete the activity: ☐ Option A. Online Completion 1. Access the Measure It Right Challenge using the QR code or the link: https://tinyurl.com/GMMeasureItRight 2. Read and answer all parts of the activity. 3. Review your answers before submitting them online. 4. Take a screenshot of your submission confirmation or score page (if applicable) for documentation.  ☐ Option B. Printed or Manual Completion 1. If you have limited or no internet access, complete the activity using the printed Learning Activity Sheet. 2. Read each item carefully and write your answers on a separate sheet of paper. 3. Show complete solutions for all items involving scientific notation and standard form conversions. 4. Label your answers clearly according to the Part and item number. 5. Keep your completed work in your portfolio or submit it following your teacher's instructions. Activity Proper Part I: Identify the Appropriate Metric Unit 1. measuring the length of a football field 2. weighing a newborn baby 3. determining the volume of a soft drink can 4. measuring the thickness of a sheet of paper 5. calculating the distance between two cities 6. timing a sprint race 7. measuring the wavelength of visible light 8. measuring the mass of a grain of rice 9. measuring the temperature of a patient. 10. measuring the mass of a hormone in a blood sample Part II: Convert to Scientific Notation Express each of the following numbers in scientific notation: <table border="0"> <tbody> <tr> <td>1. 0.000451</td><td>5. 0.00000032</td></tr> <tr> <td>2. 78,000</td><td>6. 0.0000071</td></tr> <tr> <td>3. 56,000,000</td><td>7. 0.000000000053</td></tr> <tr> <td>4. 123,000</td><td>8. 0.00000000379</td></tr> </tbody> </table> Part III: Convert to Standard Notation Rewrite each of the following numbers in standard form: <table border="0"> <tbody> <tr> <td>1. 1.45×10^{-4}</td><td>5. 9.3×10^{-5}</td></tr> <tr> <td>2. 2.32×10^6</td><td>6. 5.6×10^4</td></tr> <tr> <td>3. 3.81×10^{-7}</td><td>7. 1.2×10^{-2}</td></tr> <tr> <td>4. 4.67×10^3</td><td>8. 7.8×10^8</td></tr> </tbody> </table> Part IV: Word Problems a. Solve each problem. Express the given quantity in scientific notation. Then identify a more appropriate metric unit that may also be used to describe the quantity. A laboratory technician measures the mass of a small chemical sample as 0.000045 gram. Express the mass in scientific notation. Then identify a more appropriate metric unit for describing this small quantity. b. A hydroelectric dam has a power generation capacity of approximately 5,200,000 watts. Express this amount in scientific notation. Then identify a larger metric unit that may also be used for this measurement.	1. 0.000451	5. 0.00000032	2. 78,000	6. 0.0000071	3. 56,000,000	7. 0.000000000053	4. 123,000	8. 0.00000000379	1. 1.45×10^{-4}	5. 9.3×10^{-5}	2. 2.32×10^6	6. 5.6×10^4	3. 3.81×10^{-7}	7. 1.2×10^{-2}	4. 4.67×10^3	8. 7.8×10^8
1. 0.000451	5. 0.00000032																
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4. 123,000	8. 0.00000000379																
1. 1.45×10^{-4}	5. 9.3×10^{-5}																
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3. 3.81×10^{-7}	7. 1.2×10^{-2}																
4. 4.67×10^3	8. 7.8×10^8																
V. Reflection:	Put a ✓ in each box that applies to you. ☐ I can choose appropriate metric units for different measurements. ☐ I can convert numbers to scientific notation.																

	☐ I can convert scientific notation to standard form. ☐ I can explain why scientific notation is useful. ☐ I need more practice with some of these skills. Complete the sentences below. The most interesting thing I learned today is _____ One skill I am proud of is _____ One thing I still want to learn more about is _____
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen or pencil, paper, Calculator (optional)
III. Instructions:	<i>Read each question carefully and write your answers completely.</i> - Write the correct answer for each item. - If needed, show your solution (especially for conversions). - If the question asks for an explanation, write 1–2 clear sentences. - Use the correct units and scientific notation when required. - Make sure your answers are neat and easy to understand. - What metric unit would be most appropriate for measuring the thickness of a smartphone screen protector? Justify your answer. - The average diameter of a red blood cell is approximately 0.000007 meters. Express this measurement in scientific notation and identify a more appropriate metric unit. - A bacterium has a length of 0.0000025 meters. Express the measurement in scientific notation. Why is scientific notation useful for representing very small measurements? - A hydroelectric dam has a power generation capacity of 5,200,000 watts. Express this measurement in scientific notation. Identify a larger metric unit that may also be used to express this quantity. - A water treatment facility can store 15,000 liters of water. Would it be more practical to express this capacity as 15,000 L or 15 kL? Explain your answer. - Why is scientific notation useful when expressing very large or very small measurements? Provide an example.
V. Reflection:	Do as indicated. - One thing I learned today: - What helped me understand the lesson: - Something I found difficult: - What I will do better next time: - How I feel about my learning today (circle one): ☐ Confident ☐ Okay ☐ Not sure ☐ Need help
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	For Online Completion - Mobile phone, tablet, laptop, or desktop computer - Calculator (optional) - Scratch paper or notebook for computations For Printed or Manual Completion - Pen or pencil - Notebook or answer sheet - Calculator (optional)
III. Instructions:	Choose your task. Put a ✓ in the box beside the task you will complete. ☐ Remediation (Reinforcement Tasks) Complete this if you need additional practice with the lesson concepts. ☐ Online Completion - Access the activity using the link or QR code provided. https://tinyurl.com/GMMeasureItRight - Answer all parts of the activity. ☐ Printed or Manual Completion



- Answer all parts of the activity on the Learning Activity Sheet or on a separate sheet of paper.
- Show complete solutions when required.

☐ **Enrichment (Measurements in My Community Prototype)**

Complete this if you are ready to apply the lesson concepts to your prototype design.

REMEDIATION: Reinforcement Tasks

A. State the most appropriate metric unit for each item. Justify your answer briefly.

1. Length of a pencil
2. Mass of a loaf of bread
3. Capacity of a bottle of shampoo
4. Width of a fingernail
5. Time needed to boil an egg

B. Convert the following numbers to scientific notation by completing the missing information in each step.

1. Convert 0.00045 to scientific notation.
 - a. In which direction should the decimal point move to obtain a number between 1 and 10?
 - b. How many places should the decimal point be moved to?
 - c. What is the scientific notation of the number?
2. Convert 700,000,000,000 to scientific notation.
 - a. In which direction should the decimal point move to obtain a number between 1 and 10?
 - b. How many places should the decimal point be moved to?
 - c. What is the scientific notation of the number?

C. Convert the following numbers to standard form by completing the missing information in each step.

1. Convert 4×10^7 to standard form.
 - a. In which direction should the decimal point move?
 - b. How many places should the decimal point be moved to?
 - c. What is the standard form of the number?
2. Convert 1.53×10^{-4} to standard form.
 - a. In which direction should the decimal point move?
 - b. How many places should the decimal point be moved to?
 - c. What is the standard form of the number?

D. Answer the following real-life problems

1. A laminating film has a thickness of 0.000125 meters. Express the measurement in scientific notation.
2. The estimated number of cells in the human body is approximately 3.72×10^{13} . Express the number in the standard form.

ENRICHMENT: Measurements in My Community Prototype

1. Review your IPT Prototype Development Plan (Appendix B, Section IV: Key Measurements and Design Specifications).
2. Identify three key components or parts of your prototype that are critical to its design or function.
3. For each component:
 - a. Record the component/part name.
 - b. Record the measurement or quantity as specified in your prototype.
 - c. Indicate the unit of measurements used.
 - d. Explain why this unit is appropriate for the component.
4. Present your work visually through a one-page infographic, poster, or slide, showing the connection between measurements and prototype function. Include sketches or references to the prototype components if possible.

Enrichment Task Rubric

Criteria	Excellent (4)	Proficient (3)	Developing (2)	Beginning (1)
Identification and	Correctly identifies	Correctly identifies	Identifies only one	Measurements

	Accuracy of Measurements (40%)	and records all three prototype measurements with accurate quantities and units.	and records most measurements with minor inaccuracies.	or two measurements, or with noticeable errors.	missing, inaccurate, or unrelated to the prototype.
	Justification and Relevance (30%)	Provides clear explanations of why each measurement and unit is appropriate, and how it supports the prototype's function.	Provides generally clear explanations, but some connections are missing.	Provides limited or partially correct explanations; weak connection to function.	No justification or connection to prototype function.
	Presentation and Clarity (30%)	Work is organized, visually clear, and easy to interpret; includes sketches or references to components.	Work is organized and mostly clear; minor issues with readability or visuals.	Work is somewhat organized; difficult to interpret in parts.	Work is poorly organized, unclear, or missing visual references.

IV. Reflection: Think about the activity you just completed. Instead of answering, create a question that shows your curiosity or something you want to understand better about metric units, scientific notation, or measurements.

1. My Question:

2. Why I am curious about this?: Write 1–2 sentences explaining why this question came to your mind.

3. How I might find the answer: Write 1–2 ideas for how you could explore or investigate this question on your own.

*****END OF LESSON 3.1*****

Lesson 3.2. Unit Conversion Within and Across Systems

Objectives:	At the end of the lesson, you should be able to: 1. differentiate the metric and English systems of measurement, including their units for length, mass, volume, time and temperature. 2. convert measurements between different units within the metric and within English system 3. convert measurements from the metric system to the English system and vice versa 4. solve practical problems involving unit conversions, such as cooking recipes, travel distances, and scientific experiments.
Key Ideas and Examples	CONVERTING METRIC UNITS The Metric System is based on powers of 10. This means that units are related by multiples or fractions of ten, making conversions systematic and consistent. Recall the metric prefix scale: The diagram shows a horizontal number line with powers of 10 from 10^{-9} to 10^9. Above the line, prefixes are listed: nano- (n), micro- (μ), milli- (m), centi- (c), deci- (d), base unit (10⁰), deca- (da), hecto- (h), kilo- (k), mega- (M), and giga- (G). A green double-headed arrow spans from 10^{-9} to 10^9, with 'SMALLER QUANTITIES' on the left and 'LARGER QUANTITIES' on the right. Units farther apart on the metric prefix scale require larger powers of 10 in conversions. Conversion factors allow us to move from one unit to another while keeping the measurement equivalent. Example • Convert 4.5 kilometers to centimeters. Given unit: kilometers; desired unit: centimeters Conversions: $1 \text{ km} = 10^3 \text{ m}$; $1 \text{ cm} = 10^{-2} \text{ m}$

First conversion factor

Second conversion factor

$$4.5 \text{ km} \left(\frac{10^3 \text{ m}}{1 \text{ km}} \right) \left(\frac{1 \text{ cm}}{10^{-2} \text{ m}} \right)$$

$$4.5 \text{ km} \left(\frac{10^3 \text{ m}}{1 \text{ km}} \right) \left(\frac{1 \text{ cm}}{10^{-2} \text{ m}} \right) = \frac{4.5(10^3) \text{ cm}}{10^{-2}} = 4.5(10^5) \text{ cm} = 450,000 \text{ cm}$$

CONVERTING ENGLISH UNITS

The English System uses fixed relationships between units. These values are not based on powers of 10, so conversions are done using a reference table of equivalent units.

CUSTOMARY UNIT CONVERSIONS		
Common Conversions You Should Know		
LENGTH	MASS	CAPACITY
1760 yd = 1 mile  =  1760 yards = 1 mile	1 short ton = 2000 lb  =  1 short ton = 2000 pounds	1 gal = 4 qt  =  1 gallon = 4 quarts
1 yd = 3 ft  =  1 yard = 3 feet	1 lb = 16 oz  =  1 pound = 16 ounces	1 qt = 2 pt  =  1 quart = 2 pints
1 ft = 12 in  =  1 foot = 12 inches		1 pt = 2 c  =  1 pint = 2 cups

Example:

- Convert 2.5 gallons to cups

Given unit: gallons; desired unit: cups

Conversions: 1 gal = 4qt;

1 qt = 2pt;

1 pt = 2c

First conversion factor Second conversion factor Third conversion factor

$$2.5 \text{ gal} \left(\frac{4 \text{ qt}}{1 \text{ gal}} \right) \left(\frac{2 \text{ pt}}{1 \text{ qt}} \right) \left(\frac{2 \text{ c}}{1 \text{ pt}} \right)$$

$$2.5 \text{ gal} \left(\frac{4 \text{ qt}}{1 \text{ gal}} \right) \left(\frac{2 \text{ pt}}{1 \text{ qt}} \right) \left(\frac{2 \text{ c}}{1 \text{ pt}} \right) = \frac{2.5(4)(2)(2)c}{1(1)(1)} = 40 \text{ c}$$

CONVERTING BETWEEN THE METRIC AND ENGLISH SYSTEMS

In real-life situations, measurements are not always in the same system. Sometimes we convert **Metric to English** or **English to Metric**. To do this, we use conversion factors so that units cancel properly, and the value remains the same.

COMMON CONVERSIONS

ENGLISH TO METRIC

QUANTITY	ENGLISH (US CUSTOMARY)	METRIC (SI)
 LENGTH	1 inch (in)	2.54 centimeters (cm)
	1 foot (ft)	centimeters (cm)
	1 yard (yd)	meters (m)
	1 mile (mi)	kilometers (km)
 MASS (WEIGHT)	1 ounce (oz)	28.35 grams (g)
	1 pound (lb)	kilograms (kg)
	1 short ton (2000 lb)	kilograms (kg)
 CAPACITY (VOLUME)	1 fluid ounce (fl oz)	29.57 milliliters (mL)
	1 cup (c)	milliliters (mL)
	1 pint (pt)	milliliters (mL)
	1 quart (qt)	liters (L)
	1 gallon (gal)	liters (L)

Key Idea: To convert between systems, multiply by conversion factors that cancel the given unit and produce the desired unit.

You do not need to memorize all conversions. You only need to:

- know at least one basic conversion
- use it to recall or derive other equivalent values
- connect it with previously learned metric and English conversions

Example

- **Determine the Metric equivalent of 1 foot.**

Given unit: foot (ft) ; desired unit: centimeter (cm)

Conversions: 1 ft = 12 in; 1 in=2.54 cm

$$\begin{aligned}
 & \text{1 ft} \left(\frac{12 \text{ in}}{1 \text{ ft}} \right) \left(\frac{2.54 \text{ cm}}{1 \text{ in}} \right) \\
 & \text{1 ft} \left(\frac{12 \cancel{\text{ in}}}{1 \cancel{\text{ ft}}} \right) \left(\frac{2.54 \text{ cm}}{1 \cancel{\text{ in}}} \right) = \frac{12(2.54) \text{ cm}}{1(1)} \\
 & = 30.48 \text{ cm}
 \end{aligned}$$

ENGLISH-METRIC TEMPERATURE CONVERSION

Temperature is different from other measurements because the Celsius and Fahrenheit scales do not start at the same zero point. This means we cannot use simple multiplication conversion factors. Instead, we use formulas.



**ENGLISH-TO-METRIC SPECIAL CASE:
TEMPERATURE CONVERSION (°C ↔ °F)**

From °F to °C: $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times \frac{5}{9}$	From °C to °F: $^{\circ}\text{F} = \left(^{\circ}\text{C} \times \frac{9}{5}\right) + 32$
--	---

Example:

- **Convert 95°F to °C**

$$\begin{aligned}
 ^{\circ}\text{C} &= (^{\circ}\text{F} - 32) \times \frac{5}{9} \\
 &= (63) \left(\frac{5}{9}\right) \\
 ^{\circ}\text{C} &= 35
 \end{aligned}$$

Answer: 95°F = 35°C

Video Lesson:	Crash Chemistry Academy (2018). "metric unit conversions shortcut: fast, easy how-to with examples". Video. https://www.youtube.com/watch?v=TY4Yoyur3X4 ketzbook. 2016. "Unit conversion the easy Way (Dimensional analysis)." Video. YouTube. https://www.youtube.com/watch?v=HRe1mire4Gc . The Organic Chemistry Tutor. 2016. "Celsius to Fahrenheit to Kelvin Formula Conversions - Temperature Units C to F to K." Video. YouTube. https://www.youtube.com/watch?v=6qiYcyhI158
I. Activity 1:	Practicing Learned Skills
II. Materials:	Online Option • Smartphone, tablet, or computer • Calculator (optional) Printed/Manual Option • Pen and paper • Scratch paper for solutions • Calculator (optional)
III. Instructions:	Choose one of the following ways to complete the activity: ☐ Option A. Online Completion 1. Access the digital version of the activity using the QR code or the link: https://tinyurl.com/GMEscapeChallenge 2. Read and answer all parts of the activity. 3. Review your answers before submitting them online. 4. Take a screenshot of your submission confirmation or score page (if applicable) for documentation. ☐ Option B. Printed or Manual Completion 1. If you have limited or no internet access, complete the activity using the printed Learning Activity Sheet. 2. Read each item carefully and write your answers on a separate sheet of paper. 3. Show complete solutions for all items involving scientific notation and standard form conversions. 4. Label your answers clearly according to the Part and item number. 5. Keep your completed work in your portfolio or submit it following your teacher's instructions. The Great Conversion Escape Challenge Solve all stations by completing the required conversions. Follow each station's clue instruction carefully, show your solutions, and determine the correct Clue Digit. Combine all Clue Digits to form the final escape code. Station 1: Metric Mission • Convert: 1. 7.2 kilometers → millimeters 2. 450 milligrams → nanograms 3. 3.5 milliliters → kiloliters • Clue Instruction: Add the exponents of all three items to get the clue digit. Clue Digit = _____ Station 2: English Explorer • Convert: 1. 3 quarts → pints 2. 7 pints → cups 3. 2 gallons → quarts • Clue Instruction: (Answer to Item 1)+(Answer to Item 3)-(Answer to Item 2) Clue Digit = _____ Station 3: Cross-System Bridge • Convert and round your final answer to two (2) decimal places. 1. 2 feet → centimeters 2. 3 yards → meters 3. 5 pounds → kilograms • Clue Instruction: Add the ones digit of all three answers to get the Clue Digit.



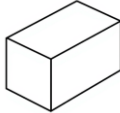
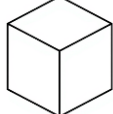
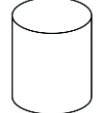
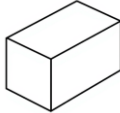
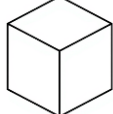
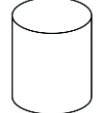
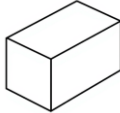
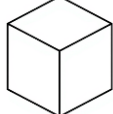
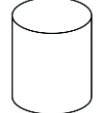
	• Clue Digit = _____ Station 4: Real-World Conversions • Solve. Round your final answer to two (2) decimal places 1. The trip is 120 miles. Convert to kilometers. 2. A box of flour weighs 2.5 kilograms. Convert to pounds. 3. A container holds 1 gallon of water. Convert to liters. • Clue Instruction: Identify the smallest numerical answer among the three conversions. Use the ones digit of that answer as the Clue Digit. • Clue Digit = _____ Station 5: Temperature Vault • Convert: 1. $68^{\circ}\text{F} \rightarrow ^{\circ}\text{C}$ 2. $86^{\circ}\text{F} \rightarrow ^{\circ}\text{C}$ 3. $30^{\circ}\text{C} \rightarrow ^{\circ}\text{F}$ • Clue Instruction: Add the three converted temperatures together. Use the hundreds digit of the sum as the Clue Digit. Clue Digit = _____ Final Escape Code
V. Reflection:	Reflect on your learning by answering the following: 1. What I liked most about the activity: 2. What I found challenging: 3. What I learned from solving the problems: 4. What I still want to improve or understand better:
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper, Calculator (optional)
III. Instructions:	<i>Answer each question carefully. Show complete solutions and include the correct units in your final answers. Use appropriate conversion factors for all items. Write your answers neatly and double-check your computations before submitting.</i> Part A. Hiking Trip to Mount Pulag Mount Pulag is the highest peak in Luzon, Philippines, standing at 2,928 meters above sea level. You are preparing for a hiking trip. 1. Convert the elevation of Mount Pulag from meters to feet. 2. The trail is 15 kilometers long. Convert this distance to miles. 3. Your backpack weighs 3 kilograms. Convert this weight to pounds. 4. You carry 5 liters of water. Convert this volume to milliliters. 5. The temperature at the campsite is 41°F. Convert this temperature to degrees Celsius. Part B. Community Event Preparation You are organizing a community event and need to prepare refreshments and supplies for a community celebration. 1. The recipe requires 5 gallons of fruit juice. Convert this volume to liters. 2. Another recipe requires 3 quarts of fruit juice. Convert this amount to cups. 3. A serving station requires a tablecloth measuring 2 yards long. Convert this length to meters. 4. A package of decorative materials weighs 16 ounces. Convert this weight to grams. 5. A banner support pole measures 8 feet high. Convert this height to centimeters.
V. Reflection:	Reflect on your learning experience by answering the following: 1. What I found easy or difficult in solving the problems: 2. What strategy I used to solve the conversion problems: 3. What I learned from this activity:
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	For Online Completion • Mobile phone, tablet, laptop, or desktop computer • Calculator (optional) For Printed or Manual Completion

	• Pen or pencil • Notebook or answer sheet • Calculator (optional)				
III. Instructions	Choose <i>your task</i>. Put a ✓ in the box beside the task you will complete. ☐ Remediation (Reinforcement Tasks) – Complete this if you need additional practice with the lesson concepts. ☐ Online Completion • Access the activity using the link or QR code provided. https://tinyurl.com/GMConversionQuest • Answer all parts of the activity. ☐ Printed or Manual Completion • Answer all parts of the activity on the Learning Activity Sheet or a separate sheet of paper. • Show complete solutions and include correct units. ☐ Enrichment (Measurements in My Community Prototype) – Complete this if you are ready to apply measurement concepts to improve and adjust your community prototype design. REMEDIATION: Conversion Quest <i>Instructions:</i> 1. Solve all problems at each station. 2. Show complete solutions. 3. Answer all items correctly before moving to the next station. 4. Collect the letter clue after each station. 5. Record all letter clues and form the hidden word. 6. Write the final answer in the space provided. <table border="0" style="width: 100%;"> <tr> <td style="vertical-align: top; width: 50%;"> Station 1: Metric → Metric (T) 1. Convert 7.2 kilometers to meters. 2. Convert 450 milligrams to kilograms. 3. Convert 0.85 megameters to centimeters. 4. Convert 3.5 milliliters to kiloliters. 5. Convert 2.4 gigagrams to micrograms </td><td style="vertical-align: top; width: 50%;"> Station 2: English → English (U) 1. Convert 3 quarts to pints. 2. Convert 7 pints to cups. 3. Convert 2 gallons to quarts. 4. Convert 16 ounces to pounds. 5. Convert 2 short tons to pounds. </td></tr> <tr> <td style="vertical-align: top;"> Station 3: Metric ↔ English (I) 1. Convert 6 feet to meters. 2. Convert 3 yards to meters. 3. Convert 15 miles to kilometers. 4. Convert 3 pounds to kilograms. 5. Convert 5 gallons to liters. </td><td style="vertical-align: top;"> Station 4: Temperature (N) 1. Convert 68°F to °C. 2. Convert 86°F to °C. 3. Convert 30°C to °F. 4. Convert 100°C to °F. 5. Convert 50°F to °C. </td></tr> </table> Hidden Word: _____ ENRICHMENT: Adapting My Community Prototype In real-world problem solving, you often need to adjust solutions to fit changing circumstances, such as available resources, environmental conditions, or the number of users. In this activity, you will apply scale factors, measurement adjustments, and unit conversions to your prototype, helping you see how each modification affects its function, capacity, and overall effectiveness. 1. Review your group's Prototype Adaptability Plan (Appendix C). 2. Identify a realistic scenario that requires your prototype to be scaled up or scaled down. 3. Select at least three design features or components that need to be modified to address the chosen scenario. 4. Complete Section IV. Scale, Measurement, and Conversion Computations by: a. recording the original measurement and unit;	Station 1: Metric → Metric (T) 1. Convert 7.2 kilometers to meters. 2. Convert 450 milligrams to kilograms. 3. Convert 0.85 megameters to centimeters. 4. Convert 3.5 milliliters to kiloliters. 5. Convert 2.4 gigagrams to micrograms	Station 2: English → English (U) 1. Convert 3 quarts to pints. 2. Convert 7 pints to cups. 3. Convert 2 gallons to quarts. 4. Convert 16 ounces to pounds. 5. Convert 2 short tons to pounds.	Station 3: Metric ↔ English (I) 1. Convert 6 feet to meters. 2. Convert 3 yards to meters. 3. Convert 15 miles to kilometers. 4. Convert 3 pounds to kilograms. 5. Convert 5 gallons to liters.	Station 4: Temperature (N) 1. Convert 68°F to °C. 2. Convert 86°F to °C. 3. Convert 30°C to °F. 4. Convert 100°C to °F. 5. Convert 50°F to °C.
Station 1: Metric → Metric (T) 1. Convert 7.2 kilometers to meters. 2. Convert 450 milligrams to kilograms. 3. Convert 0.85 megameters to centimeters. 4. Convert 3.5 milliliters to kiloliters. 5. Convert 2.4 gigagrams to micrograms	Station 2: English → English (U) 1. Convert 3 quarts to pints. 2. Convert 7 pints to cups. 3. Convert 2 gallons to quarts. 4. Convert 16 ounces to pounds. 5. Convert 2 short tons to pounds.				
Station 3: Metric ↔ English (I) 1. Convert 6 feet to meters. 2. Convert 3 yards to meters. 3. Convert 15 miles to kilometers. 4. Convert 3 pounds to kilograms. 5. Convert 5 gallons to liters.	Station 4: Temperature (N) 1. Convert 68°F to °C. 2. Convert 86°F to °C. 3. Convert 30°C to °F. 4. Convert 100°C to °F. 5. Convert 50°F to °C.				

	b. identifying the scale factor or conversion needed; c. showing the required computation; and d. recording the revised measurement and unit. 5. Explain how the revised measurements affect the function, capacity, size, efficiency, or intended use of the prototype. 6. After completing your manual computations, use a built-in phone unit converter or other available digital unit converter, when available, to verify your results. 7. Compare the digital results with your manual calculations and revise any errors that you identify. 8. Present your completed computations and explanation to your group or class.
IV. Reflection:	Reflect on your learning experience by answering the following: 1. What I learned from this activity: 2. What strategy I used to solve the problems or complete the tasks: 3. What I still need to improve in my understanding of conversions and measurements:

*****END OF LESSON 3.2*****

Lesson 3.3: Determining the Surface Area of Three-Dimensional Objects

Objectives:	At the end of the lesson, learners should be able to: 1. identify the basic 3D figures such as prisms, pyramids, cylinders, cones, and spheres and their different parts using blueprints and models. 2. select and apply the appropriate surface area formula for each type of solid. 3. compute the surface area of individual 3D objects accurately using given measurements. 4. solve real-life problems involving surface area of 3D objects or solid figures and their combinations.																														
Key Ideas and Examples:	Surface area is the total area of all the outer faces and surfaces of a three-dimensional object. It helps estimate the amount of material needed to cover, package, construct, or manufacture an object. By imagining a solid figure as a net, the areas of all its faces and surfaces can be added to obtain its total surface area. Each solid figure has a corresponding formula that serves as a shortcut for this process. Common Three-Dimensional Figures <table> <tr> <th>Solid Figure</th><th>Definition</th><th>Illustration</th><th>Surface Area Formula</th></tr> <tr> <td>Rectangular Prism</td><td>A solid with six rectangular faces. Opposite faces are congruent and parallel.</td><td></td><td>$SA = 2lw + 2lh + 2wh$ where l = length, w = width and h = height</td></tr> <tr> <td>Cube</td><td>A special rectangular prism with six congruent square faces and equal edge lengths.</td><td></td><td>$SA = 6s^2$ where s = edge length</td></tr> <tr> <td>Cylinder</td><td>A solid with two congruent circular bases connected by one curved surface.</td><td></td><td>$SA = 2\pi r^2 + 2\pi rh$ where r = radius and h = height</td></tr> <tr> <td>Cone</td><td>A solid with one circular base and one curved surface that tapers to a vertex.</td><td></td><td>$SA = \pi r^2 + \pi rs$ where r = radius and s = slant height</td></tr> <tr> <td>Sphere</td><td>A perfectly round solid with one continuous curved surface.</td><td></td><td>$SA = 4\pi r^2$ where r = radius</td></tr> <tr> <td>Square Pyramid</td><td>A solid with one square base and four triangular faces meeting at a common vertex.</td><td></td><td>$SA = b^2 + 2bs$ where b = base edge length and s = slant height</td></tr> </table>			Solid Figure	Definition	Illustration	Surface Area Formula	Rectangular Prism	A solid with six rectangular faces. Opposite faces are congruent and parallel.		$SA = 2lw + 2lh + 2wh$ where l = length, w = width and h = height	Cube	A special rectangular prism with six congruent square faces and equal edge lengths.		$SA = 6s^2$ where s = edge length	Cylinder	A solid with two congruent circular bases connected by one curved surface.		$SA = 2\pi r^2 + 2\pi rh$ where r = radius and h = height	Cone	A solid with one circular base and one curved surface that tapers to a vertex.		$SA = \pi r^2 + \pi rs$ where r = radius and s = slant height	Sphere	A perfectly round solid with one continuous curved surface.		$SA = 4\pi r^2$ where r = radius	Square Pyramid	A solid with one square base and four triangular faces meeting at a common vertex.		$SA = b^2 + 2bs$ where b = base edge length and s = slant height
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Illustrative Examples:**1. Rectangular Prism (Balikbayan Box)**

An OFW is preparing a balikbayan box measuring 60 cm long, 40 cm wide, and 50 cm high. How much cardboard is needed to make the entire box?

Given: $l = 60\text{cm}$; $w = 40\text{cm}$; $h = 50\text{cm}$

Formula: $SA = 2lw + 2lh + 2wh$

Solution:

$$SA = 2lw + 2lh + 2wh$$

$$SA = 2(60)(40) + 2(60)(50) + 2(40)(50)$$

$$SA = 4800 + 6000 + 4000$$

$$SA = 14\,800\text{ cm}^2$$

Final Answer: $14,800\text{ cm}^2$ of cardboard

2. Cube (School Souvenir Box)

The school's entrepreneurship club produces cube-shaped souvenir boxes for a local trade fair. Each box has an edge length of 20 cm. Determine the amount of decorative paper needed to cover one box.

Given: $s = 20\text{cm}$

Formula: $SA = 6s^2$

Solution:

$$SA = 6s^2$$

$$SA = 6(20)^2$$

$$SA = 6(400)$$

$$SA = 2\,400\text{ cm}^2$$

Final Answer: $2,400\text{ cm}^2$ of decorative paper

3. Cylinder (Barangay Water Tank)

A barangay plans to repaint a cylindrical water tank with a radius of 2 m and a height of 6 m. Determine the total surface area to be painted. Use $\pi = 3.14$.

Given: $r = 2\text{m}$; $h = 6\text{m}$

Formula: $SA = 2\pi r^2 + 2\pi rh$

Solution:

$$SA = 2\pi r^2 + 2\pi rh$$

$$SA = 2(3.14)(2)^2 + 2(3.14)(2)(6)$$

$$SA = 2(3.14)(4) + 2(3.14)(12)$$

$$SA = 25.12 + 75.36$$

$$SA = 100.48\text{ m}^2$$

Final Answer: 100.48 m^2 of paint

4. Cone (Rice Mill Hopper)

A rice mill uses a conical hopper with a radius of 1.5 m and a slant height of 4 m. Determine the amount of metal sheet needed to fabricate the hopper. Use $\pi = 3.14$.

Given: $r = 1.5\text{m}$; $s = 4\text{m}$

Formula: $SA = \pi r^2 + \pi rs$

Solution:

$$SA = \pi r^2 + \pi rs$$

$$SA = 3.14(1.5)^2 + 3.14(1.5)(4)$$

$$SA = 3.14(2.25) + 3.14(6)$$

$$SA = 7.07 + 18.84$$

$$SA = 25.91\text{ m}^2$$

Final Answer: 25.91 m^2 of metal sheet

5. Sphere (Municipal Water Reservoir)

A local government plans to apply a protective coating to a spherical water reservoir with a radius of 1 m to prevent corrosion. Determine the total surface area that will be coated.

Given: $r = 1\text{m}$

Formula: $SA = 4\pi r^2$

Solution:

	$SA = 4(3.14)(1)^2$ $SA = 4(3.14)(1)$ $SA = 12.56 \text{ m}^2$ Final Answer: 12.56 m² of protective coating 6. Square Pyramid (Waiting Shed Roof) A community is constructing a pyramid-shaped roof for a waiting shed. The roof has a square base with a side length of 5 m and a slant height of 4 m. Determine the amount of roofing material needed. Given: $b = 5\text{m}$; $s = 4\text{m}$ Formula: $SA = b^2 + 2bs$ Solution: $SA = b^2 + 2bs$ $SA = 5^2 + 2(5)(4)$ $SA = 25 + 40$ $SA = 65 \text{ m}^2$ Final Answer: 65 m² of roofing material is needed
Video Lesson:	<i>The Surface Area of Three-Dimensional Objects Solid Figures.</i> YouTube video. Accessed June 10, 2026. https://www.youtube.com/watch?v=VM1rYkTejZ0 .
I. Activity 1:	Practicing Learned Skills
II. Materials:	Calculator, Ruler or measuring tape (if using models), Worksheet
III. Instructions:	Surface Area Design Project Objective: Determine the surface area of various three-dimensional objects and composite solids using appropriate formulas and measurements. Scenario: Your group has been hired as a project design team for an eco-tourism park. Before construction begins, the management needs an estimate of the materials required for different structures. Your task is to determine the surface area of each structure and recommend the amount of material needed - Read and analyze each structure in the eco-tourism park. Structure 1: Information Booth The information booth is shaped like a rectangular prism measuring 6 m long, 4 m wide, and 3 m high. Task: Determine the amount of exterior wall material needed to cover the booth. Structure 2: Water Storage Tank The park uses a cylindrical water tank with a radius of 2 m and a height of 5 m. Task: Determine the amount of metal sheet needed to construct the tank. Structure 3: Decorative Garden Feature A decorative structure in the garden is shaped like a cone with a radius of 1.5 m and a slant height of 4 m. Task: Determine the amount of material needed to cover the structure. Structure 4: Observation Globe The park plans to install a spherical observation globe with a radius of 2.5 m. Task: Determine the surface area of the globe. Structure 5: A Monument Roof A square pyramid-shaped monument roof has a square base measuring 4 m on each side and a slant height of 3 m. Task: Determine the amount of glass needed to cover the roof. Structure 6: Viewing Tower A viewing tower consists of: a cylinder with radius 2 m and height 8 m a cone roof with radius 2 m and slant height 3 m Task: - Determine the total exterior surface area of the structure. - Clearly identify which surfaces should be included and excluded from the computation.

	Structure 7: Visitor Center The visitor center consists of a rectangular prism with a square pyramid roof. Dimensions: Rectangular Prism: Length = 12 m, Width = 8 m, Height = 5 m Square Pyramid Roof: Base Edge = 8 m, Slant Height = 6 m Task: 1. Determine the total exterior surface area of the structure. 2. Clearly identify which surfaces should be included and which surfaces should be excluded from the computation. 2. Identify the solid figure or composite solid represented. 3. Select the appropriate surface area formula and determine the surface area. 4. Record your complete solution and final answer in the worksheet. Investigation Worksheet <table><tr><th>Structure</th><th>Solid Figure</th><th>Materials Needed</th><th>Formula Used</th><th>Surface Area</th></tr><tr><td>Information Booth</td><td></td><td></td><td></td><td></td></tr><tr><td>Water Storage Tank</td><td></td><td></td><td></td><td></td></tr><tr><td>Decorative Garden Feature</td><td></td><td></td><td></td><td></td></tr><tr><td>Observation Globe</td><td></td><td></td><td></td><td></td></tr><tr><td>A Monument Roof</td><td></td><td></td><td></td><td></td></tr><tr><td>Viewing Tower</td><td></td><td></td><td></td><td></td></tr><tr><td>Visitor Center</td><td></td><td></td><td></td><td></td></tr></table>	Structure	Solid Figure	Materials Needed	Formula Used	Surface Area	Information Booth					Water Storage Tank					Decorative Garden Feature					Observation Globe					A Monument Roof					Viewing Tower					Visitor Center				
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V. Reflection:	Complete each statement. 1. I feel most confident about _____ because _____. 2. The item I found most challenging was _____ because _____. 3. To improve my understanding of determining the surface area of various three-dimensional objects, I need to _____.																																								
I. Activity 2:	Assessing Learning Outcomes																																								
II. Materials:	Task cards, Pen and paper, calculator																																								
III. Instructions:	Task 1: Identify and Match (5 pts) <i>Instructions:</i> Match each solid figure with the correct surface area formula. <table><tr><th>Solid Figure</th><th>Formula Choices (Write the letter)</th></tr><tr><td>1. Cube</td><td>a. $2\pi r(r + h)$</td></tr><tr><td>2. Cylinder</td><td>b. $6s^2$</td></tr><tr><td>3. Rectangular Prism</td><td>c. $\pi r^2 + \pi rs$</td></tr><tr><td>4. Sphere</td><td>d. $2(lw + lh + wh)$</td></tr><tr><td>5. Cone</td><td>e. $4\pi r^2$</td></tr></table> Task 2: Compute! (15 pts) <i>Instructions:</i> Solve the following. Show your solutions clearly. 1. Find the surface area of a rectangular prism with length = 5 cm, width = 3 cm, and height = 4 cm. 2. Find the surface area of a cylinder with a radius = 7 cm and height = 10 cm. Use $\pi = 3.14$. 3. A cone has a radius of 6 cm and a slant height of 10 cm. Find its surface area. Task 3: Real-Life Word Problem (10 pts) <i>Instructions:</i> 1. Solve and explain in a sentence.	Solid Figure	Formula Choices (Write the letter)	1. Cube	a. $2\pi r(r + h)$	2. Cylinder	b. $6s^2$	3. Rectangular Prism	c. $\pi r^2 + \pi rs$	4. Sphere	d. $2(lw + lh + wh)$	5. Cone	e. $4\pi r^2$																												
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	You are wrapping a gift shaped like a cube with 8 cm edges. How much decorative paper do you need to fully cover the gift (excluding waste)? 2. Show your formula, computation, and final answer with units. Task 4: Analyze & Explain (5 pts each) Instructions Reflect and respond in 2–3 sentences. - Why is it important to know the surface area of objects in real life? Give one example and explain how surface area helps in that situation. - Why is slant height used when determining the surface area of a cone instead of vertical height?
V. Reflection:	Reflect on what you learned from the activity. Answer the following questions in 2–3 sentences each. - Which problem was the most challenging? Why? - Why is selecting the correct formula important? - How can accurate volume calculations help in planning and resource management?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	Pen and paper, Calculator
III. Instructions:	For Remediation Improving Understanding about Surface Area of 3D Shapes (Solid) Task 1: Surface Area Puzzle Match Objective: Match 3D solids to their surface components and formulas. Instructions: - You'll receive a set of cut-out puzzle cards: 3D Shape name - Surface Area formula - Net (unfolded drawing) - Real-life object example - Your task is to match the 4 pieces per shape and tape them together. - Check with a partner or your teacher for feedback. Task 2: Build It & Label It Objective: Use paper nets to construct 3D shapes and label faces. Instructions: - Cut and assemble paper nets (cube, rectangular prism, cone, cylinder). - Count the number of flat and curved surfaces. - On each face, write the shape name (e.g., rectangle, circle) and its area formula (e.g., $A = lw$). - Use your labeled model to guide surface area computation. Task 3: Formula Fill-in Practice Objective: Practice applying formulas step-by-step with scaffolding. Instructions: - Use guided worksheets with partially filled formulas for surface area. Example: $SA = 2\pi r(r + h)$ Given: $r = \underline{\hspace{1cm}}$, $h = \underline{\hspace{1cm}}$ Step 1: Plug in the values Step 2: Multiply and simplify Final Answer: $\underline{\hspace{2cm}}$ cm^2 - Repeat for different solids with increasing independence. Task 4: "My Surface Area Story" – Real-Life Application Writing Objective: Connect surface area to everyday situations. Instructions: - Choose one real-life object (e.g., box, can, ball). - Describe the object's 3D shape.

	3. Write how to compute its surface area and why it might be important (e.g., painting, wrapping). 4. Draw and label the object if it is helpful. For Enhancement Enhancement Tasks – Surface Area of Solid Figures Task 1: Design a Packaging Box Objective: Apply surface area knowledge in product design. Instructions: 1. Imagine you are designing a box for a new product (snack, toy, gift, etc.). 2. Decide the 3D shape of the box (e.g., cube, rectangular prism, cylinder). 3. Determine the dimensions based on your product size. 4. Calculate the surface area of the box. 5. Design the outer layout (color, label, logo) that would cover the surface area. Task 2: Solve a Real-World Problem Combo Objective: Compute total surface area of combined solid figures. Instructions: 1. Analyze the following figure: a cylindrical water tank with a hemispherical top. Radius = 5 m, Height of cylinder = 12 m 2. Find the total surface area (exclude base of hemisphere if attached to cylinder). 3. Show all steps and formulas used. Task 3: Write a Surface Area Word Problem Objective: Create and solve your own word problem involving surface area. Instructions: 1. Write a real-life scenario involving a 3D object (e.g., a tent, a swimming pool, and a lamp). 2. Include dimensions and what the person needs to find (paint, wrap, cover). 3. Solve your problem and present it to a partner or group for peer checking. Task 4: Math + Art Integration – Surface Area Sculpture Objective: Design a sculpture using 3D shapes and compute total surface area. Instructions: 1. Create a model or sketch of a 3D "sculpture" using at least 3 different solid figures. 2. Label the dimensions of each part. 3. Calculate the total surface area. 4. Decorate or explain how the surface area relates to covering the sculpture.
IV. Reflection:	Answer the following questions. 1. How can knowledge of volume help in construction, engineering, agriculture, and business? 2. How can accurate volume calculations help avoid waste and improve efficiency? 3. Complete the statement: "I realized that understanding the volume of three-dimensional objects helps me because _____ _____."
*****END OF LESSON 3.3*****	

Lesson 3.4. Calculating the Volume of Prisms, Pyramids, Cylinders, Cones, Spheres and Combinations of These.	
Objectives:	By the end of the lesson, you will be able to: 1. identify the required measurements (e.g., base area, height, radius, and diameter) needed to determine the volume of a given solid figure; 2. select and apply the appropriate volume formula for prisms, pyramids, cylinders, cones, and spheres; and 3. solve routine, contextualized, and real-life problems involving the volume of individual and composite solid figures, and express answers using the appropriate cubic units.
Key Ideas and Examples:	

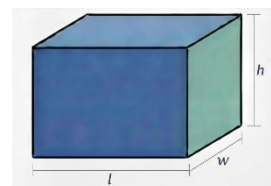
A. Volume of Prisms

A prism is a three-dimensional solid with two congruent and parallel bases. The volume of a prism is determined by multiplying the area of its base by its height. Thus, the volume depends on both the size of the base and the distance between the two bases.

Example: A shipping company uses a rectangular storage crate with a length of 8 cm, width of 5 cm, and height of 4 cm. Determine the volume of the crate.

$$\begin{aligned}\text{Base Area} &= lw \\ \text{Base Area} &= 8(5) = 40 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Volume} &= Bh \\ \text{Volume} &= 40(4) = 160 \text{ cm}^3\end{aligned}$$



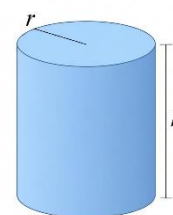
Source: Canva. "Canva AI Thread." Accessed May 30,

B. Volume of Cylinders

A cylinder is a solid figure with two congruent circular bases connected by a curved surface. Similar to a prism, its volume is obtained by multiplying the area of the base by its height. Since the base is circular, the area of the base is represented by πr^2 .

Example: A cylindrical water container has a radius of 3 cm, and a height of 10 cm. Determine its volume.

$$\begin{aligned}\text{Volume} &= 3.14(3)^2(10) \\ \text{Volume} &\approx 282.6 \text{ cm}^3\end{aligned}$$



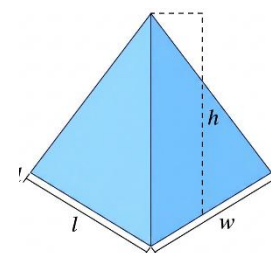
Source: Canva. "Canva AI Thread." Accessed May 30,

C. Volume of Pyramids

A pyramid is a solid figure with a polygonal base and triangular faces that meet at a common vertex. The volume of a pyramid is one-third of the volume of a prism having the same base area and height.

Example: A pyramid-shaped monument has a base area of 36 cm² and a height of 9 cm. Determine its volume.

$$\begin{aligned}\text{Volume} &= \frac{1}{3}(36)(9) \\ \text{Volume} &= 108 \text{ cm}^3\end{aligned}$$



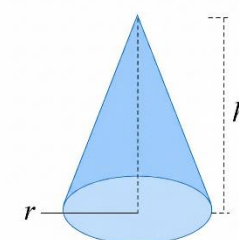
Source: Canva. "Canva AI Thread." Accessed May 30,

D. Volume of Cones

A cone is a solid figure with a circular base and a vertex. Like a pyramid, the volume of a cone is one-third of the volume of a cylinder with the same base area and height.

Example: An ice cream cone has a radius of 3 cm and a height of 9 cm. Determine its volume.

$$\begin{aligned}\text{Volume} &= \frac{1}{3}(3.14)(3)^2(9) \\ \text{Volume} &\approx 84.78 \text{ cm}^3\end{aligned}$$



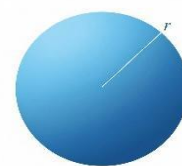
Source: Canva. "Canva AI Thread." Accessed May 30,

E. Volume of Spheres

A sphere is a perfectly round three-dimensional figure in which all points on its surface are equidistant from its center. Its volume depends solely on its radius.

Example: A basketball has a radius of 3 cm. Determine its volume.

$$\begin{aligned}\text{Volume} &= \frac{4}{3}(3.14)(3)^3 \\ \text{Volume} &\approx 113.04 \text{ cm}^3\end{aligned}$$



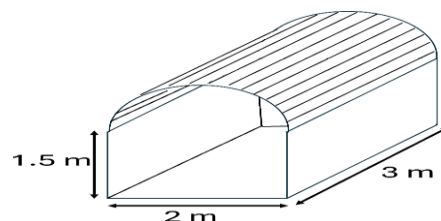
Source: Canva. "Canva AI Thread." Accessed May 30,

Video Lesson:	Geometry - Volume of 3D Shapes https://www.youtube.com/watch?v=NTwyvkh11MA																		
I. Activity 1:	Practicing Learned Skills																		
II. Materials:	Station Cards, Calculator, Pencil/Pen, Mastery Tracker Sheet																		
III. Instructions:	Volume Drill and Skill Stations Objective: Calculate the volume of prisms, pyramids, cylinders, cones, and spheres using correct substitution and mathematical operations. 1. Form groups of 4–5 members. 2. Rotate through the five skill stations. 3. Read and solve the problem assigned at each station. 4. Identify the solid figure involved and select the appropriate volume formula. 5. Show complete solutions in your notebook. 6. Record your answers on the Mastery Tracker Sheet. 7. Compare and discuss your answers with your group members. 8. Review your work. If you find an error or a better solution, revise your answer and briefly explain the change. 9. Complete the Challenge Task after finishing all stations. Station 1: Prisms Focus: Apply the volume formula for prisms. An architectural firm designs a rectangular concrete foundation for a small structure. The foundation has a length of 12 m, width of 8 m, and depth of 2 m. ► Determine the volume of concrete required for the foundation. Station 2: Cylinders Focus: Apply the volume formula for cylinders. A municipal water system includes a cylindrical storage tank with a radius of 4 m and a height of 10 m. Use $\pi = 3.14$. ► Determine the capacity of the tank in cubic meters. Station 3: Cones Focus: Apply the volume formula for cones. An engineer is designing a conical hopper used to store grain. The hopper has a radius of 3 m and a height of 8 m. Use $\pi = 3.14$. ► Determine the volume of grain the hopper can hold. Station 4: Pyramids Focus: Apply the volume formula for pyramids. A gazebo in a public park is designed with a roof shaped like a square pyramid. The base measures 12 m by 12 m and the height is 9 m. ► Determine the volume enclosed by the roof structure. Station 5: Spheres Focus: Apply the volume formula for spheres. A manufacturing company produces spherical storage tanks with a radius of 5 m. Use $\pi = 3.14$. ► Determine the volume of one storage tank. Challenge Task A water treatment facility uses a cylindrical filtration chamber with a radius of 3 m, and a height of 8 m. Attached above, it is a conical section with the same radius and a height of 4 m. ○ Determine the volume of the cylindrical section. ○ Determine the volume of the conical section. ○ Determine the total volume of the filtration structure. Mastery Tracker Sheet <table border="1"> <thead> <tr> <th>Station</th><th>Solid Figure</th><th>Formula Used</th><th>Answer</th><th>Remarks</th></tr> </thead> <tbody> <tr> <td>1</td><td>Prism</td><td></td><td></td><td></td></tr> <tr> <td>2</td><td>Cylinder</td><td></td><td></td><td></td></tr> </tbody> </table>				Station	Solid Figure	Formula Used	Answer	Remarks	1	Prism				2	Cylinder			
Station	Solid Figure	Formula Used	Answer	Remarks															
1	Prism																		
2	Cylinder																		

	3	Cone			
	4	Pyramid			
	5	Sphere			
	Challenge	Composite Solid			
	Think: After finding the total volume, explain how the volume of the composite solid would change if the height of the conical section were doubled while the other dimensions remained the same.				
V. Reflection:	Reflect on what you learned from the activity. Answer the following questions in 2–3 sentences each. 1. Which volume formula do you use most often? 2. How does determining the volume of an object help estimate its capacity? 3. In what real-life situations can you apply this concept?				
I. Activity 2:	Assessing Learning Outcomes				
II. Materials:	pen and paper, calculator				
III. Instructions:	1. Read each item carefully.				
	2. Show complete solutions for all computation and application problems. Use $\pi = 3.14$ whenever necessary				
	Part A: Formula Identification				
	Match each solid figure in Column A with its corresponding volume formula in Column B.				
	Column A		Column B		
	1. Square Pyramid		A. $V = \frac{4}{3}\pi r^3$		
	2. Rectangular Prism		B. $V = lwh$		
	3. Cylinder		C. $V = \frac{1}{3}\pi r^2h$		
	4. Sphere		D. $V = \pi r^2h$		
	5. Cone		E. $V = \frac{1}{3}Bh$		
	Part B. Solving for Volume				
	Show complete solutions.				
	1. A rectangular prism has a length of 8 cm, width of 5 cm, and height of 4 cm. Determine its volume.				
	2. A cylindrical water container has a radius of 3 cm, and a height of 10 cm. Determine its volume.				
	3. A square pyramid has a base area of 36 cm ² and a height of 9 cm. Determine its volume.				
	4. A cone has a radius of 3 cm, and a height of 9 cm. Determine its volume.				
	5. A sphere has a radius of 3 cm. Determine its volume.				
	Part C. Real-Life Application				
	Solve the following problems.				
	1. A municipal water tank is cylindrical in shape with a radius of 4 m, and a height of 10 m. Determine its capacity in cubic meters.				
	2. A gazebo in a public park has a roof shaped like a square pyramid. The base measures 12 m by 12 m and the height is 9 m. Determine the volume enclosed by the roof.				
	3. A water treatment facility uses a cylindrical chamber with a radius of 3 m, and a height of 8 m. Attached above, it is a cone with the same radius and a height of 4 m.				
	a.	Determine the volume of the cylindrical chamber.			
	b.	Determine the volume of the conical section.			
	c.	Determine the total volume of the structure.			
V. Reflection:	Reflect on what you learned from the activity. Answer the following questions in 2–3 sentences each. 1. Which problem was the most challenging? Why? 2. Why is selecting the correct formula important?				

	3. How can accurate volume calculations help in planning and resource management?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	pen and paper calculator any EdTech apps useful for designing 3D objects
III. Instructions:	For Remediation <i>Instructions:</i> 1. Read each problem carefully. 2. For Problems 1 and 2, use the Guided Problem-Solving Template to identify the solid figure, write the appropriate volume formula, substitute the given values, and solve the problem. 3. For Problems 3–5, apply the same procedure independently and show complete solutions. 4. Express all answers using the appropriate unit of measure. 5. Review your work before submitting it. Guided Problem-Solving Template Step 1: Identify the solid figure. My Answer: _____ Step 2: Write the appropriate volume formula. My Answer: _____ Step 3: Draw and label the figure using the given dimensions Step 4: Substitute the given values into the formula. My Work: _____ Step 5: Solve. My Answer: _____ Step 6: Write a complete statement. The volume of the _____ is _____. Problems 1. A water tank is shaped like a rectangular prism. Its length is 5 m, width is 2 m, and height is 3 m. How much water can it hold? 2. The fish tank is 50 cm long, 30 cm wide, and 40 cm high. How much water can the tank hold when it is completely full? 3. A toy block is shaped like a cube with each side measuring 8 cm. What is the volume of the toy block? 4. A cylindrical water tank has a radius of 5 m, and a height of 10 m. Determine the volume of the tank. Use $\pi = 3.14$. 5. A party hat is shaped like a cone with a radius of 6 cm, and a height of 15 cm. Determine the volume of the space inside the hat. Use $\pi = 3.14$. For Enrichment 1. Read each problem carefully. 2. Identify the solid figure or combination of solid figures involved. 3. Select and apply the appropriate volume formula(s). 4. Show complete solutions and computations. 5. Express all answers using the appropriate unit of measure. 6. For the Design Challenge, include a labeled sketch, complete computations, and a brief justification of your design. 7. Review your work before submitting it. Problems 1. The storage box is 2.5 m long, 1.8 m wide, and 1.2 m high. It is filled with smaller boxes, each measuring $0.3 \text{ m} \times 0.3 \text{ m} \times 0.3 \text{ m}$. How many small boxes can fit inside the storage box? 2. A freezer produces ice cubes that are 5 cm on each side. If a container has a volume of 2 liters, how many ice cubes are needed to fill the container? 3. A conical water tank has a height of 12 m, and a base diameter of 8 m. Water is being filled into the tank at a rate of 2 m^3 per minute.

	a. Determine the total volume of the tank. b. Determine the time required to fill the tank completely. 4. A cylindrical silo has a radius of 4 m and a height of 15 m. If it is filled with grain and each cubic meter of grain weighs 720 kg: a. Determine the volume of the silo. b. Determine the total weight of the grain. 5. A water tank is formed by attaching a hemisphere to the top of a cylinder. The cylinder has a radius of 3 m, and a height of 5 m. Determine the total volume of the tank. Design Challenge: Playground Tunnel A playground designer is building a tunnel. The base of the tunnel is a rectangular prism (length = 3 m, width = 2 m, height = 1.5 m), and on top of it is a half-cylinder (radius = 1 m, same length). 1. What is the total volume of the tunnel? 2. Present your solutions on a poster with: a. Sketches of the shapes involved b. Step-by-step computations c. Final answers with units d. Real-life explanation of the result
IV. Reflection:	Answer the following questions. 1. How can knowledge of volume help in construction, engineering, agriculture, and business? 2. How can accurate volume calculations help avoid waste and improve efficiency? 3. Complete the statement: "I realized that understanding the volume of three-dimensional objects helps me _____ because _____."



*****END OF LESSON 3.4*****

Lesson 3.5: Scale and Distance: Interpreting Maps, Blueprints, and Sketches

Objectives:	At the end of the lesson, you should be able to: 1. identify and read scales on maps, sketches, or blueprints; 2. measure distances accurately using available tools; 3. convert measurements from scale to actual distance; 4. compare distances between points or objects to make decisions or evaluations; and 5. construct a scaled sketch, blueprint, or layout using accurate measurements and scale.
Key Ideas and Examples:	SCALE DRAWINGS A scale drawing shows an object that is either reduced or enlarged while keeping its true proportions. For a drawing to be considered a scale drawing, it must maintain: • Proportionality – All parts must remain proportional. If one length is changed, all other lengths must change by the same amount. • Relative Position – Objects must keep their relative sizes and positions. If one object is longer in real life, it should also appear longer in the drawing. • Correspondence – Every part of the drawing must match a real object or feature. Examples: • A map of a barangay showing roads and houses in smaller size but correct shape and position • A floor plan of a classroom showing desks, doors, and windows in correct proportion • A blueprint of a house showing rooms and walls in reduced but proportional size • A sketch of a park layout showing paths, trees, and playground areas in

correct relative sizes

READING AND INTERPRETING SCALES

A scale shows the relationship between a measurement on a drawing (map, blueprint, or sketch) and the actual measurement in real life. It helps represent large distances or objects in a smaller, manageable form while keeping correct proportions.

Scales are commonly represented in three ways:

1. **Unit Scale.**

A unit scale compares two different units of measurement.

Example: 1 cm = 100 m

This means that every 1 centimeter on the map represents 100 meters in real life.

2. **Ratio Scale.**

A ratio scale compares measurements using the same unit.

Example: 1: 10,000

This means that 1 unit in the drawing represents 10,000 of the same units in real life.

3. **Graphic Scale (Scale Bar).**

A graphic scale is a line or bar drawn on a map or blueprint that represents distance.

Example:



You can use a ruler to measure the scale bar and estimate real distances without complex calculations.

APPLYING SCALE TO MEASURE DISTANCE

Scale is used to determine actual distances from a drawing, map, or blueprint.

To find the real distance, multiply the measured length on the drawing by the scale relationship.

1. **Multiplication for conversion:**

If the scale is given (e.g., 1 cm = 100 m), multiply the map distance by the scale value.

Example: $5 \text{ cm} \times \frac{100 \text{ m}}{1 \text{ cm}} = 500 \text{ m}$

2. **Proportional reasoning:**

If the scale is written as a ratio (e.g., 1:10,000), multiply the map distance by the scale ratio.

Example: $4 \text{ cm} \times \frac{10000}{1} = 40,000 \text{ cm} = 400 \text{ m}$

3. **Using a ruler for direct measurement:**

Measure the distance on the map using a ruler and compare it with the scale bar to estimate the real distance.

Example:

From the map, the distance between the **School** and the **Evacuation Center** is measured using a ruler and compared with the graphic scale.

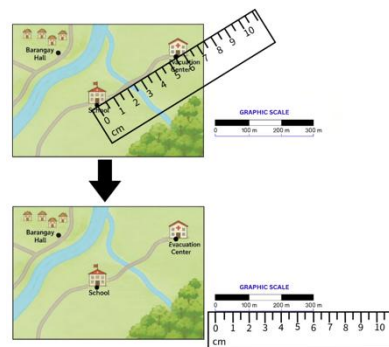
- Measured map distance: **about 5.5 cm**
- Using the graphic scale: **1 block = 100 meters**

To determine the actual distance:

$$5.5 \text{ cm} \approx 5.5 \text{ blocks}$$

$$5.5 \times 100 = \mathbf{550 \text{ meters}}$$

Final Answer: The actual distance from the School to the Evacuation Center is approximately **550 meters**.



SCALE FACTOR

The scale factor shows how much a drawing is enlarged or reduced compared to the

	actual object. It is used when the scale is not directly given. $\text{Scale Factor} = \frac{\text{Actual Distance}}{\text{Drawing Distance}}$ Examples: - A road measures 2 cm on a map and represents 200 m in reality. Convert to the same unit: 200 m = 20,000 cm $\text{Scale Factor} = \frac{20,000 \text{ cm}}{2 \text{ cm}} = 10,000$ Therefore, the scale factor is 10,000 and the scale can be written as: 1:10,000 - A classroom measures 4 cm on a floor plan and 8 m in reality. Convert to the same unit: 8 m = 800 cm $\text{Scale Factor} = \frac{800}{4} = 200$ Therefore, the scale factor is 200 and the scale can be written as: 1:200
Video Lesson:	Cognito. 2022. "GCSE Maths - Using Scales on Maps and Scale Diagrams (2026/27 exams)." Video. <i>YouTube</i> . https://www.youtube.com/watch?v=6XpBX-7cDPE .
I. Activity 1:	Practicing Learned Skills
II. Materials:	Pen and paper, Ruler, Graph paper (for drawing tasks), and Calculator (optional)
III. Instructions:	Part A: Identifying and Reading Scales - A map shows a scale of 1:5000. Explain what this scale means in practical terms. - A floor plan has a scale of 1 cm = 2 m. If a wall measures 6 cm on the drawing, what is its actual length? Part B: Measuring Distances Accurately - Using a ruler, you measure two points on a blueprint as 7 cm apart. The scale is 1:100. Calculate the actual distance. - A route on a scaled map is measured at 12 cm. If the map's scale is 1 cm = 250 m, determine the actual distance. Part C: Converting Measurements from Scale to Actual Distance - A park is drawn as 8 cm long on a sketch. The scale is 1 cm = 50 m. How long is the park in reality? - A road measures 3.5 cm on a map. If the map's scale is 1:20,000, calculate the actual distance in meters. Part D: Comparing Distances to Make Decisions - Two routes lead to the evacuation center: - Route A: 6 cm on a map, passes through a flooded area - Route B: 9 cm on a map, safe route The map scale is 1 cm = 100 m. - Calculate the actual distances for both routes. - Which route would you recommend? Explain using distances and safety considerations. - A blueprint shows two rooms: Room 1 is 4 cm long; Room 2 is 5 cm long. The scale is 1 cm = 3 m. Determine which room is longer in actual meters. Part E: Constructing Scaled Sketches or Blueprints - Draw a simple classroom layout on a piece of graph paper. The classroom measures 8 m × 6 m in reality. Use a scale of 1 cm = 1 m. Label the walls and at least two objects (desk, door). - A rectangular garden is 20 m × 12 m. Using a scale of 1:50, create a scaled sketch showing the garden, a tree, and a bench. Include the measurements.
V. Reflection:	Reflect on your learning by answering the following: What I learned about scale and distance:

	2. What I found challenging in solving scale problems: 3. How I can apply scale in real-life situations (maps, plans, or designs):
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Pen and paper, Ruler, Calculator (optional), Graph paper (for drawing tasks), Colored pencil or marker (for labeling routes and objects) Additional Learning Support (Optional Video Tutorials) • https://tinyurl.com/CanvaDesignTutorial • https://tinyurl.com/GDrawingsTutorial • https://tinyurl.com/PowerpointDrawingTutorial
III. Instructions:	Initial Prototype Design with Scaling The Integrative Performance Task (IPT) is the end-of-term project where you will develop a disaster-preparedness innovation that addresses a community's needs and can be adapted to different situations and conditions. As an initial step toward this project, you will create the first scaled version of your proposed innovation. Use the Key Measurements and Design Specifications you completed in Lesson 3.1 to guide your design. This is your opportunity to apply scale, measurement, and spatial reasoning skills in developing a visual model that may serve as the foundation for your final IPT output. <i>Instructions:</i> 1. Work with your assigned IPT group (3–5 learners). 2. Select one disaster-related innovation scenario, or propose another approved by your teacher. 3. Review your Key Measurements and Design Specifications from Lesson .1. You may refine these measurements if needed before using them for this activity. 4. Brainstorm your initial design as a group, considering all components, compartments, or features. 5. Select an appropriate scale for your sketch, blueprint, or layout. 6. Construct an initial scaled sketch, blueprint, or layout of your proposed innovation, using related evaluation, reflections, and generalizations from Activity C.2 to guide improvements and design decisions. 7. Clearly label all features, dimensions, and the scale used. 8. Measure at least three features or components in your design and convert the measurements from scale to actual dimensions. 9. Compare distances or dimensions and discuss how they affect space usage, accessibility, efficiency, or safety. 10. Write a brief design justification explaining: a. How the design maximizes space, accessibility, or safety b. How scale influenced your layout and feature placement c. How the design addresses the scenario requirements 11. Submit your completed visual model with scaling. 12. Be prepared to present your initial prototype design to the class, highlighting how you applied scale and measurements in your design. Rubrics: (To be presented by the teacher.)
V. Reflection:	Reflect on your learning by answering the following: 1. What I learned about scale and measurement in designing a prototype: 2. What challenges I encountered while creating my scaled design and how I addressed them: 3. How I can apply scale and measurement skills in real-life design or community-based problems:
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	Pen and paper, Ruler, Calculator (optional), Graph paper (for drawing tasks), Colored pencil or marker (for labeling routes and objects) Additional Learning Support (Optional Video Tutorials) • https://tinyurl.com/CanvaDesignTutorial • https://tinyurl.com/GDrawingsTutorial

	• https://tinyurl.com/PowerpointDrawingTutorial
III. Instructions:	Choose your task. Put a ✓ in the box beside the task you will complete. ☐ Remediation (Guided Prototype Design with Scaling) – You will complete this if you need guided support in improving and correcting your scaled prototype using feedback, proper measurements, and proportionality. ☐ Enrichment (Developing an Adaptability Plan) – You will complete this if you are ready to extend your understanding by analyzing how your prototype can be modified through scaling up or scaling down under different real-world conditions. REMEDIATION: Guided Prototype Design with Scaling If you had difficulty creating your scaled sketch, blueprint, or layout in Activity C.3, this activity will help you strengthen your understanding of scale, measurements, and proportionality. Use the guidance provided to improve your design and create a more accurate scaled representation. 1. Review your feedback and your Key Measurements from Lesson .1. 2. Select one feature or component that needs improvement. 3. Redraw the selected part using an appropriate scale and correct measurements. 4. Label all measurements, dimensions, and scale used. 5. Check if your drawing is proportional and accurate. 6. Submit your revised output for teacher feedback before continuing. 7. Revise your design based on feedback until it is correct. 8. Finalize and submit your completed scaled design. ENRICHMENT: Developing an Adaptability Plan The Integrative Performance Task (IPT) requires you to design a disaster-preparedness innovation that can adjust to different conditions. In this activity, you will examine how your prototype can be improved by changing its size, layout, or measurements based on real-world needs. 1. Work with your IPT group (3–5 learners). 2. Review your Initial Prototype Design with Scaling from Activity C.3. 3. Identify parts of your design that may change under different conditions. 4. Choose one scenario: Scale Up or Scale Down. 5. Explain why the change is needed (e.g., space, materials, budget, environment, users). 6. Compute how at least one measurement changes when scaled. 7. Draw a revised version of your prototype showing updated dimensions and layout. 8. Label key features, measurements, and scales clearly. 9. Explain how the changes affect the function and usability of your design.
IV. Reflection:	Reflect on your learning by answering the following: 4. What I learned about scale and measurement in designing a prototype: 5. What challenges I encountered while creating my scaled design and how I addressed them: 6. How I can apply scale and measurement skills in real-life design or community-based problems:
*****END OF LESSON 4.5*****	
Lesson 3.6. Solving Real-World Problems Involving Currency, Time, and Temperature Conversions	
Objectives:	At the end of the lesson, you should be able to: 1. identify the information and conversion factors needed for currency, time, and temperature conversions; 2. interpret exchange rates, time zones, and temperature scales in given situations; 3. perform currency, time, and temperature conversions using appropriate mathematical procedures; 4. solve problems involving currency, time, and temperature conversions in real-world contexts;

	5. apply currency, time, and temperature conversions to authentic situations such as travel planning, business transactions, project management, recipe adjustments, weather reports, scientific experiments, storage conditions, and energy efficiency.
Key Ideas and Examples:	A. Currency Conversion Different countries use different currencies for financial transactions. When engaging in international travel, online purchases, educational payments, or remittance transactions, money often needs to be converted from one currency to another through currency conversion. Currency conversion is the process of changing the value of one currency into its equivalent value in another currency. This process uses an exchange rate, which is the value of one currency in relation to another currency. Exchange rates indicate how much of one currency can be exchanged for a specific amount of another currency. Understanding currency conversion and exchange rates is important for budgeting, making financial decisions, and accurately calculating costs in international transactions. Important Note: Exchange rates may change due to economic and global factors. For this activity, fixed exchange rates are provided to help you focus on the conversion process. Example: A learner is preparing to participate in an international youth leadership program in Singapore and needs to exchange PHP15,000 into Singapore dollars (SGD) to cover meals, local transportation, entrance fees, and other travel-related expenses. Suppose the exchange rate is: 1 Singapore Dollar (SGD) = PHP41.00 Determine the amount of Singapore dollars the learner will receive. Solution: To convert Philippine pesos (PHP) to Singapore dollars (SGD), divide the amount in pesos by the exchange rate. $\text{Amount in Foreign Currency} = \frac{\text{Amount in Philippine Peso}}{\text{Exchange Rate}}$ $\text{Singapore Dollar} = \frac{15,000}{41}$ $\text{Singapore Dollar} = \text{SGD}365.85$ Answer: The learner will receive approximately SGD365.85. B. Time Zone Conversion Countries around the world observe different local times based on their geographical locations. A time zone is a region that follows the same standard of time. Time differences between locations are determined using Coordinated Universal Time (UTC), a global time standard used as the reference point for world timekeeping. A UTC offset indicates the number of hours a location is ahead of or behind UTC. Time conversion is the process of determining the equivalent local time in another location by comparing their respective UTC offsets. Examples of UTC offsets include: 1. Philippines – UTC+8 2. Japan – UTC+9 3. Australia (Sydney) – UTC+10 4. United Kingdom – UTC+0 5. Germany – UTC+1 6. United States (New York) – UTC–5 (<i>standard time</i>) 7. United Arab Emirates – UTC+4 Understanding time zones, UTC offsets, and time conversion is important for coordinating international travel, academic programs, virtual meetings, and communication with individuals in different parts of the world. Example:

A learner from the Philippines will attend an online scholarship interview with an institution in New York City. The interview is scheduled for 10:00 AM New York City time (Eastern Time, UTC-4). Since the learner will attend the interview from the Philippines, which follows Philippine Time (UTC+8), determine the equivalent local time in the Philippines to ensure timely attendance.

Solution:

To determine the equivalent local time in the Philippines, compare the time zones of New York City and the Philippines.

Given:

1. New York City: UTC-4
2. Philippines: UTC+8
3. Interview Schedule: 10:00 AM (New York City Time)

Step 1: Determine the time difference.

Compute the difference between the two time zones:

$$\text{UTC}+8 - \text{UTC}-4 = 12 \text{ hours}$$

This means that the Philippines is 12 hours ahead of New York City.

Step 2: Adjust the scheduled interview time.

Since the Philippines is ahead, add 12 hours to the interview time in New York City:

$$10:00 \text{ AM} + 12 \text{ hours} = 10:00 \text{ PM}$$

Answer:

The equivalent local time in the Philippines is 10:00 PM Philippine Time. Therefore, the learner should attend the online scholarship interview at 10:00 PM.

C. Temperature Conversion

Temperature is a measure of the degree of hotness or coldness of an object or environment. It is commonly used in weather forecasting, food preparation, food storage, scientific investigations, and energy management. Different countries and fields of study use different temperature scales; therefore, understanding temperature conversion is important for interpreting and comparing temperature measurements. The three commonly used temperature scales are **Celsius (°C)**, **Fahrenheit (°F)**, and **Kelvin (K)**. The Celsius scale is widely used in the Philippines and many other countries, while the Fahrenheit scale is commonly used in the United States. The Kelvin scale is the standard unit of temperature used in scientific and laboratory work because it is based on absolute zero, the lowest possible temperature.

The reference points of the three temperature scales are shown below:

Temperature	Celsius	Fahrenheit	Kelvin
Freezing point of water	0°C	32°F	273.15 K
Boiling point of water	100°C	212°F	373.15 K

Temperature conversion is the process of determining the equivalent value of a temperature from one scale to another. The following formulas are commonly used:

To convert Celsius to Fahrenheit

$$F = (9/5)(C + 32)$$

To convert Fahrenheit to Celsius

$$C = (5/9)(F - 32)$$

To convert Celsius to Kelvin

$$K = C + 273.15$$

To convert Kelvin to Celsius

$$C = K - 273.15$$

When converting temperatures, it is important to:

- Identify the given temperature scale.
- Select the appropriate conversion formula.
- Substitute the given value correctly.
- Perform the mathematical operations carefully.
- Express the final answer using the correct unit (°C, °F, or K).

Understanding temperature conversion is important when preparing food using international recipes, interpreting weather reports, maintaining proper food storage

	conditions, conducting scientific experiments, working in laboratories, and managing energy-efficient appliances. Example 1 A Senior High School student is preparing a dessert using an online recipe from the United States. The recipe requires baking at 350°F, but the oven available at home uses the Celsius scale. Determine the equivalent baking temperature in Celsius. $C = \frac{5}{9}(F - 32)$ $C = \frac{5}{9}(350 - 32)$ $C = \frac{5}{9}(318)$ $C \approx 176.67^{\circ}\text{C}$ Final Answer: 176.67°C Example 2 A school canteen manager stores fresh vegetables in a refrigerator maintained at 5°C, but the imported refrigerator displays the temperature in Fahrenheit. Determine the equivalent temperature in Fahrenheit. $F = \frac{9}{5}C + 32$ $F = \frac{9}{5}(5) + 32$ $F = 9 + 32$ $F = 41^{\circ}\text{F}$ Final Answer: 41°F Example 3 A Senior High School student is conducting a laboratory experiment where the required temperature is 25°C. Determine its equivalent temperature in Kelvin. $K = C + 273.15$ $K = 25 + 273.15$ $K = 298.15\text{ K}$ Final Answer: 298.15 K
Video Lesson:	Converting Time in Traveling Decision (Part 3) https://www.youtube.com/watch?v=vbBXzc6FWB8 Temperature Conversions - Fahrenheit to Celsius to Kelvin https://www.youtube.com/watch?v=G7sRfdAq4M8
I. Activity 1:	Practicing Learned Skills
II. Materials:	pen and paper, conversion chart for time zone, currency and temperature, calculator
III. Instructions:	<i>Instructions:</i> - Solve each problem independently by applying the concepts of time zone conversion, currency conversion, and temperature conversion. - Show your complete solutions, including the appropriate mathematical process and computations. - Write your final answers clearly using the correct units or currency symbols. - Use logical reasoning and accuracy in solving each problem. A. Time Zone Conversion - Academic Meeting Schedule A learner from the Philippines (UTC+8) is scheduled to attend an online academic meeting with a partner institution in Berlin, Germany (UTC+1). If the meeting is set at 3:00 PM Philippine Time, determine the corresponding local time in Berlin. - International Flight Arrival A tourist is travelling from New York, United States (UTC-5) to Paris, France (UTC+1). The flight departs at 6:00 PM New York Time and lasts 10 hours. Determine the local arrival time in Paris. B. Currency Conversion - Souvenir Purchase

	A learner participating in an educational trip to Japan plans to purchase a souvenir worth ¥3,000 Japanese Yen (JPY). Suppose the exchange rate is: 1 US Dollar (USD) = ¥150 Determine the cost of the souvenir in US dollars. 4. Travel Budget Conversion A family is preparing for a trip to France and has a travel budget of 500 Euros (EUR). Suppose the exchange rates are: 1 Euro (EUR) = PHP61.87 1 US Dollar (USD) = PHP56.25 Determine the equivalent travel budget in US dollars. A. Temperature Conversion 5. A learner is preparing to participate in an international robotics and technology competition. The event guidelines indicate that electronic equipment should be stored at a temperature of 14°F during transport. Determine the equivalent temperature in Celsius to help the learner maintain the proper storage condition of the equipment. 6. A Senior High School TechPro learner is developing a computer system unit as part of a hardware servicing project. The equipment manual recommends maintaining the testing room at 24°C to prevent overheating of computer components. Determine the equivalent temperature in Fahrenheit.
V. Reflection:	Reflect on your learning by answering the following 1. Which mathematical concept was the easiest for you to apply? 2. How can conversion concepts help people make informed decisions? 3. In what situations can you use these concepts in your daily life?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper, conversion chart for time zone, currency and temperature, calculator
III. Instructions:	My International Exchange Program Plan Scenario: You have been selected to represent your school in a five-day International Youth Exchange Program. As part of the preparation, you are required to develop a travel plan that demonstrates your ability to use currency, time, and temperature conversions in making practical decisions. Travel Information: Destination: Seoul, South Korea Travel Budget: PHP 50,000 Exchange Rate: 1 PHP = 24.30 KRW Currency Service Charge: 2% Departure: 8:00 AM (Manila Time, UTC+8) Travel Duration: 4 hours Destination Time Zone: Seoul (UTC+9) Forecast Temperature: 12°C 1. Read the given information carefully and identify the conversion factors needed to complete the task. 2. Use appropriate mathematical procedures to perform the required currency, time, and temperature conversions. 3. Show your complete solutions, including formulas, computations, and final answers using the correct units and symbols. 4. Use your answers to complete the travel preparation plan. 5. After completing the task, answer the processing questions. Task 1: Travel Budget Preparation Determine the amount of Korean Won (KRW) you will receive after the 2% service charge is deducted. Task 2: Travel Schedule Coordination Determine: a. Your local arrival time in Seoul.

	b. If an orientation program starts at 3:00 PM Seoul time, what time should your family in the Philippines watch the online orientation? Task 3: Weather Preparation Convert the forecast temperature of 12°C into Fahrenheit (°F). Based on your answers, explain how the weather information can help you prepare for the trip. Task 4: Travel Preparation Reflection In three to five sentences, explain how currency, time, and temperature conversions helped you prepare for your participation in the International Youth Exchange Program.
V. Reflection:	Reflect on your learning by answering the following 1. Which problem was the most challenging? Why? 2. Why is it important to perform conversions accurately? 3. How can these concepts help people in travel, business, science, and communication?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials	pen and paper, conversion chart for time zone, currency and temperature, calculator
III. Instructions:	For Remediation <i>Instructions:</i> 1. Read each situation carefully. 2. Complete the guide questions before solving the problem. 3. Use the appropriate formula or mathematical operation. 4. Show your computations and write your final answer using the correct unit or symbol. 1. Currency Conversion A learner plans to visit Japan and wants to exchange P5,000 into Japanese Yen (JPY). The exchange rate is 1 PHP = 2.10 JPY. Complete the guide below. a. Amount to be exchanged: _____ b. Exchange rate: _____ c. To convert Philippine Pesos to Japanese Yen, should you: • Multiply • Divide d. Complete the equation: JPY = _____ × _____ e. Solve and write the final answer. Final Answer: _____ 2. Time Zone Conversion A learner in Manila (UTC+8) wants to call a relative in Dubai (UTC+4). It is 6:00 PM in Manila. Complete the guide below. a. Time in Manila: _____ b. Manila is ____ hours ahead of Dubai. c. To find the time in Dubai, should you: • Add the time difference • Subtract the time difference d. Complete the equation: Time in Dubai = _____ - _____ e. Solve and write the final answer. Final Answer: _____ 3. International Currency Exchange A learner participating in a student exchange program has EUR 100. The exchange rate is 1 EUR = 1.08 USD. Complete the guide below. a. Amount in Euro: _____ b. Exchange rate: _____ c. To convert Euro to US Dollar, should you:

	• Multiply • Divide d. Complete the equation: USD = _____ × _____ e. Solve and write the final answer. Final Answer: _____ For Enhancement <i>Instructions:</i> 1. Read the situation carefully. 2. Analyze the given information and determine the mathematical concepts needed. 3. Solve each task using appropriate formulas and procedures. 4. Show complete solutions, including computations and final answers. 5. Use your answers to make practical decisions and justify your recommendations. International Student Travel Challenge A Senior High School learner has qualified to represent the school in an International Youth Leadership Conference. The learner must choose the most practical travel plan based on the available information. Travel Information Destination: Sydney, Australia Pocket Money: P45,000 Exchange Rates: • 1 AUD = P37.50 • 1 USD = P58.00 The learner may exchange the money directly into Australian Dollars or first convert it into US Dollars for an airport stopover. Flight Schedule: • Departure from Manila (UTC+8): 9:30 PM • Flight duration to Sydney: 8 hours • Sydney follows UTC+10 Weather Forecast: • Sydney temperature: 22°C Task 1. Currency Decision a. Determine the amount of Australian Dollars (AUD) if P45,000 is exchanged directly. b. Determine the amount of US Dollars (USD) if P45,000 is exchanged into USD. c. Based on your computations, explain which currency exchange option is more appropriate for the learner's needs and why. Task 2. Travel Schedule Analysis a. Determine the local arrival time in Sydney. b. The conference registration begins at 8:00 AM the following day. Explain whether the learner will have enough time to rest before the event. Task 3. Weather and Travel Preparation a. Convert 22°C to Fahrenheit (°F). b. Convert 22°C to Kelvin (K). c. Recommend suitable clothing or travel items based on the expected temperature and justify your answer. Task 4. Decision-Making Challenge Suppose the exchange rate suddenly changes from P37.50 to P39.00 for 1 AUD before the learner exchanges the money. a. Determine the new amount of AUD that will be received. b. Compare the original and new amounts. c. Explain how changes in exchange rates can affect travel planning and budgeting.
IV. Reflection:	Answer the following questions. 1. How can mathematics help people prepare for international travel? 2. How can conversion concepts help in business, education, and science? 3. Complete the statement.

"I realized that mathematics helps me _____
because _____."

*****END OF LESSON 3.6*****

Unit 4. Functions and Graphs

Lesson 4.1. Linear And Quadratic Functions

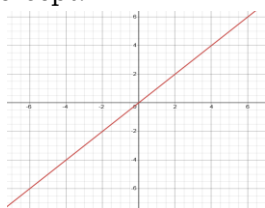
Objectives:

At the end of the lesson, you should be able to:

1. recall the definition of function, domain and range;
2. identify and classify the function types from their equations as linear and quadratic based on their forms;
3. identify and classify the function types from the given graphs by analyzing their shapes and key features to determine if they represent a linear or quadratic function;
4. sketch the graph of the following functions: linear and quadratic; and
5. apply the linear and quadratic functions to model and solve real-life problems.

Key Ideas: Examples and Illustrations;

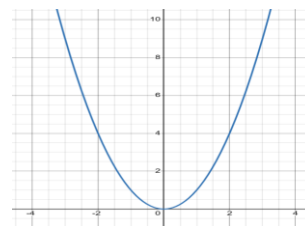
Linear Function (Line): a linear function is represented by a straight line on the graph. The general form of a linear function is $y = mx + b$, where m is the slope and b are the y-intercept.



Quadratic Function (Parabola): a quadratic function is represented by a parabola on the graph. The general form of a quadratic function is

$$y = ax^2 + bx + c,$$

where a , b , and c are constants, and $a \neq 0$.



Example 1: Determine the Slope and the Y-Intercept of the Function $y = 2x + 1$.
Instruction:

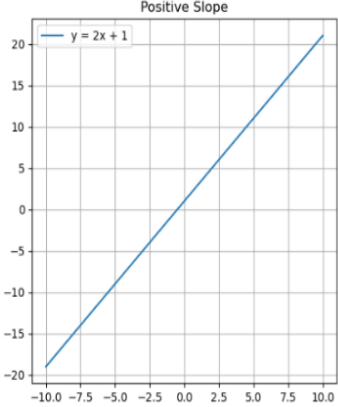
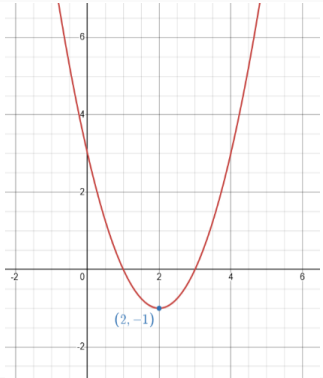
1. Determine the slope and the y-intercept of the function $y = 2x + 1$.
To find the slope and the y-intercept, compare the given function to the general form of a linear equation $y = mx + b$ where:
 - **Slope (m):** In the function $y = 2x + 1$, the coefficient of x is 2. Therefore, the slope (m) is 2.
 - **y-intercept (b):** In the function $y = 2x + 1$, the constant term is 1. Therefore, the y-intercept (b) is 1.

The equation in the form $y = mx + b$, m is the value of the slope and b is the y-intercept. This is how to find the slope and y-intercept in Example 1.

Example 2: Graph the Function $y = 2x + 1$.

Instructions:

1. Graph the function $y = 2x + 1$.
To graph the function, follow these steps:
Step 1: Identify the slope and y-intercept.
From the equation
 $y = 2x + 1$:
 - The slope (m) is 2.
 - The y-intercept (b) is 1.
 Step 2: Plot the y-intercept on the graph.
The y-intercept is the point where the line crosses the y-axis.
Plot the point $(0, 1)$ on the graph.

	Step 3: Use the slope to find another point. The slope of 2 means that for every 1 unit increase in x, y increases by 2 units. Starting from the y-intercept (0, 1), move 1 unit to the right (to x=1) and 2 units up (to y=3). Plot the point (1, 3). Step 4: Draw the line. Draw a straight line through the points (0, 1) and (1, 3). Extend the line in both directions.  Example 3: Construct the Graph of $y = x^2 - 4x + 3$ on a Large Poster Instructions: Group Formation: Divide the class into small groups of 3-5 learners. Ensure each group has a mix of abilities to promote collaborative learning. Step 1: Table of Values Ask each group to create a table of values for xx ranging from -1 to 5. This will help them plot the points accurately. Calculate y values using the equation $y = x^2 - 4x + 3$. Step 2: Plotting Points On the large poster, draw a coordinate plane. Ensure the axes are labeled and scaled appropriately. Plot the points from the table of values accurately. Step 3: Drawing the Parabola Connect the points smoothly with a curve to form the parabola. Emphasize the importance of accuracy in connecting the points. Highlight key features such as: Vertex: (x = 2, y = -1) Axis of symmetry: x = 2 X-intercepts: x = 1 and x = 3 Y-intercept: y = 3 Step 4: Labeling and Analyzing Label the graph with: Equation: $y = x^2 - 4x + 3$ Key points: (vertex, intercepts) Axis of symmetry 
Video Lesson	https://www.khanacademy.org/math/pisa-2025-math-supplement/x092b0b93d51c9e96:1/x092b0b93d51c9e96:algebraic-expressions/v/application-problem-with-graph . Accessed 9 May 2025. https://www.youtube.com/watch?v=jv8O5iKB1ks https://www.youtube.com/watch?v=XVdVFOGf3aY https://www.youtube.com/watch?v=OYOXMyFKotc https://www.youtube.com/watch?v=1WFhDqOpjpU https://www.youtube.com/watch?v=yOrXqvO2M1M https://www.youtube.com/watch?v=SLV600BpHpE https://www.youtube.com/watch?v=3B1wm-Qan30 https://www.youtube.com/watch?v=PI6IV7K3DbM
I. Activity 1	Practicing Learned Skills
II. Materials:	Calculator, Paper and pen/pencil
III. Instructions:	Work independently to graph and analyze your assigned function. This activity will help you understand how functions behave and how their equations are represented visually through graphs.

	Graph the following functions: 1. $y = 2x + 1$ 2. $y = x - 4$ 3. $y = -\frac{1}{2}x + 3$ 4. $y = x^2 - 4x + 3$ 5. $y = -x^2 + 2x + 1$ 6. $y = x^2 + x - 6$ 7. $y = -2x^2 - 3x + 1$
IV. Reflection:	Answer this question: How did comparing the graphs of linear and quadratic functions help you understand how different equations model different types of relationships and situations in real life?
I. Activity 2	Making Real-World Connections
II. Materials:	Calculator, Notebook or data sheet
III. Instructions:	For each problem: 1. <i>Identify the variables and important information.</i> 2. <i>Write the appropriate mathematical model or function.</i> 3. <i>Show complete solutions and computations.</i> 4. <i>Use correct mathematical notation and units.</i> 5. <i>Explain your answers clearly when required.</i> 6. <i>Check your computations before submitting your work.</i> <i>Problem 1: Know Your Household Consumption</i> A household pays: ➤ a fixed monthly connection fee of ₱300 ➤ ₱10 per kilowatt-hour (kWh) of electricity consumed. Tasks: ➤ Write a linear function $B(k)$ representing the monthly electricity bill based on the number of kilowatt-hours k consumed. ➤ Calculate the total monthly bill if the household consumes 150 kilowatt hours. Use the linear function: $B(k) = 300 + 10k$ <i>Problem 2: School Fundraiser</i> Scenario: Your school is organizing a fundraising event. Each ticket costs ₱100, and the total fixed expense for organizing the event is ₱5,000. Tasks: ➤ Write a profit function representing the relationship between profit and the number of tickets sold. ➤ Calculate the profit when: 50 tickets are sold 100 tickets are sold 200 tickets are sold 350 tickets are sold Use the profit function: $P(x) = 100x - 5000$ <i>Problem 3: Projectile Motion</i> Scenario: A ball is thrown into the air, and its height is modeled by the function: $h(t) = -4.9t^2 + 20t + 1.5$ where: t = time in seconds $h(t)$ = height in meters Tasks: ➤ Determine the time when the ball reaches its maximum height. ➤ Find the maximum height of the ball.

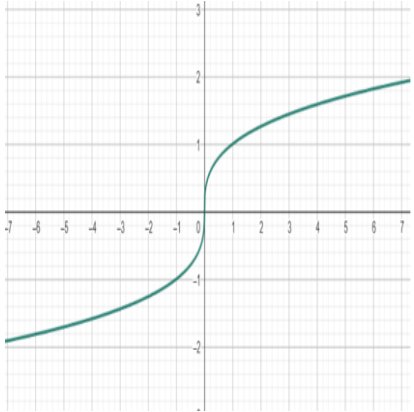
	➤ Determine when the ball hits the ground. ➤ Graph the function and describe the shape of the parabola in relation to the ball's motion. Use the vertex formula: $t = \frac{-b}{2a}$ <i>Problem 4: Halo-Halo Stand in Burnham Park</i> Scenario: A small halo-halo stand in Baguio City sells each cup for ₱40 and serves 120 customers per day. The owner plans to increase the price. For every ₱2 increase, the owner expects to lose 6 customers. Tasks: ➤ Define a variable representing the number of ₱2 price increases. ➤ Write a quadratic model representing the daily profit. ➤ Analyze the function and determine: - the maximum daily profit - the price that gives the maximum profit - the corresponding number of customers Use the quadratic model: $P(x) = (40 + 2x)(120 - 6x)$ <i>Problem 5: Vegetable Garden Optimization</i> Scenario: A farmer in Nueva Ecija plans to build a rectangular vegetable garden beside a barn using 30 meters of fencing. Since one side is against the barn, only three sides need fencing. Tasks: ➤ Define the variables. ➤ Write a quadratic function representing the area of the garden. ➤ Determine the dimensions that maximize the area. ➤ Find the maximum possible area. Use the area function: $A(x) = -2x^2 + 30x$
IV. Reflection:	Share your experience in the activity by responding to each question. 1. Which problem did you find most relevant to your daily life and why? 2. What new insights did you gain about the role of mathematics in decision-making? 3. After completing these activities, how has your appreciation of mathematics as a tool for solving real-world problems changed?
I. Activity 3	Assessing Learning Outcomes
II. Materials:	Ballpen, Calculator
III. Instructions:	Analyze and solve the following: 1. Problem: A tricycle driver charges a base fare of ₱10 plus ₱5 per kilometer traveled. Question: Write a linear function that models the total fare F for traveling x kilometers. How much would it cost to travel 8 kilometers? 2. Problem: A local festival in Cebu features a fireworks display. The height h of a firework in meters after t seconds is given by the function $h(t) = -4.9t^2 + 29.4t + 1.$ Question: How long does it take for the fireworks to reach their maximum height? What is the maximum height? 3. Problem: Revenue Optimization for a Local Concert A learner organization in Cebu City is organizing a benefit concert. Tickets are priced at ₱300, and they expect 200 attendees at that price. They estimate that for every ₱20 increase in ticket price, 10 fewer people will attend.

	1. Write a quadratic function that models the total revenue based on the number of ₱20 price increases. 2. Determine the ticket price that will maximize revenue. 3. Find the maximum possible revenue. Interpret the result in the context of the problem.
IV. Reflection:	To synthesize your gains, provide your insights to this question. Among the three situations - the tricycle fare, the fireworks display, and the benefit concert - which one best demonstrates how mathematics can be used to make informed decisions in real life? Explain your answer in 3 - 5 sentences.

*****END OF LESSON 4.1*****

Lesson 4.2. Function $y = x^3$, $y = \sqrt{x}$, and $y = \sqrt[3]{x}$

Objectives:	At the end of the lesson, learners should be able to: 1. Classify the functions $y = x^3$, $y = \sqrt{x}$, and $y = \sqrt[3]{x}$ based on their equations and graphs as cubic, square root, and cube root functions respectively. 2. Sketch the graphs of the given functions with or without the use of technology. 3. Apply the functions in solving real-life problems involving growth, motion, or geometric dimensions.																														
Key Ideas : Examples and Illustrations	A function of the form $f(x) = ax^3 + bx^2 + cx + d$ where is called a cubic function. In other words, a cubic function is a polynomial with a degree of 3 (the highest power of the variable x is 3). A square root function is a function that contains a square root with the independent variable in the radicand. The parent function for the family of square root functions is $f(x) = \sqrt{x}$. The domain of f is $x \geq 0$, and the range of f is $y \geq 0$. A cube root function is a radical function with an index of 3. The parent function of cube root functions is $f(x) = \sqrt[3]{x}$. The domain and range of f are all real numbers. The graph of $f(x) = \sqrt[3]{x}$ increases on the entire domain. Examples 1. Cubic Function Sketch the graph of the function $f(x) = x^3 - 1$ for $-2 \leq x \leq 3$. Solution Using the table of values, we get <table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>-9</td><td>-2</td><td>-1</td><td>0</td><td>7</td><td>26</td></tr></table> The graph on the right side represents function $f(x) = x^3 - 1$ 2. Square Root Function Example: Sketch the graph of the function $f(x) = \sqrt{x}$. Solution. Using the table of values, we get <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>y</td><td>1.00</td><td>1.41</td><td>1.73</td><td>2.00</td><td>2.24</td><td>2.45</td><td>2.65</td></tr></table> The graph on the right side represents function $f(x) = \sqrt{x}$,	x	-2	-1	0	1	2	3	y	-9	-2	-1	0	7	26	x	1	2	3	4	5	6	7	y	1.00	1.41	1.73	2.00	2.24	2.45	2.65
x	-2	-1	0	1	2	3																									
y	-9	-2	-1	0	7	26																									
x	1	2	3	4	5	6	7																								
y	1.00	1.41	1.73	2.00	2.24	2.45	2.65																								

	3. Cube Root Function Example: Sketch the graph of $y = \sqrt[3]{x}$. Note that $y = \sqrt[3]{x}$ can be expressed into exponential form $y = x^{\frac{1}{3}}$. By applying cube both sides, we have$y^3 = (\sqrt[3]{x})^3$ Then$y^3 = x.$ Using the table of values, we get<table><tr><td>x</td><td>-27</td><td>-8</td><td>-1</td><td>0</td><td>1</td><td>8</td><td>27</td></tr><tr><td>y</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr></table> The graph on the right side represents function $y = \sqrt[3]{x}$. Note: You may use Desmos/GeoGebra Graphing Tool to sketch the graph of the cubic, square root and cube root functions. 	x	-27	-8	-1	0	1	8	27	y	-3	-2	-1	0	1	2	3
x	-27	-8	-1	0	1	8	27										
y	-3	-2	-1	0	1	2	3										
Video Lesson	https://www.youtube.com/watch?v=XVdVFQGf3aY https://www.youtube.com/watch?v=OYOXMyFKotc https://www.youtube.com/watch?v=1WFhDqOpjpU https://www.youtube.com/watch?v=yOrXqvO2M1M https://www.youtube.com/watch?v=SLV600BpHpE https://www.youtube.com/watch?v=3B1wm-Qan30 https://www.youtube.com/watch?v=PI6IV7K3DbM																
I. Activity 1:	Practicing Learned Skills																
II. Materials:	Paper and Pen, Cellphone, GeoGebra/ Desmos Graphing Tool Apps																
III. Instructions:	Sketch the graph with or without using application software, like GeoGebra/Desmos Graphing Tool. 1. $y = 5x^3 + 3x^2 + 2x + 5$ 2. $y = 2\sqrt{-x}$ 3. $y = -x^3 + 3x$ 4. $y = \sqrt[3]{x} + 2$ 5. $f(x) = 3\sqrt[3]{x+4} + 2$																
IV. Reflection::	To synthesize your gains, provide your insights to these questions 1. Which graphs have similar shapes or curves? 2. Do any of the functions have the same domain or range? 3. How do the graphs differ in terms of shape?																
I. Activity 2:	Making Real Life Connections																
II. Materials:	Paper and Pen																
III. Instructions:	Carefully read and analyze each problem to identify the type of function represented. Use the correct mathematical formulas to solve, making sure to show all steps of your work. Include the appropriate units with your final answers and be ready to share and discuss your solutions during class. Problem 1: If the volume V of a cube is related to the side length s by $V=s^3$, how can you determine s given V? Which function would you use? Given the Volume V is 25 cm^3, what is the side length? Problem 2: In a small coastal town in Zamboanga, Aling Nena runs a sari-sari store that sells local brands of <i>sardinas</i>. One of the best-selling items is a cylindrical tin of sardines from a brand called “Masarap Sardines”. The label on the can says it contains Volume: 350 cm^3 of sardines. Aling Nena measures the height of the can and finds it to be 10 cm.																

	She wants to stack the cans efficiently on her shelf and needs to know the radius of the base of each can, to calculate how much space she needs. Problem 3: Miguel, a Grade 9 student from Cebu, wants to surprise his younger sister with a gift. He chooses a small box shaped like a cube, which he plans to wrap with a colorful gift wrapper bought from the nearby market. The box has a volume of 64 cm^3, and Miguel wants to know how much wrapping paper he needs to cover the entire surface of the box without leaving any side exposed.
IV. Reflection:	To synthesize your gains, provide your insights to these questions 1. What mathematical concepts or functions did you use in solving the problems? 2. How does the cube root help in finding the side length of a cube? 3. Why is it important to identify the information given before solving a problem? 4. How are mathematical functions applied in real-life situations such as packaging and household consumption? 5. What difficulties did you encounter while solving the problems, and how did you overcome them?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Paper and Pen
III. Instructions:	Read each item carefully. Answer the questions that follow. Show complete solutions when necessary. Part I. Classification of Functions Identify whether each function is Cubic Function, Square Root Function, or Cube Root Function. 1. $f(x) = x^3 - 5x + 1$ 2. $g(x) = \sqrt{x} + 4$ 3. $h(x) = \sqrt[3]{x-2}$ 4. $y = -x^3 + 6$ 5. $y = \sqrt[3]{x} + 7$ Part II. Sketching Functions (Rubric to be presented by the teacher) Sketch the following functions and label important features such as intercepts and direction of the graph. 1. $y = x^3$ 2. $y = \sqrt{x}$ 3. $y = \sqrt[3]{x}$ Part III. Real-Life Applications Instructions: Provide what is asked. Problem 1: A cube-shaped water tank has a volume of 125 m^3. a. What function is represented by the relationship between volume and side length? b. Determine the side length of the tank. c. Explain why the function is appropriate. Problem 2: The area of a square garden is 196 m^2. a. Which function can be used to determine the side length? b. Find the side length. Problem 3: A storage box has a side length of 6 cm. a. What type of function models the relationship between side length and volume? b. Find its volume. c. Explain the mathematical model used.
IV. Reflection:	To synthesize your gains, provide your insights to these questions:

	After completing this assessment, how has your understanding of functions changed, and how do you think these mathematical concepts can help you solve problems in your daily life or future career?
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INTEGRATIVE PERFORMANCE TASK (FIRST TERM)	
Goal :	1. Creation and presentation of a Community Innovation Advocacy Pitch that proposes a prototype-based solution to a disaster-related problem affecting the barangay. 2. Development of a proposed solution that responds to a disaster-related problem affecting the barangay while considering its geography, culture, experiences, existing practices, and community needs. Examples may include: • Flooding • Storm surges • Landslides • Drought • Water shortages • Poor evacuation conditions • Waste management issues during disasters • Other disaster-related concerns relevant to the barangay 3. Application of scale interpretation through: • Measurement, conversion, and proportional reasoning • Scale use in sketches, blueprints, diagrams, and layouts • Accurate representation of real-world spaces and distances 4. Evaluation of the proposed innovation based on: • Practicality • Safety • Affordability (conceptual cost) • Suitability for intended users and barangay needs 5. Demonstration of: • Mathematical reasoning, particularly measurement, conversion, scale interpretation, and representation • Scientific understanding and application of Physics concepts • Communication and advocacy skills • Socio-cultural and community awareness 6. Justification of design decisions considering: • Space and scale • Budget estimate (conceptual) • Materials (conceptual) • Number of beneficiaries • Community needs and conditions 7. Delivery of an advocacy pitch with a clear call to action that encourages the audience to support, improve, adopt, implement, pilot, or sponsor the proposed solution 8. Development of: • Self-awareness through reflection on experiences, strengths, and areas for improvement • Effective collaboration and social interaction • Responsible decision-making • Readiness for future studies, career goals, community involvement, and real-life situations
Role :	Learners will take on the role of community innovation advocates. In this role, they will study a local community problem, design a prototype, prepare scaled models and measurements, analyze cost and feasibility, create communication materials, and present an advocacy pitch to persuade others to support, improve, adopt, scale up, or scale down the innovation. This role reflects real-life responsibilities because learners perform tasks similar to young innovators, community researchers, project designers, financial planners, communicators, and advocates. They must use evidence, collaborate effectively, and make responsible decisions to address real community needs.
Audience :	The intended audience includes: teachers from the integrated learning areas; invited stakeholders such as barangay officials, SK officials, community leaders, or other community representatives; and classmates. Teachers and invited stakeholders will serve as the panel of evaluators, while classmates will serve as the peer audience during the advocacy pitch. The peer audience and panel make the task more authentic and meaningful because learners must present and defend their proposed innovation before people who can evaluate its relevance, feasibility, and responsiveness to community needs and

	conditions. This encourages learners to support their recommendations with evidence and consider how community context, existing practices, safety, sustainability, and cost influence the design and implementation of their proposed innovation.
Situation :	Your barangay faces disaster-related problems that affect the safety, well-being, resources, or daily lives of its residents. Examples may include flooding, storm surges, landslides, drought, water shortages, poor evacuation conditions, or other community concerns. Existing coping, preparedness, or adaptation practices of the barangay may help address these problems, but they may also have limitations. How can you, as a team of community innovation advocates, collaborate to design a feasible and sustainable innovation that addresses a disaster-related problem in the barangay, using Physics concepts, measurements, conversions, and scale, while considering its geography, culture, experiences, existing practices, and the needs of residents? To answer this question, you will investigate the problem, analyze existing practices, develop a prototype-based innovation, evaluate its feasibility, and communicate your recommendations through an advocacy pitch. This task allows you to apply what you have learned in different subjects to a real community concern while considering feasibility, safety, sustainability, cost, and community needs.
Product/ Performance :	Learners will deliver a Community Innovation Advocacy Pitch supported by an Advocacy Brochure. The pitch will present a disaster-related barangay problem, existing coping/adaptation practices, proposed innovation, feasibility findings, and a call to action. The brochure will serve as an audience-friendly summary of the project. The following outputs will serve as evidence of learning: 1. Prototype Development Plan 2. Prototype Adaptability Plan 3. Feasibility and Comparison Report 4. Advocacy Brochure 5. Advocacy Pitch 6. Individual Monitoring Sheet and Work Gallery 7. Reflection Guide Before starting the project, review Appendix A (Innovator's Roadmap) to understand the project phases, sequence of tasks, expected outputs, and learning areas integrated throughout the project. LEARNER INSTRUCTIONS 1. Form your group and assign specific roles based on each member's strengths, interests, and skills. Roles may include leaders, researchers, designers, computation handlers, finance handlers, science explainers, writers, and documenters. Roles may be combined or adjusted depending on group size. Discuss and agree on each member's responsibilities to ensure meaningful participation and collaboration throughout the project. Each member must actively contribute to the project and participate in the final advocacy pitch. 2. Review Appendix A (Innovator's Roadmap) to understand the project phases, sequence of tasks, and expected outputs. 3. Accomplish the Individual Monitoring Sheet (Appendix F) throughout the project to document assigned tasks, accomplishments, means of verification, and supporting evidence. Update the sheet regularly as you complete project activities. Attach evidence in the Work Gallery, such as photos, sketches, computations, drafts, meeting notes, and short narratives. 4. Brainstorm possible disaster-related problems affecting your barangay. Discuss which problems are observable, relevant, and feasible to address through an innovation-based solution. As a group, select one problem to investigate and address throughout the project. Describe the problem by explaining who is affected, where it occurs, and why it is important to address. Record your responses in the Description of Problem subsection of Section I. Barangay Problem of the Prototype Development Plan (Appendix B). 5. Describe how the identified problem is connected to the barangay's geography, culture, way of life, traditions, experiences, or history. Understanding these factors can help you better understand the problem and design a solution that is appropriate, acceptable, and sustainable for the barangay and its residents. Record your responses in the Community Context subsection of Section I of the Prototype Development Plan (Appendix B). 6. Investigate the existing coping, preparedness, or adaptation practices used by the barangay or residents in addressing the identified problem. Gather information through observations, interviews, barangay documents, reports, plans, or other credible sources. Complete Section II. Existing Coping/Adaptation Practices of the Prototype Development Plan (Appendix B) by identifying the

	supporting evidence, describing the existing practice, and analyzing its strengths and limitations. 7. Develop a proposed innovation that addresses the identified barangay problem based on your analysis of the problem, community context, and existing coping/adaptation practices. The innovation should respond to the limitations of existing practices and be appropriate, acceptable, and sustainable for the barangay and its residents. Complete Section III. Proposed Innovation of the Prototype Development Plan (Appendix B) by describing the innovation concept, identifying the intended beneficiaries, explaining how the innovation responds to the community context identified in Section I, and discussing the science or Physics concept(s) that support its design and function. 8. Prepare a scaled sketch, blueprint, layout, or diagram of the proposed innovation. Clearly indicate the scale used and label the dimensions and measurements needed to understand the design. Attach the visual model and complete Section IV. Visual Model with Scaling of the Prototype Development Plan (Appendix B). 9. Identify the key measurements, quantities, units, and design specifications used in the proposed innovation. Complete the Key Measurements and Design Specifications table in Section IV of the Prototype Development Plan (Appendix B) by indicating the component or part, measurement or quantity, unit, and purpose or function of each item. 10. Complete the Prototype Adaptability Plan (Appendix C). Real-world solutions often need to be adjusted to meet changing community needs, resources, or environmental conditions. Using the information gathered about the barangay's geography, culture, livelihood, population, available resources, or environmental conditions, select a realistic scale-up or scale-down scenario (e.g., larger or smaller service area, increased or decreased number of users, limited budget, availability of materials, or changing environmental conditions). Revise the measurements as needed, show the required scale and conversion computations, and explain how the changes affect the design and its intended use. 11. Analyze the feasibility of the proposed innovation using the Feasibility and Comparison Report (Appendix D). Estimate the costs and expected benefits of the proposed innovation and determine whether it is practical to implement in the barangay. Consider factors such as efficiency, safety, sustainability, accessibility, available resources, and community needs. Support your analysis with appropriate computations, tables, graphs, or other evidence when needed. 12. Using the results of your investigation and feasibility analysis, compare the existing coping/adaptation practices and the proposed innovation. Evaluate their effectiveness, cost, efficiency, safety, sustainability, accessibility, and community impact. Record your findings in the Solution Comparison section of the Feasibility and Comparison Report (Appendix D) and justify whether the proposed innovation should be adopted, improved, or reconsidered. 13. Create an Advocacy Brochure using the required contents listed in Appendix E. Present the community problem, community context, existing coping/adaptation practices, proposed innovation, visual model with scaling, key measurements and design specifications, adaptability plan, Physics explanation, financial feasibility, solution comparison, Filipino paninindigan, and call to action in a clear, organized, and audience-friendly format. 14. Submit the Prototype Development Plan, Prototype Adaptability Plan, Feasibility and Comparison Report, and Advocacy Brochure to the evaluation panel at least three (3) days before the advocacy pitch. 15. Deliver a group advocacy pitch in which every member speaks and presents a part related to their assigned role. Present the identified barangay problem, community context, existing coping/adaptation practices, proposed innovation, feasibility and comparison findings, Filipino paninindigan, and call to action. Support your recommendations with evidence from your investigation and analysis. Explain how the proposed innovation addresses community needs and why it should be considered for adoption, improvement, implementation, support, or sponsorship. The pitch will be presented to a panel composed of subject teachers and invited stakeholders, such as barangay officials, SK officials, community leaders, or other relevant representatives. After the presentation, participate in a question-and-answer session and respond to the panel's questions based on your assigned role and project findings. 16. After the advocacy pitch, complete the Reflection Guide (Appendix G). Reflect on your role, contributions, collaboration, challenges, learning, and personal growth throughout the project. Submit the completed Reflection Guide together with the Individual Monitoring Sheet and Work Gallery (Appendix F).
Standards	: The Community Innovation Advocacy Pitch will be assessed through multiple measures to capture learners' knowledge, skills, collaboration, and reflection. The Panel-Rated

	Rubric evaluates Community Context and Socio-Cultural Responsiveness, Physics Application and Scientific Justification, Scale, Measurement, Conversion, and Representation, Communication and Advocacy, Collaboration and Healthy Relationships, and Integration of Learning Areas and Advocacy Brochure. The Engagement, Reflection, and Personal Growth Assessment evaluates learners' role performance, project engagement, self-reflection, and personal growth. Invited stakeholders may also provide feedback on the relevance, responsiveness, feasibility, potential impact, and persuasiveness of the proposed innovation. Learners will have successfully completed the task when they present a community-based innovation that is relevant, feasible, evidence-based, and responsive to local needs; demonstrate accurate application of scientific, mathematical, communication, and socio-cultural concepts; collaborate effectively with others; and communicate a clear and persuasive advocacy message supported by appropriate evidence and reasoning.
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